

Bill of material (BOM)

Sl. No.	Material Code	Description	PR	Qty	Spec
1	TC9765132581	DSH 6"300RF-INTEG TCV 1"600RF-HART	7000107199	1	TC65132
2	TC9765132603	DSH 4"300RF-NON INTEG TCV 1"600RF-HART	7000107340	1	TC65132
3	TC9765132522	SET OF POSITIONERS	7000107342	2	TC65132
4	TC9765132530	SPARE AIR FILTER REGULATOR-DS	7000107343	2	TC65132
5	TC9765132590	DSH 6"300RF-INTEG TCV 1"600RF-SPARE	7000107344	1	TC65132
6	TC9765132611	DSH 4"300RF-NON INTEG TCV 1"600RF-SPARE	7000107345	1	TC65132
7	TC9765132700	DSH 24"300RF-NON INTEG TCV 1"600RF-HART	7000108823	1	TC65132
8	TC9765132719	DSH 24"300RF-NON INTEG TCV 1"600#-SPARE	7000108826	1	TC65132

NOTE:

- Overall L1 Evaluation, Vendor has to quote for all the above items as one set any deviation in this regard is not acceptable, refer purchase specification.
- Quoted price should be for all the items in BOM with total quantity, including packing, freight, insurance with GST in Gem portal.
- Vendor To Submit Detailed QAP Including All Inspection & Test Requirements As Per Material Specification, Applicable Standards
- Delivery: FOR Destination
- Guarantee/warranty: As per purchase specification attached.
- Open Tender will be floated in GEM with a condition that if offers received from other than PMD/CAV. Vendor acceptance is subject to TC Engg/end customer approval for further evaluation.
- Terms & conditions as per GEM ATC.

The following documents are hereby enclosed.

- (i) **BHEL technical specification & GEM ATC.**
- (ii) **PQC**
- (iii) **QAP - HYQA_SQP_TC_DESUPERHEATER**
- (iv) **Special contract conditions (SCC)**
- (v) **Timeline annexure**

****Please submit the above duly filled in, signed and stamped.**

Signature of the supplier with seal

SPECIAL CONTRACT CONDITIONS (SCC)

Sl. No	Terms & Conditions	Supplier confirmation	Remarks
1	Third party inspection by BHEL empaneled TPI and inspection charges to BHEL account. Inspection calls shall be raised by supplier in the below portal cqir.bhel.in or mail us at vvrk@bhel.in & nagesht@bhel.in BHEL empaneled inspection agencies as below (i) M/ s.TUV SUD or M/ s.INTERTEK		Non Deviatable
2	Delivery : 20 weeks from the date of PO (GEM PO).		
3	All items are to be supplied in line with BHEL specification. Deviation, if any shall be brought out clearly in the technical offer.		To be Specified
4	All the original documents like CQIR report, Test Certificates, IBRs (if any), in line with Quality plan and signed by TPI must be submitted before dispatch of materials to vvrk@bhel.in and originals must be submitted along with the materials.		Non Deviatable
5	Warranty: Refer purchase specification Unless otherwise specified, guarantee period shall be 12 months from the date of commissioning or 18 months from the date of supply/ replacement whichever is earlier. A guarantee period as described above shall apply afresh to replaced, repaired or re-executed parts of a delivery.		Non Deviatable
6	Payment term: As per GEM ATC attached.		

Additional Notes:

- (i) Tender evaluation shall be on overall L1 basis
- (ii) All the bidders will be proposed to BHEL (TC Engg) for approval. Offers of approved vendors will only be considered for further evaluation.
- (iii) **Penalty clause:** In the event of delay in supply of goods, penalty of 0.5% of the contract value of delayed quantity per week or part of the week of delayed period as pre-estimated damages not exceeding 10% of the contract value of delayed quantity without any controversy/ dispute of any sort whatsoever.. Penalty amount so determined along with applicable GST thereon shall be recovered.
- (iv) **Price Validity:** Unless otherwise specified, offer shall be valid for a period of 90 days from the date of bid opening (Technical bid /Part-I in case of two part bid).
- (v) Supplier has to submit GAD and project specific Quality Plan for approval after placement of PO.
- (vi) Delivery: FOR Destination
- (vii)** Consignee Address: BHEL RC Puram, Hyd-32.
- (viii) All the other terms are as per ATC GEM.

Date:

Signature of the supplier with seal

Annexure-A				
Major Activity timelines shall be considered for indigenous purchases				
S.NO	Actiivity	Agency	Timeline	Acceptance/Remarks
1	PO acknowledgement	Vendor	04 days from receipt of PO	
2	Submission of Drawings and QP	Vendor	15 days from receipt of PO	
3	Approval of Drawings and QP	BHEL/Customer	07 days from receipt of Rev 00 Drawings and QP	
4	Submission of Revised Drawings and QP (if any)	Vendor	07 days from receipt of commented Drawings and QP	
	Approval of Revised Drawings and QP	BHEL/Customer	07 days from receipt of Revised Drawings and QP	
4	Raising of Inspection Call	Vendor	07 days before the proposing Inspection Date.	
5	Inspection completion	BHEL/Third party inspection agency	07 days from inspection call date	
6	Despatch Instructions	BHEL	07 days from receipt of final inspection clearance report to BHEL.	
7	Reciept of Material	Vendor	15 days from Despatch instructions	

Date:

Signature of the supplier with seal

TD-201 Rev No. 00 Form No.	 HYDERABAD	PRODUCT STANDARD HYDERABAD		TC 65132									
				Rev No. 03									
				Page 1 of 28									
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED, It must not be used directly or indirectly in any way detrimental to the interest of the company.		STANDARD SPECIFICATION FOR DESUPERHEATER & TCV, PRDS 1. GENERAL 1.1. Scope 1.1.1. This specification, together with the data sheets describes the requirements for the design, materials, fabrication, documentation, nameplate marking, inspection, testing and shipping of desuperheater assembly (along with steam Stud connections, 1 meter pipe & liner(if required)) and PRDS. 1.1.2. The related standards referred herein and mentioned below shall be of the latest edition of the following codes, standard practices and publications, unless otherwise specified. ASME: American Society of Mechanical Engineer. i. B 1.20.1: Pipe Threads, General Purpose (Inch) ii. B 16.5: Pipe Flanges and Flanged Fittings iii. B 16.20: Metallic Gaskets for Pipe Flanges, Ring Joint, Spiral wound and jacketed iv. B 16.34: Valves — flanged, Threaded and Welding End v. B 16.47-B: Large Diameter Steel Flanges ANSI/FCI: American National Standard Institute/Fluid Control Institute FCI 70-2: Control Valve Seat Leakage ASTM: American Society for Testing and Materials EN: European Standards i. 10204: Inspection Documents for Metallic Products. IBR: Indian Boiler Regulation API: American Petroleum Institute i. 6D: Pipelines valves (Gate valve, Plug, Ball and Check valves) ii. 598: Inspection Requirements. iii. 609: Double Flanged, Lug and wafer type, Butterfly valves IS/IEC: Indian Standards/International Electro-Technical Commission i. IS/IEC 60079: Electrical Apparatus for Explosive Gas Atmosphere ii. IS/IEC 60529: Degree of Protection Provided by Enclosures iii. IEC 60534-3-2: Face-to-Face dimension of rotary valves except butterfly valves iv. IEC 60534-4: Seat leakage v. IEC 60534-8.3: Industrial process control valves, noise consideration, control valve aerodynamic noise prediction method vi. IEC 60534-8.4: Industrial process control valves, noise consideration, prediction of noise generated by hydrodynamic flow vii. IEC 61508: Functional Safety of Electrical/Electronic/Programmable Electronic Safety related Systems viii. IEC-61511: Functional safety instrumented system for the process industry sector ix. IEC-61804: Function Blocks (FB) for Process Control: Electronic Device Description ISA: International Society of Automation - i.75.01.01: Flow equation for sizing control valves. - ii.75.02: Control valve capacity test procedure. - iii.TR75.04: Control Valve positioner stability - iv.75.05.01: Control Valve terminology - v.75.07: Laboratory measurement of aerodynamic noise generated by control valves											
		<table border="1" style="width: 100%;"> <tr> <td style="width: 15%;">Ref. Doc</td> <td style="width: 30%;">Revisions :</td> <td style="width: 30%;">Prepared:</td> <td style="width: 20%;">Approved:</td> <td style="width: 5%;">Date:</td> </tr> <tr> <td></td> <td>Refer to record of revisions :</td> <td style="text-align: center;">SANDHYA</td> <td style="text-align: center;">RAM</td> <td style="text-align: center;">14.03.20</td> </tr> </table>				Ref. Doc	Revisions :	Prepared:	Approved:	Date:		Refer to record of revisions :	SANDHYA
Ref. Doc	Revisions :	Prepared:	Approved:	Date:									
	Refer to record of revisions :	SANDHYA	RAM	14.03.20									

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 2 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		vi.75.08.01: Face-to-Face, dimensions for integral flanged Globe-style control valve bodies (ASME Class 125,150,300 and 600) vii.75.08.02: Face-to-Face, dimensions for flangeless control valve (ASME Class 150,300 and 600) viii.75.08.04: Face-to-Face dimensions for Butt-weld end globe style control valve (ASME Class 4500) ix.75.08.05: Face-to-Face dimensions for Butt-weld end globe style control valve (ASME Class 150, 300, 600, 900, 1500 and 2500) x.75.08.06: Face-to-Face dimensions for flanged Globe-style control valve bodies (ASME Class 900, 1500 and 2500) xi.75.11.01: Inherent flow characteristic and Range ability of control valves xii.75.19.01: Hydrostatic testing of control valves xiii.75.22: Face to Centre-line dimensions for flanged Globe-style Angle control valve bodies (ASME Class 150,300 and 600) xiv.75.25.01: Test Procedure for Control valve response measurement for step inputs xv.TR 75.25.02: Control valve response measurement for step inputs xvi.75.13: Method of evaluating the performance of positioners with analog input signals and pneumatic output xvii.75.17: Control valve aerodynamic noise prediction xviii.RP75.23: Considerations for evaluating control valve cavitation MSS: Manufacturer's Standardization Society. SP25 Standard Markings System for Valves, Fittings, Flanges and Unions OSHA Occupational Safety and Health Authority 1.1.3. ORDER OF PRECEDENCE: In the event of any conflict between this standard specification, job specifications/data sheets, statutory regulations, related standards, codes etc., the following order of priority shall govern: - Statutory regulations. - Job Specifications / Data Sheets. - Standard specification. - Codes and standards. 1.1.4. In addition to meeting purchaser's specifications in totality, vendors' extent of responsibility shall also include the following: - Sizing and selection of desuperheater, pressure reduction and desuperheating station as per purchaser's datasheets. - Detailed design of desuperheater and PRDS including mechanical design of its components and internals like stem, plug assembly, atomizing nozzle, sleeve, seat assembly, bonnet, packing etc. - Purchaser's data sheets indicate the minimum requirements of material of construction for the body, trim (i.e. stem, plug and seat), atomizing nozzles, sleeve, packing, gasket etc. Alternate superior material of construction shall also be acceptable provided vendor assumes complete responsibility for all the parts of the desuperheater/ PRDS so as to be compatible with the process fluid and its operating conditions. - Checking for cavitation, flashing and noise generated and provide suitable trim and treatment to limit these within appropriate limits.		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 3 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		1.2. BIDS 1.2.1. Vendor's quotation shall be strictly as per the bidding instructions to vendor attached with the Material Requisition. Illustration: a) BHEL RFQ shall contain a list of BHEL material codes / Purchase Requisition numbers (PRs) for desuperheater, TCV & PRDS and its spares and / or special requirements like IBR etc. b) Vendor shall offer its quotation in the following manner in the Price Schedule included with RFQ. c) Indicate unit price and total price for the material portion including MAIN ITEMS and SPARE ITEMS (as per RFQ). 1.2.2. Whenever a detailed technical offer is required, vendor's quotation shall include the following: a) Compliance to the specifications. b) A detailed specification sheet for each item shall be furnished, which shall provide information regarding type, principle of operation, material of construction, flow co-efficient (Cv) data, % opening, rangeability, leakage class, capacity and end connection etc. of the desuperheater and its accessories for each desuperheater. The material specification and the units of measurement for various parts in vendor's specification sheets shall be to the same standards as those indicated in purchaser's data sheets. c) A detailed sizing sheet for each desuperheater/PRDS including pressure reduction control valve and coolant control valve, as required shall be submitted. The sizing sheet shall also include the following information for each desuperheater/PRDS; i. Minimum Velocity Requirement ii. Maximum Turndown iii. Minimum coolant pressure, flow and temperature requirement iv. Maximum coolant pressure requirement v. % opening of desuperheater, temperature control valve and pressure control valve at minimum/maximum condition. vi. Expected Outlet steam temperature with respect to saturation temperature. vii. Full open flows for pressure reducing valve and temperature control valve considering maximum inlet pressure/temperature. d) Vendor to indicate minimum upstream and downstream pipe straight length requirements and location of temperature sensor downstream of desuperheater. e) A copy of approval from local statutory authority, as applicable, such as Petroleum and Explosives Safety Organization(PESO)/ Chief Controller of Explosives (CCE), Nagpur or Director General of Mining Safety (DGMS) in India, for the electronic instruments installed in electrically hazardous area along with: i. Test certificate from recognized test house like Central Institute of Mining and Fuel Research(CIMFR)/Electronics Regional Testing Laboratory (ERTL) etc. for flameproof enclosure and/or intrinsic safety, as specified in the data sheets, as per relevant Indian standard for all Indian manufactured equipment or for items requiring DGMS approval.		
Ref. Doc				

TD-201 Rev No.00		Form No.	बी एच ई एल BHEL	PRODUCT STANDARD		TC 65132
				HYDERABAD		Rev No. 03
						Page 4 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.				ii. Certificate of conformity from agencies like LCIE, Baseefa, FM, PT, CSA, UL etc. for compliance to ATEX / IEC directives or other equivalent standards for all equipment manufactured outside India etc. f) Deviations on technical requirements shall not be entertained. In case vendor has any valid technical reason, they must include a list of deviations tag number wise, summing up all the deviations from the purchaser's data sheets and other technical specifications along with the technical reasons for each of these deviations. g) Catalogues giving detailed technical specifications, model decoding details and other related information for each model of desuperheater/PRDS covered in the bid. h) Schematic drawing indicating scope of supply of desuperheater/ PRDS, coolant control valves, pressure reducing valves, strainers etc. as applicable along with end connections and ratings shall be furnished. 1.2.3. All items, as offered, shall be field proven and should have completed trouble free satisfactory operations individually for a period of minimum 4000 hours on the bid due date in the similar application with the process conditions similar to those as specified in the purchaser's data sheets. Items with prototype design or items not meeting provenness criteria specified above shall not be offered.		
				1.3. DRAWING AND DATA 1.3.1. Detailed drawings, data, catalogues and manuals required shall be submitted by the vendor as per Vendor Data Requirements (VDR) attached with the requisition. 1.3.2. Final documentation consisting of design data, installation manual, operation and maintenance manual etc. shall be submitted by vendor after placement of purchase order which shall include the following as a minimum: a) Specification sheet for each desuperheater, PRDS, related control valves and other accessories, as applicable. b) Certified drawing sheets for each desuperheater, PRDS, related control valves and their accessories, tag number wise, as applicable, which shall provide the following information: i. Overall dimensions in millimeters. ii. Clearance space required for maintenance work in millimeters. iii. Material of construction. iv. Weight of complete assembly. c) Sizing sheet for each desuperheater/PRDS/control valve. d) Copy of type test certificate, as applicable. e) Copy of test certificates for all the tests indicated in this specification. f) Installation procedure for desuperheater/ PRDS/ Control valves. g) Dimensions of clearance space for maintenance work, weight of completely assembled valve, internal constructional details, materials of construction of actuator, wiring diagram, installation procedure, orientation with respect to flow direction marked on the body, sizing codes, approvals & certifications. h) Copy of the type test certificates and the test certificates for all tests indicated in Inspection Test Plan. i) Calibration, configuration and maintenance procedures including replacement of its parts / internals wherever applicable.		
Ref.	Doc					

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 5 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		j) Control valve operating signatures whenever smart positioners are specified. k) Device Descriptor (DD) Files for configuring Smart positioner parameters. 2. DESIGN AND CONSTRUCTION 2.1. GENERAL 2.1.1. Vendor shall offer type of desuperheater/PRDS as per the type specified in the purchaser's data sheets, ensuring that the selected desuperheater meets the process requirements specified in data sheets, particularly the pressure and temperature characteristics of the process fluid at the outlet of the desuperheater. In case, vendor offers desuperheater of a type other than that specified, they must indicate the specific technical reason along with their offer. 2.1.2. The coolant injection of desuperheater/PRDS shall be solely responsible for coolant atomization and shall ensure uniform atomization of the coolant fluid without requiring separate atomizing fluid. 2.1.3. Liner a) Vendor must review the design of desuperheater/PRDS with respect to the specified process conditions and recommend liner if necessary, to improve the system turndown i.e. to increase the steam velocity at the point of water injection. b) Vendor shall review and provide Thermal liner, if necessary, for the protection of steam line, particularly where the difference between the coolant temperature and process fluid temperature exceeds 250°C and/or where pipe thickness exceeds 12mm. c) The diameter, length, material and design of the liner shall be decided by the vendor. The liner material as selected shall be suitable for the process conditions specified in the purchaser's data sheets. 2.1.4. The flow direction of the process fluid shall be clearly marked preferably stamped or cast on the desuperheater/ PRDS/ control valve body. In case where direction is stamped on the body, vendor must ensure that stamped portion is unambiguously clear and visible. 2.1.5. The desuperheater, PRDS, pressure reducing valve and coolant valve body rating shall be equal to or better than the flange rating specified in the data sheet. 2.1.6. Material of construction a) The material of construction of body and trim of desuperheater, PRDS and control valves shall be as specified in the purchaser's data sheet. The term trim covers those parts of desuperheater, PRDS and control valve assembly, excluding the body, which are exposed to and are in contact with the line fluid consisting of but not limited to the seat ring, plug, stem, atomizing nozzles, coolant sleeve, plug guide, guide bush and cage. b) For the internals where no material of construction is explicitly indicated in the purchaser's data sheet, vendor shall select the material based on their standard design but ensuring suitability of the material for the specified process conditions. c) Whenever purchaser's data sheet specifies trim material as hardened SS or stellited SS, vendor may select alternate material as Nitrided plated SS /solid satellite/440C/17-4 PH SS provided the selected material is compatible with the specified process conditions.		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 6 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		d) Vendor shall ensure that selected material, surface finish, hardness and clearances for the parts, which slide against each other, shall avoid galling effect. In addition, valve stem surface finish shall be selected considering type of packing used. e) Unless specified otherwise in purchaser's data sheet, the material of construction of desuperheater or BFW control valve Trim/Internals shall be as follows: Seat: Stellite Plug: Nitrided SS / Stellite SS / Solid Stellite Desuperheater Nozzles: Stellite Desuperheater Internals: SS316 Cage: SS Nitrided / 17-4 PH SS f) Unless otherwise specified, the Nut-bolt and gasket material shall be as follows: i. Nut & Bolt: For Pressure rating up to 1500#, Stud Bolt & Nut shall be ASTM A193 Gr. B7 & ASTM A194 Gr. 2H respectively. For pressure rating of 2500#, bolt & Nut shall be as per ASTM A194 Gr.4 & ASTM A193 Gr. B16 respectively. Nuts / bolts shall be hot dip galvanized. ii. Gasket: For line rating up to 600#. gasket shall be Sp. Wnd + SS316 + grafil + I-ring. For line pressure rating of 900# and 1500#, Gasket shall be Octagonal Ring of rating 900# or 1500# respectively of MOC Soft iron. For line pressure of 2500#, Gasket shall be 5% Cr 0.5% Mo (130 BHN MAX) Oct-Ring, 2500#. g) The body and trim design and material of construction shall be such that the unit is capable of withstanding excessive stresses produced due to pressure reduction and thermal quenching because of coolant. h) The plug and seat assembly shall be of sufficient hardness to avoid any wear and tear of these components. i) The material of construction of control valves body shall be as specified in the data sheet / variant table. No material shall be substituted by vendor without specific written approval from purchaser. j) Control valve body, bonnet, bottom flange, line flanges and other pressure containing assemblies shall be of the same material of construction as specified for valve body in the purchaser's data sheets. k) The bonnet flange and bottom flange shall have metallic spiral wound gaskets suitable for the specified service. The gaskets with asbestos bearing fillers shall not be used. l) Vendor shall be responsible for selecting proper material for the internal parts of control valve. All such materials shall have the same or better specification than specified in data sheet / variant table. m) The material of construction of silencers, diffuser plates, diffuser plate assembly etc. shall be as per the body material specified in the data sheet / variant table, as a minimum. n) Minimum acceptable materials of construction to be considered by vendor for valve components are already specified in datasheet / variant table. Carbon content shall not exceed 0.25% for all pipes, fittings, flanges etc. that may		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 7 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		require welding. Moreover, for flanges the Sulphur and phosphorous shall also be limited to 0.05% each. o) However, alternate superior materials may be provided if they meet all process conditions (operating, design, fluid properties, corrosion, etc.), testing (ASME, API, radiography, etc.) and statutory requirements (ASME, API, IBR, NACE, etc.) specified by purchaser. Accordingly, the following alternate materials can be accepted: Body & bonnet: ASTM A 105, ASTM A 216 gr. WCB / WCC in place of carbon steel. 2.1.7. END CONNECTIONS a) Unless otherwise specified, the following shall govern: i. Threaded end connections shall be to NPT as per ASME B1.20.1. ii. Flanged end connections shall be as per ASME B 16.5 / B 16.47B / AWWAA C207 CL.D. iii. Flange face finish shall be as per ASME B16.5. The face finish shall be as follows: 125 AARH: 125 to 250 micro inch AARH. 63 AARH: 32 to 63 micro inch AARH. iv. Grooves of ring type joint flanges shall be octagonal as per ASME B 16.20. 2.1.8. Vendor shall supply two (2) numbers of strainers for installation in coolant line. Unless specified otherwise, these strainers shall be installed in 1:1(one main and one standby) configuration by Purchaser in coolant line. 2.1.9. The coolant control valve shall be of leakage class V as per ANSI/ FCI 70-2. 2.2. DESUPERHEATER 2.2.1. ATOMIZING SPRAY NOZZLE TYPE DESUPERHEATER 2.2.1.1. Atomizing spray nozzle type of desuperheater shall have insertion type of design and shall consist of a coolant injection assembly with nozzle / nozzles inserted perpendicular to the process line. The length of insertion shall be such that the uniform vaporization of coolant shall take place throughout the cross section of the desuperheater internally avoiding impingement of cooling media on pipe walls. 2.2.1.2. The desuperheater shall be either single nozzle or multi nozzle design as per the vendor standard design. However, vendor shall ensure that design of offered type of desuperheater shall avoid problems related to larger coolant droplet size and poor atomization. 2.2.1.3. In case of multi nozzle desuperheater design, the number of nozzles required in the atomizer shall be decided based on the process requirements specified in purchaser's data sheets. 2.2.1.4. The desuperheater shall have an integral control valve, which shall regulate amount of coolant in to the desuperheater. This valve along with a temperature transmitter and temperature controller shall form part of the temperature control loop, which shall be able to control the temperature within 6 to 7°C of the saturation temperature. Unless otherwise specified, temperature element / transmitter and temperature controller are beyond vendor's scope of supply. 2.2.1.5. Whenever specified or necessitated by the desuperheater design, a separate temperature control valve shall be used. Special Trim like multipath / multistage trim shall be selected when necessary for coolant control valve.		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 8 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		2.2.1.6. The inherent characteristics of the integral coolant control valve shall be equal percentage / modified linear. 2.2.1.7. Integral coolant control valve shall have leakage class V as per ANSI/ FCI 70-2, as a minimum. 2.2.1.8. End Connection: a) Unless specified otherwise in purchaser's data sheet, the desuperheater shall be supplied with following end connections: Body connection: 3" FLGD or higher as per variant table / datasheet. Coolant connection: 1" FLGD or higher as per variant table / datasheet. b) The rating and facing of end connection flanges shall be as specified in data sheet. 2.2.2. VENTURI TYPE DESUPERHEATER 2.2.2.1. Venturi type of desuperheater shall be a line-mounted desuperheater wherein coolant is preheated in the circulating chamber around the water diffuser tube and is introduced in many small jets to achieve atomization. Such a desuperheater shall be capable of installation in horizontal as well as in vertical line. 2.2.2.2. This type of desuperheater shall require a separate temperature control valve, which along with temperature controller shall form part of temperature control loop. This desuperheater shall be able to control the outlet temperature within 5°C to 6°C of the saturation temperature of the line fluid. Unless specified otherwise, temperature element / transmitter and temperature controller are beyond the vendor's scope of supply. 2.2.2.3. End connection: a) Venturi type desuperheater shall generally be offered with wafer type of design suitable for mounting in between line flanges. The line reducers/expanders if required shall not be in vendor scope of supply. Long stud-bolts and set of gaskets required for the installation of desuperheater shall be supplied by the vendor. b) Unless otherwise specified in purchaser's data sheets, the end connections for the coolant shall be 1" flanged. 2.2.3. LIFTING PLUG (VARIABLE ORIFICE) TYPE DESUPERHEATER 2.2.3.1. Lifting type of desuperheater shall be in-line mounted desuperheater, which shall desuperheat the process fluid flowing through it by the coolant, which is injected into the low pressure area created by a lifting plug in the direction of flow over the seat area. The design shall ensure that the injection of the coolant into a low-pressure area and the subsequent turbulence in the process stream shall result in complete atomization and vaporization of the coolant by the superheated line fluid, which shall subsequently desuperheat. 2.2.3.2. This desuperheater shall require a separate control valve, which along with temperature controller shall form part of temperature control loop. This desuperheater shall be capable of controlling outlet temperature within 5°C to 6°C of the saturation temperature of the line fluid.		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 9 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		2.2.3.3. Vendor to ensure that the maximum pressure drop across the desuperheater shall be less than allowable pressure drop indicated in data sheets. 2.2.3.4. The desuperheater body shall be flanged and shall be mounted in the line always in vertical position. 2.2.3.5. Unless specified otherwise, the material of construction of plug shall be SS316 while plug seating and seat shall be SS 316 stellited. 2.3. PRESSURE REDUCTION AND DESUPERHEATING STATION (PRDS) 2.3.1. Whenever specified, PRDS shall have pressure reduction along with the desuperheating of process fluid as a complete assembly. Type of PRDS (as per clause 2.3.2 and 2.3.3 of this specification) shall be offered as per Purchaser's datasheet. 2.3.2. COMBINED PRESSURE REDUCTION AND DESUPERHEATING UNIT (CPRDS): 2.3.2.1. Combined pressure reduction and desuperheating unit shall reduce pressure and temperature of the process fluid simultaneously in one common unit. The design shall ensure that desuperheating is achieved by injecting coolant in the center of the line flow i.e. in an area of high velocity, minimum pressure and very high turbulence through a coolant sleeve and atomizing injection nozzles thus contributing to very efficient mixing and very high evaporation of the coolant. The pressure reduction shall be achieved by multipath / multistage trim assembly design. 2.3.2.2. In case noise level is sufficiently high, vendor shall consider special PRDS design with low noise trim. In case noise still exist, diffuser plates suitable for design condition shall be provided at the outlet of PRDS to limit the noise level below the specified limits. The stud-bolts, nuts and gasket material shall be as per section 2.1.6-f. Design shall ensure that the performance of PRDS is insensitive to the negative effect of coolant evaporation because of presence of diffuser. 2.3.2.3. The rangeability of PRDS shall be atleast 50 to 1 even when diffuser plates or silencer is provided to reduce the noise level. 2.3.2.4. Unless otherwise specified, the PRDS shall have leakage class IV as per ANSI/ FCI 70-2. 2.3.2.5. Inherent flow characterization of PRDS shall be modified linear, in general. 2.3.2.6. The unit shall be capable of controlling temperature within 6° C of saturation temperature of process fluid. A separate control valve shall form part of temperature control loop consisting of temperature transmitter and temperature controller. Unless otherwise specified, temperature element / transmitter and temperature controller shall be beyond vendor's scope of supply. 2.3.2.7. The pressure control shall be achieved by a downstream pressure sensor / transmitter and pressure controller, which regulates the pressure regulator of PRDS through its integral actuator. Unless specified otherwise, pressure transmitter and pressure controller shall be beyond vendor's scope of supply. 2.3.2.8. End Connection: a) PRDS shall have flanged end connections and shall be installed directly in the line. b) Unless otherwise specified, coolant connection shall be 1" flanged.		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 10 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		2.3.3. SEPARATE PRESSURE REDUCTION AND DESUPERHEATING UNIT (SPRDS) 2.3.3.1. PRDS with separate pressure reduction and desuperheater assembly construction shall be offered when specifically indicated in the Purchaser's datasheet. Such a unit shall be supplied as a complete assembly along with line reducers/expanders and interconnecting piping. The pipe section between pressure reducing valve and desuperheater (material same or different than that is indicated in purchaser's specifications) shall also be supplied by vendor. 2.3.3.2. Complete assembly shall be supplied with flanged end connection including control valve and desuperheater, which shall also have flanged end connections. The type and rating of flanged ends shall be as specified in purchaser's data sheets. 2.3.3.3. In the pressure reducing and desuperheating assembly dedicated pressure reduction control valve shall be located upstream of the desuperheater and shall form part of pressure control loop. Unless specified otherwise, pressure transmitter and pressure controller shall be beyond the vendor's scope of supply. 2.3.3.4. PRDS assembly shall be sized as per the process conditions specified in the purchaser's data sheet. However, the pressure drop indicated in purchaser's data sheet may be readjusted across desuperheater and control valve keeping battery limit pressure drop unchanged. Desuperheater end connection rating if required to be upgraded because of pressure drop readjustment shall be taken care of by vendor. 2.3.3.5. The pressure reducing control valve trim design shall be special low noise type. Control valve with external diffuser may be offered along with the special trim when special trim alone is not sufficient to reduce the noise level below the specified limits and the pressure drop across the diffuser does not affect the pressure control loop sensitivity / controllability. Diffuser shall be suitable for design condition. 2.3.3.6. All other desuperheater requirements shall be similar to the type of desuperheater selected alongside the pressure-reducing valve. 2.4. SIZING CRITERIA 2.4.1. The sizing of the desuperheater / PRDS shall be carried out as per the process conditions specified in the purchaser's data sheets. 2.4.2. Vendor shall consider the following points while sizing the desuperheater, unless specifically indicated otherwise in purchaser's datasheets: - Coolant flow shall be calculated for the worst conditions specified i.e. high coolant temperature for maximum process flow and low coolant temperature for minimum process flow condition, similarly high inlet coolant pressure for minimum coolant flow and low coolant pressure for maximum coolant flow. - Highest and lowest available pressure difference between steam pressure and coolant pressure for minimum flow and maximum flow respectively. - Required turndown ratios i.e. coolant turndown ratios i.e. minimum coolant flow to maximum coolant flow (nozzle turndown) and turndown ratio of the fluid to be desuperheated (System turndown). - Required coolant flow to fluid to be desuperheated. 2.4.3. The flow coefficient (Cv) of the desuperheater shall be selected such that preferably it operates between 20% to 80% of the lift under minimum and maximum conditions.		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 11 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		However, the desuperheater with lift less than 10% and/or higher than 90% shall not be selected even under worst operating conditions. 2.4.4. Vendor must ensure that minimum fluid velocity shall not fall below the allowable minimum fluid velocity of the selected desuperheater. If such a condition exists, vendor shall propose a reduction in the pipe diameter or a liner to keep velocity within limits considering allowable pressure drop specified in the purchaser's data sheet. 2.5. TRIM 2.5.1. The term 'trim' covers those parts of valve assembly (excluding the body, bonnet and bottom flange) which are exposed to and are in contact with the line medium consisting of but not limited to the seat ring, valve stem, valve plug, valve plug guide, guide bushing and cage. 2.5.2. In case of rotary type of control valves like butterfly, ball, segmental ball, rotary plug, eccentric disc and rotary disc, the term trim covers disc / ball, seat ring, shaft and bearing. 2.6. TRIM DESIGN 2.6.1. Control valve manufacturer / vendor shall be responsible for trim selection and trim design of the control valve. However, it must meet the following minimum requirements: 2.6.2. Control valve trim design shall suit the type of guiding required for the application. Trim design shall be quick replacement type. 2.6.3. The trim design and material of construction shall be selected to minimize the risk of galling particularly in case of cage guided valves. Vendor shall select proper material pairs, surface finish, hardness and clearances wherever possibility of galling exist. 2.6.4. Under extreme temperature conditions, vendor shall consider increased clearances at room temperature and seal welding of threaded seat rings etc., Hard facing of trim shall be used in high temperatures. For very low temperature application, material used shall have adequate cold impact strength. 2.6.5. For globe / angle / 3-way type of control valves, stem and plug shall be detachable and shall be attached together by suitable threaded design secured with a pin to avoid plug rotation during operation. 2.6.6. Whenever cavitation conditions are expected, vendor shall select a special anticavitation trim design and shall use trim material of sufficiently high hardness. 2.6.7. Whenever the possibility of aerodynamic noise in a control valve exists under any operating condition specified in data sheet / variant table, vendor shall select a special low noise trim for that application. 2.6.8. The plug inherent characteristics shall be as indicated in the data sheet / variant table. However, following shall be followed unless otherwise specified: 2.6.9. Control valves with low noise and anti-cavitation trim design shall have modified equal percentage or linear characteristics. 2.6.10. Whenever linear or modified equal percentage characteristics are specified, equal percentage characteristics shall also be acceptable. Characterized positioner cam design to meet specified inherent control valve characteristics shall not be offered. 2.7. TRIM MATERIAL		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 12 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		2.7.1. Whenever stellited trim is specified in the data sheet / variant table following material of construction for the trim parts shall be acceptable. - Plug and seat Stellite sheathing / hard alloy coating or solid stellite. - Cage Nitrided or stellite sheathing / hard alloy coating or solid stellite. - Guide bushing Stellite coating / sheathing or solid stellite - Valve stem Hardened SS / Inconel X-760. 2.7.2. In case of steam, water, air and other non-hydrocarbon utility services 17-4 PH SS or 440C is also acceptable. Special material requirements, if required, shall supersede the above mentioned requirements. 2.8. GUIDING 2.8.1. Single seated globe and angle type control valves shall have heavy top plug guiding or cage guided design. The cage shall provide a continuous plug guiding. 2.8.2. Double seated valves shall have top and bottom or cage guiding and shall be of the pressure balanced type. 2.8.3. Whenever cage type control valves are offered, top guided or top and bottom guided control valves can be offered provided it meets all other process and functional requirements. But whenever top or top and bottom type of guiding is required, cage type control valves shall not be offered. However, if the service requires the anti-cavitation trim or low noise trim then in place of top or top and bottom guiding the cage type control valve can be accepted only after reviewing the control valve sizing. 2.9. LEAKAGE CLASS 2.9.1. Leakage class as per ANSI FCI 70.2 shall be specified in the purchaser's data sheet for each control valve. Where no leakage class is specified, the same shall be considered as Class V. Higher leakage class shall be provided wherever indicated in purchaser's data sheet. 2.9.2. For Control valve experiencing cavitation, in case vendor's calculation show occurrence of cavitation, flashing or noise under any of the specified process condition, vendor shall select proper type of valve trim to suit such process conditions. 2.10. SIZING 2.10.1. The control valve capacities in terms of flow coefficient (Cv) shown in the datasheet / variant table. have been calculated based on the formulae given in the standard ISA 75.01 "Control valve sizing equations". All factors used while arriving at the sizing are also as per ISA 75.01. Vendor must resize the control valves considering various factors specific to the offered valves. 2.10.2. Datasheet / variant table indicates calculated flow co-efficient values at normal operating conditions. Vendor shall calculate this co-efficient as per the offered control valve and select the size considering valve openings as under: - At maximum flow less than 85% open - At normal flow typically 75% open - At minimum flow more than 25% open 2.10.3. Rangeability of globe type of valves shall be 30:1, as a minimum. Vendor shall ensure that the actual rangeability of offered valves shall meet these requirements. For other types of offered control valves, vendor to specify that the available rangeability		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 13 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		of the offered valves meet the requirements specified in the datasheet / variant table. 2.11. NOISE 2.11.1. Vendor shall examine each control valve for noise generation possibilities. The noise level shall be calculated as per ISA 75.17 or any other equivalent standard. 2.11.2. Noise generated by control valve during operation shall be limited to OSHA specified levels i.e. the maximum allowable noise shall be less than 85 dBA, when measured at a distance governed by ISA 75.17. 2.11.3. If the predicted noise level is found to exceed 85 dBA SPL, control valve shall be treated for noise. Source treatment for noise shall be resorted to. When source treatment for noise is not sufficient to reduce the noise level below 85 dBA, vendor shall provide path treatment like diffuser plate/silencer etc. in addition to source treatment so as to reduce the level below 85dBA. Whenever additional path treatment is recommended, the maximum differential pressure across the silencer / diffuser plate shall not exceed 40% of the specified differential pressure. 2.11.4. Vendor shall also furnish noise calculations with and without the use of these devices and the noise abatement achieved in individual components. 2.11.5. Diffuser plate shall be inserted between control valve body and pipe flanges. Vendor shall supply long studs and nuts along with a pair of gaskets on the downstream side with material. In cases where the diffuser plate size is higher than the valve body size or there is a need of multiple diffuser plates, vendor shall supply the complete diffuser assembly with flanged end connections matting with the control valve flange and downstream line size. 2.11.6. The silencer whenever offered for noise abatement shall always be supplied with flanged end connections. 2.12. ACTUATOR a) The actuator of the desuperheater and control valves shall be sized for the shutoff differential pressure indicated in the data sheet. b) The actuator shall be pneumatically operated with its casing made up of pressed steel or anodized aluminium. Non-metallic actuator casing shall not be offered. The actuator operating range shall be 0.2 to 1.0 kg/cm² preferably. c) In general, spring opposed diaphragm type of actuators are preferred. Only when this type of actuator becomes extremely unwieldy, piston and cylinder type with single acting spring return can be considered. The actuator casing and diaphragm shall be designed for minimum twice the maximum air supply pressure. d) Whenever double acting spring less type of actuator is unavoidable, all accessories like pilot valves, booster relays, non-return valve, pressure gauge, volume tank etc., shall be provided to achieve desired failure action. The volume tank shall be sized considering full stroking of the desuperheater / vale for three complete cycles. The volume tank shall be of carbon steel / SS construction and sized as per ASME section VIII with design pressure specified in the data sheet. Accessories like pressure relief valves and tubing shall be of 316 stainless steel construction. Other accessories can be of Cast Aluminium.		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 14 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		e) Actuator spring shall be manufactured out of corrosion resistant steel and shall be nickel-plated. Alternately vendor standard coating is also acceptable. The spring shall be enclosed in the actuator casing. f) Each desuperheater actuator and valve actuator shall be provided with stem position indicator with scale calibrated from 0 to 100% in steps of 10%. g) While sizing the actuator, vendor must ensure that the sizing factors indicated below are fully complied. Higher sizing factor may be considered if found necessary by vendor. i. For leakage class IV and below, the actuator shall be sized considering actuator thrust more than 1.3 times the total force induced by shut-off conditions specified in the data sheet and the force required to overcome packing friction. Vendor shall utilize this factor as 1.5 in case the desuperheater/PRDS is operating between 80% to 90% or 10% to 20% in any of the specified conditions. ii. For leakage class V and above, the actuator shall be sized considering actual thrust more than 1.7 times the total force induced by specified shut-off conditions in the purchaser's data sheet and the force required to overcome packing friction. h) The stroke time with positioner for open / close shall be equal to valve body size in seconds. i) Actuator shall be sized for the shut-off differential pressure indicated in datasheet / variant table. j) The actuator shall be designed to move the valve to the failure position specified in the datasheet / variant table. k) Actuator casing shall be made of pressed steel. Non-metallic actuator casings shall not be offered. l) Springs shall be corrosion-resistant and shall be cadmium or nickel-plated. Alternately vendor standard coating shall also be acceptable if accepted by customer. These shall be of the enclosed type. The compression of the springs shall be adjustable. m) In general, an actuator operating range of 0.2-to 1.0 kg/cm²g is preferred. However, when vendor standard actuator model is not able to meet the specified shutoff pressure, higher actuator operating range may be offered. n) In general, spring opposed diaphragms type actuators shall be used. Only when this type of actuator becomes extremely unwieldy, based on the data specified in the datasheet / variant table, should a piston and cylinder type of actuator be considered. o) Whenever piston and cylinder actuator is considered, single acting spring return type shall be used. p) Whenever double acting spring less type of actuator is unavoidable, all accessories like pilot valves, booster relays, non-return valve, pressure gauge, volume tank etc. shall be provided to ensure desired action on air failure. The volume tank shall be sized considering full stroking of the valve for THREE complete cycles. The volume tank shall be of carbon steel, epoxy painted or stainless steel construction (refer datasheet) and sized as per ASME Section VIII with design pressure of 10.5 kg/cm²g gas a minimum. Accessories like pressure relief valves, pressure gauge and tubing shall be of 316/316L Stainless		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 15 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		steel construction. Provision shall be provided for purchaser's pressure transmitter connection. Acceptance for double acting spring less actuator is subject to acceptance by customer. q) The actuator casings and diaphragms shall be designed for minimum twice the maximum pneumatic operating pressure of the control valve. r) Valve stem position indicator shall be provided for every control valve. The position indicator scale shall be marked from 0 to 100% in steps of 10%. s) In general, side-mounted hand wheels will be used. Hand wheels shall provide manual t) control in both opening and closing directions independent of spring action. Hand-wheels shall be of non-rising type suitable for accurate valve positioning. The hand-wheel actuator shall be sized to provide the required thrust for valve positioning with 178N force applied to the hand-wheel. u) Actuator orientation shall be as per purchaser's requirements, in general. When, no requirements are indicated by purchaser; vendor shall provide recommended actuator orientation. It shall be possible to change this orientation at site for the offered valve actuator combination in case it is found necessary. v) ACTUATOR SIZING i. Vendor shall be fully responsible for the sizing and selection of the correct actuator for the specified control valve. Any change in actuator model number necessary to meet specification requirements after placement of order shall be taken care of by vendor without any price or time implication. While sizing the actuator, vendor shall ensure that the actuator is able to develop sufficient thrust to properly seat the control valve plug / disc at the actuator air pressure (lowest specified pressure) and shut-off conditions specified in the data sheet / variant table. ii. While sizing the actuator, vendor must ensure that the sizing factors indicated below are fully complied. Higher sizing factor may be considered if found necessary by vendor. iii. For control valves with leakage class IV and below, the actuator shall be sized considering actuator thrust more than 1.5 times the total force induced by shut-off conditions specified in the data sheet and the force required to overcome packing friction. iv. Vendor shall consider instrument air pressure of 3.5 Kg/Cm2g for actuator sizing. v. For control valves with leakage class V and above, the actuator shall be sized considering actual thrust more than 1.7 times the total force induced by specified shut-off conditions in the purchaser's data sheet and the force required to overcome packing friction. vi. Vendor shall indicate the recommended actuator orientation. In addition, other allowable orientations shall also be provided along with the offer. The orientation of the actuator may be changed after order. The vendor shall do the same without any price or time implication.		
Ref. Doc				

TD-201 Rev No.00		Form No.	बी एच ई एल BHEL	PRODUCT STANDARD		TC 65132
				HYDERABAD		Rev No. 03
						Page 16 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.				2.13. BONNET AND PACKING		
				i. PACKING BOX		
				a) The packing box shall be flanged bolted to the bonnet and shall meet the requirements specified in datasheet / variant table. In general, the gland packing shall be fire safe type.		
				b) Generally low friction type packing like braided Teflon will be used wherever operating conditions permit. For high temperature application (>200°C), Grafoil or equivalent gasket suitable for the specified service shall be selected. Asbestos based packing material shall not be used.		
				c) For application in vacuum service, vendor to provide inverted packing design suitable for vacuum service. For pressure-cum-vacuum service, the control valve shall have dual packing design suitable for the application. Dual packing design shall also be provided for control valves in toxic service.		
				d) Wherever environmental packing design is specified vendor shall supply special packing design suitable to minimize fugitive emission.		
				ii. BONNET		
				a) The bonnet shall be flanged bolted to the body. Threaded bonnets are not permitted.		
				b) Wherever the operating temperature of the fluid is above 200°C, extension or radiation finned bonnet shall be provided.		
				c) For the operating temperature below 0°C, vendor shall provide extended bonnet design. For valves in cryogenic application bonnet extension shall be as per BS-6364 as a minimum.		
				iii. STEM		
				a) The stem surface finish shall be fine. Extra fine surface finish shall be provided wherever the packing material is PTFE.		
				b) The stem / shaft shall be designed for the maximum thrust of the actuator without any measurable deflection.		
				c) The valve stem / shaft shall be connected to the actuator stem / shaft by suitable arrangement to avoid backlash problem.		
				d) The stem shall not be welded with the plug.		
				iv. The desuperheater valve bonnet shall be flanged bolted to the body. Bonnet with threaded connections is not permitted. Extended or finned bonnet shall be provided for low temperature or high temperature applications.		
				v. Generally low friction type of packing is preferred. Wherever external lubrication is required for gland packing lubrication isolation valve shall be provided along with the lubricator. Vendor shall indicate the lubricator stick material and supply the same for one year of continuous operational use.		
				vi. The desuperheater/valve shall have its gland packing box drilled and tapped to 3/4" NPTF for connecting external lubricator, when external lubricator is not provided, the hole shall be plugged.		
				vii. Asbestos or asbestos bearing material gaskets shall not be used.		
				2.14. ACCESSORIES		
				2.14.1. POSITIONER		
				2.14.1.1. Positioners shall be of smart type as specified in the purchaser's datasheet. They shall be direct acting, with an adjustable gain unless otherwise specified.		
Ref.	Doc					

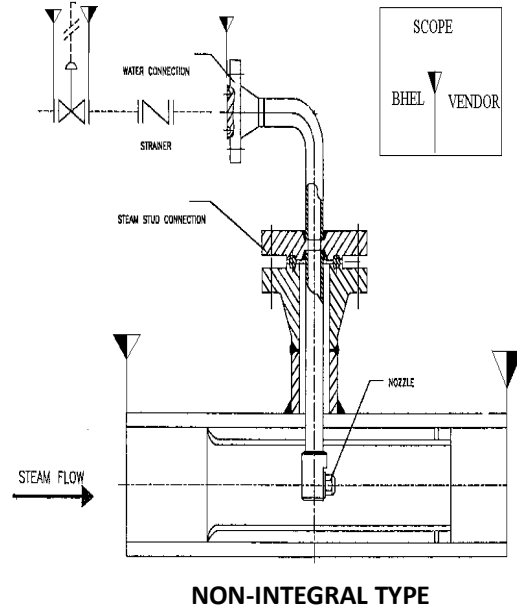
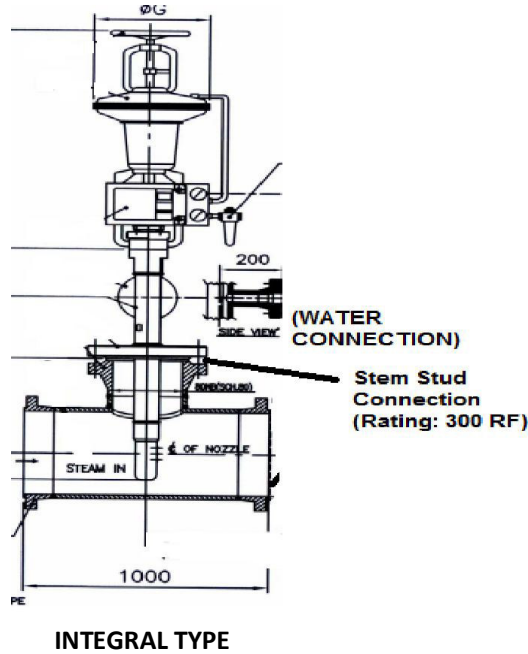
TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 17 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		The positioner shall be a two-wire device, which shall operate on two-way digital communication mode. All engineering, configuration, diagnostic and maintenance related data shall be provided by the positioner. 2.14.1.2. Every positioner shall have two pressure gauges mounted on it, one each for air supply and for positioner output to actuator. 2.14.1.3. Pneumatic connections shall be ¼" NPTF or ½" NPTF and cable entry shall be 1/2" NPTF. If connection for cable entry is different than that specified, suitable adapters shall be provided. 2.14.1.4. Positioners shall be side-mounted on the control valve and shall be contactless & linkage less type. The positioner pressure control valve & electronics shall be housed in separate compartment, i.e. positioner electronics shall be isolated from pressure part. 2.14.1.5. Positioners shall be dual certified for flameproof & intrinsic safe explosion protection, suitable for IEC Zone – 1, IIC. 2.14.1.6. Positioner shall be provided with inbuilt position transmitter, 4-20mA with cable entry of ½" NPTF. 2.14.1.7. The positioner sensor and sensing mechanism shall be rugged and shall not be affected by the line / valve vibration. The performance of the positioners shall be immune to above vibration. 2.14.1.8. The positioner's output and input range shall be field adjustable without any hardware modification. The output from the positioners shall be available for both single acting as well as double acting actuator. 2.14.1.9. Positioner shall be operable, configurable and accessible through HART /FF compatible hand held configurator. 2.14.1.10. Positioner shall be a two-wire device, which shall operate on two-way digital communication mode. All engineering, configuration, diagnostic and maintenance related data shall be provided by the positioner. 2.14.1.11. The smart positioner shall provide HART of latest version and shall be capable of implementing commands from Instrument Asset Management System / hand-held HART configurator. 2.14.1.12. Positioner shall meet the following performance characteristics: - - Overall control accuracy shall be better than $\pm 0.5\%$ of span. - Repeatability shall be less than $\pm 0.25\%$ of span. - Hysteresis shall be less than $\pm 0.5\%$ of span. - Vendor shall supply the valve's operating signatures in the form of hard copy and soft copy for each control valve supplied with smart positioners. The necessary software for advanced control valve diagnostics like seat ring condition, gland packing condition, actuator leakage etc. shall also be included. 2.14.1.13. Positioner shall have metallic casing and cover either of stainless steel or of anodized aluminum. 2.14.1.14. Vendor shall supply positioner DTM files for integration in customer IAMS. 2.14.1.15. The positioner shall be supplied with soft copy of valve signature. 2.14.1.16. HART positioners HART positioner with diagnostic capabilities shall be supplied whenever specified in the purchaser's data sheets. These shall meet the following additional requirements:		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 18 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		a) The HART positioner shall provide HART protocol of latest version and shall be capable of implementing commands from Instrument Asset Management System / hand-held configurator. 2.14.1.17. Fieldbus Positioners Positioners with Fieldbus output shall also meet the following requirements: a) All positioners must satisfy the requirements of the fieldbus registration laboratory with applicable checkmark like Fieldbus foundation, Profibus Nutzerorganisation e.v (PNO) or as specified in the purchaser's data sheets. b) All positioners shall have analog output (AO) and controller blocks (PID). c) All positioner must be interoperable and shall have valid interoperability test clearance like ITK latest version for foundation fieldbus or equivalent for profibus PA, as applicable. d) The fieldbus positioners shall support peer-to-peer communication with two wire communicator and bus powered supply. e) Fieldbus positioner as offered shall not be polarity sensitive. f) The fieldbus positioners in hazardous area shall be certified as per entity concept or shall be FISCO approved as per the requirements specified in the purchaser's specification. g) Internal software shall be configured by the vendor including the following information: • Serial Number • Device tag (Tag No.) • Process Description/service h) Positioner shall be capable of supporting incremental Device Description (DD) for extra functionality and/or software revisions in Device memory. i) The positioner shall be suitable to operate with commercially available Instrument asset management software and shall support the following features, as a minimum: i. It shall allow multi-master for configuration, calibration, diagnosis and maintenance. The primary could be a host computer and secondary could be a hand held communicator. ii. It shall be capable of implementing universal commands. It shall be possible to communicate all commands of commercially available instrument asset management system to/ from smart positioner. j) Fieldbus positioners shall support EDDL requirements as per IEC61804/ FDT/ DTM requirements, as specified in specifications. 2.14.2. LINE STRAINERS 2.14.2.1. Vendor shall supply two line strainers per desuperheater / PRDS. The mesh size of the strainer shall be decided by the vendor as per desuperheater nozzle and coolant control valve trim size, as applicable. 2.14.2.2. The material of construction of the strainer shall be as follows: 2.14.2.3. Mesh Material: SS316 2.14.2.4. Body Material: Same as Control Valve body material or desuperheater body material specified in the data sheet. 2.14.2.5. The body size of the strainer shall be of the same size as coolant line size. Unless specified otherwise it shall be 1".		
Ref. Doc				

TD-201 Rev No.00	Form No.		PRODUCT STANDARD HYDERABAD	TC 65132 Rev No. 03 Page 19 of 28
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		2.14.2.6. Unless specified otherwise in datasheet, the end connection of the strainer shall be flanged only with rating as per the coolant end connection specified in the data sheet. 2.14.2.7. The strainers shall include provisions for maintenance of elements and shall be supplied with block and bypass arrangement. 2.14.3. AIR FILTER REGULATOR 2.14.3.1. Vendor shall supply air filter regulator with each positioner complete with an integral output gauge. 2.14.3.2. Air filter regulator shall be sized considering the air supply pressure and flow required to meet requirements specified in the purchaser's data sheets. 2.14.3.3. Filter material shall be SS. Filter size shall be maximum 5 microns. However, lower filter mesh size shall be considered to suit the electro pneumatic positioner vendor's requirement. 2.14.3.4. Air Filter regulator MOC shall be SS316. 2.15. PAINTING & FINISH 2.15.1. The external surface of the desuperheater/PRDS being supplied shall first be thoroughly cleaned by sand blasting or other suitable means, before painting. 2.15.2. Carbon steel external surfaces other than flange faces shall be given one coat of red oxide zinc chromate primer of single pack. The composition shall be of modified phenolic alkyd pigmented with red oxide and zinc chromate. 2.15.3. The valve body shall be painted in general as below. Customer specific painting colour & painting specification shall be provided after PO placement. Vendor shall follow the same without any price or delivery implications. • Carbon steel body - Light grey • Alloy steel body - Canary yellow • Stainless steel body - Natural • Above 150°C - Aluminum paint (white colour) The actuator shall be painted as below: • Direct action (open on air failure) valves - Green • Reverse acting (close on air failure) valve - Red 2.15.4. Items like silencers, diffuser plate assemblies etc., shall be painted as per respective control valve body requirements. 2.16. CABLE GLAND, PLUGS: a) All electrical entries shall be plugged with SS316 plugs, Ex-d & Ex-e certified for IEC zone 1, IIC, T3. b) Cable glands are in BHEL scope. c) Vendor to provide suitable adapter / y-type cable splitter (for positioner), duly plugged for all cable entries. 3. NAMEPLATE 3.1. Each desuperheater shall have a stainless steel nameplate attached firmly to it at a visible place furnishing the following information: a) Tag No. as per purchaser's datasheet b) Body size in inches and ANSI Rating c) Designed inlet/ outlet conditions of desuperheater		
Ref. Doc				

TD-201 Rev No.00 Form No.		PRODUCT STANDARD HYDERABAD		TC 65132															
				Rev No. 03															
				Page 20 of 28															
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.		d) Designed inlet/ outlet conditions of coolant e) Stem travel in mm f) Action on air failure g) Spring Range h) Air supply pressure i) Manufacturer's model no. for desuperheater, actuator and positioner.																	
		4. INSPECTION AND TESTING 4.1. Purchaser reserves the right to inspect and witness testing of all the items at vendor's works as per Inspection Test Plan and approved quality documents. All these tests shall be completed by the vendor and test reports shall be submitted to Purchaser for scrutiny. 4.2. IGC: Inter granular corrosion: For all stainless steel fittings Inter Granular Corrosion (IGC) test shall have to be conducted as per the following: a) ASTM A 262 Practice "B" with acceptance criteria of "60 mils/year (max)". b) OR ASTM A262 Practice E: The bent specimen shall be examined under 20X magnification. The acceptance criteria is that there will be no crack or fissure in the bent specimen. The bent specimen shall also be subjected to metallographic examination at 250X magnification to ensure no crack or fissure.																	
		5. SHIPPING 5.1. The desuperheater/PRDS shall be transported to site by the vendor in fully assembled condition. However, in case some components are liable to be damaged during transit, the same shall be dismantled and supplied separately. 5.2. All parts shall be thoroughly cleaned of loose scale, dirt and other foreign material before packing. 5.3. All flange faces shall be suitably protected to avoid damage to the gasket faces and prevent entry of foreign materials. 5.4. All tapped openings shall be plugged. 5.5. Each item shall be suitably wrapped in polythene sheets, packed in wooden crates, properly tagged and secured before dispatch.																	
		6. DEVIATION FORMAT: Bidder shall submit duly filled deviation format (as given below) along with technical offer, otherwise, it will be presumed that there are no deviations from this specification. Offer without this deviation list will not be evaluated & shall be considered for rejection. If, there are no deviations, bidder shall submit signed copy of this format, mentioning "No Deviations".																	
		<table border="1"> <thead> <tr> <th rowspan="2">Sl. No.</th> <th rowspan="2">Clause No. of Spec</th> <th rowspan="2">Deviation</th> <th rowspan="2">Reason for deviation</th> <th colspan="2">Deviation category</th> </tr> <tr> <th>Product / design limitation</th> <th>Optimization</th> </tr> </thead> <tbody> <tr> <td>1</td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>				Sl. No.	Clause No. of Spec	Deviation	Reason for deviation	Deviation category		Product / design limitation	Optimization	1					
Sl. No.	Clause No. of Spec	Deviation	Reason for deviation	Deviation category															
				Product / design limitation	Optimization														
1																			
Ref. Doc																			

7. Typical Diagrams



8. **VARIANT TABLE: REFER DATASHEET ATTACHED WITH THIS SPECIFICATION FOR PROJECT / TAG DETAILS OF EACH DESUPERHEATER:**

Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner
TC9765132018	DESUPERHEATER IN MOPDT EXHAUST-MATHURA	8	300RF	A106GrB	1	NON INTEGRAL	300RF	HART
TC9765132026	DS SPARES IN MOPDT EXHAUST-MATHURA	1 SET COMPRISING OF FOLLOWING ITEMS: a. Gasket for actuator per unit. b. Gasket for valve and connections per unit (if such gasket is special and supplied by Desuperheater vendor). c. 10% or a minimum 1 no of each type, size, rating of trim set consisting of seat ring/ seal ring, plug with stem, cage (wherever applicable), packing material for each type valve to be provided as spare. d. 10% subject to minimum 1 no's) of E/p Positioners of each type and make. e. 10% (subject to minimum 1 no) of Diaphragm, 10% (subject to minimum two no's) actuator spring and of plate of 1 set of each type/range. f. 10% (subject to minimum 1 no) of Air Filter Regulator of each make. g. 10% of gland packing, O-rings, Bonnet Gaskets for each valve. h. 10% (subject to minimum two no's) of any special accessories provided along with the control valve like boosters, position transmitters (wherever applicable), proximity switches etc. i. Spray, Nozzle per unit.						
TC9765132034	DS IN MOPDT EXHAUST-MATHURA HPC	8	300RF	A106GrB	1	NON INTEGRAL	300RF	HART

TD-201 Rev No.00 Form No.		PRODUCT STANDARD HYDERABAD					TC 65132														
							Rev No. 03														
							Page 22 of 28														
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.											Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner		
											TC9765132042	DS SPARES FOR MOPDT EXHAUST-MATHURA HPC	1 SET COMPRISING OF FOLLOWING ITEMS: a. Gasket for actuator per unit. b. Gasket for valve and connections per unit (if such gasket is special and supplied by Desuperheater vendor). c. Spray, Nozzle per unit. d. 10% or a minimum 1 no of each type, size, rating of trim set consisting of seat ring/ seal ring, plug with stem, cage (wherever applicable), packing material for each type valve to be provided as spare. e. 10% subject to minimum 1 no's) of E/p Positioners of each type and make. f. 10% (subject to minimum 1 no) of Diaphragm, 10% (subject to minimum two no's) actuator spring and plate of 1 set of each type/range. g. 10% (subject to minimum 1 no) of Air Filter Regulator of each make. h. 10% of gland packing, O-rings, Bonnet Gaskets for each valve. i. 10% (subject to minimum two no's) of any special accessories provided along with the control valve like boosters, position transmitters (wherever applicable), proximity switches etc.								
											TC9765132077	DSH 6"300RF-INTEG TCV 1"300RF-HART	6	300RF	A106GrB	1	INTEGRAL	300RF	HART		
											TC9765132085	DSH 6"300RF-INTEG TCV 1"300RF-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No								
											TC9765132093	DS & TCV IN AUX STEAM LINE-WGC	4	300RF	A182F11	1	NON INTEGRAL	300RF	HART		
											TC9765132107	DS SPARES IN AUX STEAM LINE-WGC	1 SET COMPRISING OF FOLLOWING ITEMS: a. Gasket for actuator per unit. b. Gasket for valve and connections per unit (if such gasket is special and supplied by Desuperheater vendor). c. 10% or a minimum 1 no of each type, size, rating of trim set consisting of seat ring/ seal ring, plug with stem, cage (wherever applicable), packing material for each type valve to be provided as spare. d. 10% subject to minimum 1 no's) of E/p Positioners of each type and make. e. 10% (subject to minimum 1 no) of Diaphragm, 10% (subject to minimum two no's) actuator spring and of plate of 1 set of each type/range. f. 10% (subject to minimum 1 no) of Air Filter Regulator of each make. g. 10% of gland packing, O-rings, Bonnet Gaskets for each valve. h. 10% (subject to minimum two no's) of any special accessories provided along with the control valve like boosters, position transmitters (wherever applicable), proximity switches etc. i. Spray, Nozzle per unit.								
											TC9765132115	DS IN AUX STEAM LINE -HPC	4	300RF	A182F11	1	NON INTEGRAL	300RF	HART		
											TC9765132123	DS SPARES IN AUX STEAM LINE-HPC	1 SET COMPRISING OF FOLLOWING ITEMS: a. Gasket for actuator per unit. b. Gasket for valve and connections per unit (if such gasket is special and supplied by Desuperheater vendor). c. 10% or a minimum 1 no of each type, size, rating of trim set consisting of seat ring/ seal ring, plug with stem, cage (wherever applicable), packing material for each type valve to be provided as spare. d. 10% subject to minimum 1 no's) of E/p Positioners of each type and make. e. 10% (subject to minimum 1 no) of Diaphragm, 10% (subject to minimum two no's) actuator spring and of plate of 1 set of each type/range. f. 10% (subject to minimum 1 no) of Air Filter Regulator of each make. g. 10% of gland packing, O-rings, Bonnet Gaskets for each valve. h. 10% (subject to minimum two no's) of any special accessories provided along with the control valve like boosters, position transmitters (wherever applicable),								
											Ref.	Doc									

TD-201 Rev No.00 Form No.		PRODUCT STANDARD HYDERABAD					TC 65132		
							Rev No. 03		
							Page 23 of 28		
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company.									
Ref. Doc									

Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner	
		proximity switches etc. i. Spray, Nozzle per unit.							
TC9765132158	DS & TCV IN AUX STEAM LINE								
TC9765132166	DS & TCV IN WANDER EXTRACTION LINE								
TC9765132174	DS & TCV IN UNCONTROLLED EXTRACTION								
TC9765132182	DS & TCV IN TURBINE EXHAUST	12	300RF	A106GrB	1	INTEGRAL	600RF	FF	
TC9765132190	DS & TCV IN TURBINE EXHAUST-SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 1 No b. FF positioner - 1 No c. Bonnet Gasket/ Special Gasket - 1 No d. Gland Packing - 1 No e. Stem Seal O rings/ Piston o- ring – 1 No							
TC9765132204	DS&TCV IN GLAND STEAM LEAK OFF LINE	2	300RF	A106GrB	1	NON INTEGRAL	600RF	FF	
TC9765132212	DS&TCV IN GLAND STEAM LEAK OFF -SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 1 No b. FF positioner - 1 No c. Bonnet Gasket/ Special Gasket - 1 No d. Gland Packing - 1 No e. Stem Seal O rings/ Piston o- ring – 1 No							
TC9765132220	DS & TCV IN TURBINE EXHAUST-VGO	20	150RF	A106GrB	1	INTEGRAL	600RF	FF	
TC9765132239	DS & TCV IN TURBINE EXHAUST-VGO SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 1 No b. FF positioner - 1 No c. Bonnet Gasket/ Special Gasket - 1 No d. Gland Packing - 1 No e. Stem Seal O rings/ Piston o- ring – 1 No							
TC9765132247	DS IN TURBINE EXTRACTION LINE	16	600RF	A182F11	1	INTEGRAL	600RF	FF	
TC9765132255	DS IN TURBINE EXTRACTION LINE-SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 1 No b. FF positioner - 1 No c. Bonnet Gasket/ Special Gasket - 1 No d. Gland Packing - 1 No e. Stem Seal O rings/ Piston o- ring – 1 No							
TC9765132263	DS & TCV IN TURBINE EXHAUST LINE	18	150BW	A106GrB	1	NON INTEGRAL	300RF	HART	
TC9765132271	DS&TCV IN TURBINE EXHAUST LINE-SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. 10% or minimum 2 nos of each type, size, rating of Trim set consisting of seat, seat ring/seal ring, plug with stem, cage (wherever applicable), packing material. b. 20% (subject to minimum of 1) of positioners (smart or pneumatic) with links of each type and make. c. 15% (subject to minimum of 1) of Diaphragm, Actuator Spring of each type/ range. d. 20% (subject to minimum of 1) of Air Filter Regulator of each make. e. 10% (subject to minimum if 1) of Bonnet Gaskets/ Gland Packing, O rings.							
TC9765132280	DS&TCV IN MOPDT EXHAUST LINE	8	150BW	A106GrB	1	NON INTEGRAL	300RF	HART	
TC9765132298	DS&TCV IN MOPDT EXHAUST LINE SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. 10% or minimum 2 nos of each type, size, rating of Trim set consisting of seat, seat ring/seal ring, plug with stem, cage (wherever applicable), packing material. b. 20% (subject to minimum of 1) of positioners (smart or pneumatic) with links of each type and make. c. 15% (subject to minimum of 1) of Diaphragm, Actuator Spring of each type/ range. d. 20% (subject to minimum of 1) of Air Filter Regulator of each make. e. 10% (subject to minimum if 1) of Bonnet Gaskets/ Gland Packing, O rings.							

TD-201 Rev No.00 Form No.		PRODUCT STANDARD HYDERABAD					TC 65132		
							Rev No. 03		
							Page 24 of 28		
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company. Ref. Doc									
Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner	
TC9765132301	DS IN TURBINE EXHAUST	26	300RF	A106GrB	1	INTEGRAL	300RF	FF	
TC9765132310	DS SPARES IN TURBINE EXHAUST	1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 2 No b. FF positioner with links - 2 No c. Bonnet Gasket/ Special Gasket - 1 No d. Gland Packing - 1 No e. Stem Seal O rings/ Piston o- ring - 1 No f. Diaphragm - 2 No g. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material - 1 No. h. Any other Special accessories - 1 No.							
TC9765132328	DS IN MOPDT EXHAUST LINE	6	300RF	A106GrB	1	INTEGRAL	300RF	FF	
TC9765132336	DS SPARES-MOPDT EXHAUST LINE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 2 No b. FF positioner with links - 2 No c. Bonnet Gasket/ Special Gasket - 1 No d. Gland Packing - 1 No e. Stem Seal O rings/ Piston o- ring - 1 No f. Diaphragm - 2 No g. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material - 1 No. h. Any other Special accessories - 1 No.							
TC9765132344	DS IN MOPDT EXHAUST - IOCL BONGAIGAON	8	300RF	A106GrB	1	NON INTEGRAL	900RTJ	HART	
TC9765132352	DS SPARES IN MOPDT EXHAUST -IOCL BONGAI	1 SET COMPRISING OF FOLLOWING ITEMS: a. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material - 1 No. b. Positioner with links - 2 No c. Actuator Diaphragm - 2 No d. Air filter regulator - 2 No e. Bonnet Gasket/ Special Gasket - 1 No f. Gland Packing - 1 No g. Stem Seal O rings/ Piston o- ring - 1 No h. Any other Special accessories - 1 No. i. O-ring of piston act - 1No j. Desuperheater Nozzle - 1No							
TC9765132360	INTEGRAL TCV TYPE DS - TURBINE EXHAUST	12	300RF	A106GrB	1	INTEGRAL	600RF	HART	
TC9865132370		1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 2 No b. Bonnet Gasket/ Special Gasket - 1 No c. Gland Packing - 1 No d. Stem Seal O rings/ Piston o- ring - 1 No e. Actuator Diaphragm - 2 No f. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material - 1 No. g. Any other Special accessories - 1 No.							
TC9765132387	DS TURBINE GLAND SEAL LEAKAGE LINE	3	300RF	A106GrB	1	NON INTEGRAL	600RF	HART	
TC9865132397		1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 2 No b. Bonnet Gasket/ Special Gasket - 1 No c. Gland Packing - 1 No d. Stem Seal O rings/ Piston o- ring - 1 No e. Actuator Diaphragm - 2 No f. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material - 1 No. g. Any other Special accessories - 1 No.							
TC9765132409	DS IN MOPDT EXHAUST	8	150RF	A106GrB	1	INTEGRAL	600RF	HART	
TC9865132419		1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 2 No							

TD-201 Rev No.00 Form No.		PRODUCT STANDARD HYDERABAD					TC 65132											
							Rev No. 03											
							Page 25 of 28											
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly to the interest of the company.										Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner
												b. Bonnet Gasket/ Special Gasket - 1 No c. Gland Packing - 1 No d. Stem Seal O rings/ Piston o- ring – 1 No e. Actuator Diaphragm - 2 No f. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material – 1 No. g. Any other Special accessories – 1 No.						
										TC9765132417	DS IN TURBINE EXTRACTION LINE - GHCL							
										TC9865132427		SMART Positioner						
										TC9765132433	DS IN TURBINE EXHAUST - MCA1045	14	300RF	A106GrB	1.5	INTEGRAL	300RF	HART
										TC9765132441	DS IN MOPDT EXHAUST - MCA1045	4	300RF	A106GrB	1	INTEGRAL	300RF	HART
										TC9765132450	DS TCV -SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Air filter regulator - 1 No b. Bonnet Gasket/ Special Gasket - 1 No c. Gland Packing - 1 No d. Stem Seal O rings/ Piston o- ring – 1 No e. Actuator Diaphragm - 2 No f. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material – 1 No. g. Any other Special accessories – 1 No. h. SMART Positioner – 1 No						
										TC9765132468	DESUPER HEATER-TURBINE EXHAUST	18	300RF	A182F11	1	INTEGRAL	600RF	HART
										TC9765132476	DESUPER HEATER-TURBINE EXHAUST-SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket/ Special Gasket - 1 No b. Gland Packing - 1 No c. Stem Seal O rings/ Piston o- ring – 1 No d. Actuator Diaphragm - 2 No e. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material – 1 No. f. Any other Special accessories – 1 No.						
										TC9765132484	DESUPER HEATER-GLAND STEAM	3	600#	A335P11	1	NON INTEGRAL	600RF	HART
										TC9765132492	DESUPER HEATER-GLAND STEAM-SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket/ Special Gasket - 1 No b. Gland Packing - 1 No c. Stem Seal O rings/ Piston o- ring – 1 No d. Actuator Diaphragm - 2 No e. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material – 1 No. f. Any other Special accessories – 1 No.						
										TC9765132506	DESUPER HEATER-MOPDT EXHAUST	8	150RF	A106GrB	1	INTEGRAL	600RF	HART
										TC9765132514	DESUPER HEATER-MOPDT EXHAUST SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket/ Special Gasket - 1 No b. Gland Packing - 1 No c. Stem Seal O rings/ Piston o- ring – 1 No d. Actuator Diaphragm - 2 No e. Trim Set consisting of Seat, Seat ring/ Seal Ring, Plug with Stem, Cage (wherever applicable), packing material – 1 No. f. Any other Special accessories – 1 No.						
										TC9765132522	SET OF POSITIONERS	SMART HART Positioner, Exi + Exd, Zone 1 IIC T3, 50 mmPG-02 nos						
										TC9765132530	SPARE AIR FILTER REGULATOR-DS	SS316 Body, SS 5 micron element, 50 mm PG,						
TC9765132549	DS IN TURBINE LEAKOFF LINE TO LP HEADER																	
TC9765132557	SPRAY WATER TCV - SPARES																	
TC9765132565	DESUPER HEATER IN TURBINE	4	300RF	A106GrB	1	INTEGRAL	600RF	HART										

Ref.
Doc

TD-201 Rev No.00 Form No.		PRODUCT STANDARD HYDERABAD					TC 65132												
							Rev No. 03												
							Page 26 of 28												
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company. Ref. Doc																			
										Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner	
											GLAND STEAM								
										TC9765132573	DESUPER HEATER-GLAND STEAM-SPARES	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No							
										TC9765132581	DSH 6"300RF-INTEG TCV 1"600RF-HART	6	300RF	A106GrB	1	INTEGRAL	600RF	HART	
										TC9765132590	DSH 6"300RF-INTEG TCV 1"600RF-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No							
										TC9765132603	DSH 4"300RF-NON INTEG TCV 1"600RF-HART	4	300RF	A106GrB	1	NON INTEGRAL	600RF	HART	
										TC9765132611	DSH 4"300RF-NON INTEG TCV 1"600RF-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No							
										TC9765132620	DSH 20"300RF-NON INTEG TCV 1"600RF-HART	20	300RF	A106GrB	1	NON INTEGRAL	600RF	HART	
										TC9765132638	DSH 20"300RF-NON INTEG TCV 1"600#-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No							
										TC9765132646	DSH 12"300RF-INTEG TCV 1"300RF-HART	12	300RF	A106GrB	1	INTEGRAL	300RF	HART	
										TC9765132654	DSH 12"300RF-INTEG TCV 1"300RF-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No							
										TC9765132662	DSH 18"300RF-INTEG TCV 1"300RF-HART	18	300RF	A106GrB	1	INTEGRAL	300RF	HART	
										TC9765132670	DSH 18"300RF-INTEG TCV 1"300RF-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No							
										TC9765132689	DSH 3"300RF-INTEG TCV 1"300RF-HART	3	300RF	A106GrB	1	INTEGRAL	300RF	HART	
TC9765132697	DSH 3"300RF-INTEG TCV	1 SET COMPRISING OF FOLLOWING ITEMS:																	

TD-201 Rev No.00 Form No.		PRODUCT STANDARD HYDERABAD					TC 65132																																									
							Rev No. 03																																									
							Page 27 of 28																																									
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED. It must not be used directly or indirectly in any way detrimental to the interest of the company. <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 15%;">Material Code</th> <th style="width: 25%;">Material Description</th> <th style="width: 10%;">DSH size</th> <th style="width: 10%;">Flange Rating</th> <th style="width: 10%;">Material</th> <th style="width: 5%;">TCV Size</th> <th style="width: 10%;">TCV Type</th> <th style="width: 10%;">Flange rating</th> <th style="width: 10%;">Positioner</th> </tr> </thead> <tbody> <tr> <td></td> <td>1"300RF-SPARE</td> <td colspan="8"> a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No </td> </tr> <tr> <td>TC9765132700</td> <td>DSH 24"300RF-NON INTEG TCV 1"600RF-HART</td> <td>24</td> <td>300RF</td> <td>A106GrB</td> <td>1</td> <td>NON INTEGRAL</td> <td>600RF</td> <td>HART</td> </tr> <tr> <td>TC9765132719</td> <td>DSH 24"300RF-NON INTEG TCV 1"600#-SPARE</td> <td colspan="8"> 1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No </td> </tr> </tbody> </table>											Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner		1"300RF-SPARE	a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No								TC9765132700	DSH 24"300RF-NON INTEG TCV 1"600RF-HART	24	300RF	A106GrB	1	NON INTEGRAL	600RF	HART	TC9765132719	DSH 24"300RF-NON INTEG TCV 1"600#-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No							
Material Code	Material Description	DSH size	Flange Rating	Material	TCV Size	TCV Type	Flange rating	Positioner																																								
	1"300RF-SPARE	a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No																																														
TC9765132700	DSH 24"300RF-NON INTEG TCV 1"600RF-HART	24	300RF	A106GrB	1	NON INTEGRAL	600RF	HART																																								
TC9765132719	DSH 24"300RF-NON INTEG TCV 1"600#-SPARE	1 SET COMPRISING OF FOLLOWING ITEMS: a. Bonnet Gasket / Special Gasket: 01 No b. Gland Packing: 01 No c. Stem Seal O rings/ Piston O-ring: 01 No d. Actuator Diaphragm: 02 No e. Trim Set consisting of Seat, Seat ring / Seal Ring, Plug with Stem, Cage (as applicable), packing material: 01 No f. Any other Special accessories: 01 No																																														
9. ATTACHMENTS: 9.1. DESUPERHEATER DATASHEETS. 9.2. STANDARD INSPECTION AND TEST PLAN.																																																
Ref.	Doc																																															

TD-106-3 Rev No. 5 Form No.		PRODUCT STANDARD		TC 65132	
		HYDERABAD		Rev No. 02	
				Page 28 of 28	
COPYRIGHT AND CONFIDENTIAL The information on this document is the property of BHARAT HEAVY ELECTRICALS LIMITED, It must not be used directly or indirectly in any way detrimental to the interest of the company.	RECORD OF REVISIONS				
	Rev. No.	Date	Revision Details	Revised By	Approved By
	01	25.05.2023	Specification Updated and new variants added	CHHAYA	RAM
	02	07.09.2023	Variants added	CHHAYA	RAM
	03	06.02.2024	Updated as Highlighted	RAVI	RAM
	Ref.				
Doc.					

General Data	1	Tag Number					TBA																																											
	2	Service					TBA																																											
	3	PID No.					TBA																																											
	4	ITEM					DSH 6"300RF-INTEG TCV 1"300RF-HART																																											
	5	SIL Level Required		Line Number			NA			TBA																																								
Operating and Mechanical Data	6	STEAM CONDITIONS																																																
	7	INLET STEAM								OUTLET STEAM																																								
	8	Line Size		Piping Class		6"		A2A		6"		A2A																																						
	9	Line Material		Line Schedule		A106 Gr.B		STD		A106 Gr.B		STD																																						
	10	Insulation Code																																																
	11	Molecular Weight		Density		18.015				18.015																																								
	12	Flow :		Min.	Norm.	Max.	Unit	1500	2500	5000	KG/HR	*	*	*	KG/HR																																			
	13	Temp. :		Min.	Norm.	Max.	Unit	200	250	280	DEG C	150	165	170	DEG C																																			
	14	Press. :		Min.	Norm.	Max.	Unit	4	5	6	KG/CM2G	3.5	4	4.5	KG/CM2G																																			
	15	Temp. (after Desuperheater)		Flexibility																																														
	16	Design Press.		Design Temp. Max.		16.5				280																																								
	17	Body Material		Stem / Plug Material		A216 Gr.WCB				SS316																																								
	18	Size		Rating		Face		6"		300#		RF																																						
	19	Pipe Nozzle Length		Schedule		*				*																																								
	20	Insertion Length				*																																												
21	Flange Material				A105N																																													
Spray Water / Temperature Control Valve	22	WATER CONDITIONS																																																
	23	Line Size		Piping Class		1"				B2A																																								
	24	Line Material		Line Schedule		A106 Gr.B				XS																																								
	25	Insulation Code					Density																																											
	26	Flow :		Min.	Norm.	Max.	Unit	*	*	*		KG/HR																																						
	27	Temp. :		Min.	Norm.	Max.	Unit	105	105	110		DEG C																																						
	28	Press. :		Min.	Norm.	Max.	Unit	30	30	31		KG/CM2G																																						
	29	Design Temp.		Design Pressure		130				44																																								
	30	Body Material		Stem / Plug Material		A105N				SS316 (STELLITE)																																								
	31	Size		Rating		Face		1"		300		RF																																						
	32	Seat Leakage Class				IV																																												
	33																																																	
	34																																																	
	35	Area Classification Req		Safety Certification		IEC ZONE 1 IIC T3				Ex-ia																																								
	36	Instrument Type		Type																																														
	37	Packing Material		Bonnet Type		*				*																																								
	38	Bellows		Cooling Fins		Extension		NA		NA		NA																																						
	39	Air-Fail Position					OPEN																																											
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*																																							
41		Pneumatic Supply					3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)																																											
Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*																																								
	43	Fluid Connection		Electrical Connection		*				1/2"NPT																																								
	44	Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67																																								
	45	Surge Prot Device		Action Direction		YES				*																																								
Accessories	46	Handwheel		Position		YES				*																																								
	47	Locking Device		Filter		Press Reduc Valve		NA		*		*																																						
	48	Air Set		Pressure Gauge		Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA																																						
	49	Solenoid Valve		Tag No.		NA				NA																																								
Solenoid Valve	50	Elec Protect. Class		Enclosure Protection		NA																																												
	51	Position Detector		Tag No.		*				NA																																								
Position Detector	52	Elec Protect. Class		Enclosure Protection		NA				NA																																								
	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA																																								
Position Transmitter	54	Elec Protect. Class		Enclosure Protection		NA				NA																																								
	55			Weight		*				*																																								
Purchase	56	Manufacturer		Model No.		*				*																																								
	57	Client Reference		Requisition No.		NA				*																																								
	58	Material Code				TC9765132077																																												
Notes:																																																		
1) TBA: To be Assigned.																																																		
2) *: Vendor to Assign.																																																		
3) IBR Certification is applicable for Desuperheater & Temperature Control Valve.																																																		
4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST																																																		
5) Painting shall be as per attached Painting Specification																																																		
<table border="1"> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr> <td>No.</td> <td>By</td> <td>Chk</td> <td>Appr</td> <td>Date</td> <td>Revision</td> </tr> </table> INSTRUMENT SPECIFICATION Desuperheater Code: 512 Dwg. No.: Ind Doc No.: Sheet Rev.: 																																													No.	By	Chk	Appr	Date	Revision
No.	By	Chk	Appr	Date	Revision																																													

General Data	1	Tag Number				TBA								
	2	Service				TBA								
	3	PID No.				TBA								
	4	ITEM				DSH 6"300RF-INTEG TCV 1"600RF-HART								
Operating and Mechanical Data	5	SIL Level Required		Line Number		NA				TBA				
	6					STEAM CONDITIONS								
	7					INLET STEAM				OUTLET STEAM				
	8	Line Size		Piping Class		6"		A2A		6"		A2A		
	9	Line Material		Line Schedule		A106 Gr.B		STD		A106 Gr.B		STD		
	10	Insulation Code												
	11	Molecular Weight		Density		18.015				18.015				
	12	Flow :	Min.	Norm.	Max.	Unit	1500	2500	5000	KG/HR	*	*	*	KG/HR
	13	Temp. :	Min.	Norm.	Max.	Unit	200	250	280	DEG C	150	180	180	DEG C
	14	Press. :	Min.	Norm.	Max.	Unit	4	5	6	KG/CM2G	3	3.5	5	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility										
	16	Design Press.		Design Temp. Max.		16.5				320				
	17	Body Material		Stem / Plug Material		A216WCB				SS316				
	18	Size		Rating	Face	6"		300#				RF		
	19	Pipe Nozzle Length		Schedule		*				*				
	20	Insertion Length				*								
	21	Flange Material				A105N								
	22					WATER CONDITIONS								
	23	Line Size		Piping Class		1"				D2A				
	24	Line Material		Line Schedule		A106 Gr.B				XS				
	25	Insulation Code		Density										
	26	Flow :	Min.	Norm.	Max.	Unit	*		*		*		KG/HR	
	27	Temp. :	Min.	Norm.	Max.	Unit			105		170		DEG C	
	28	Press. :	Min.	Norm.	Max.	Unit			61		78		KG/CM2G	
	29	Design Temp.		Design Pressure		170				78.3				
	30	Body Material		Stem / Plug Material		A105N				SS316 (STELLITE)				
	31	Size		Rating	Face	1"		600				RF		
	32	Seat Leakage Class				IV								
	33													
	34													
	35	Area Classification		Req Safety Certification		IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type										
	37	Packing Material		Bonnet Type		*				*				
	38	Bellows	Cooling Fins	Extension		NA		NA		NA				
	39	Air-Fail Position				OPEN								
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*			
		41	Pneumatic Supply				3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)							
	Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*			
		43	Fluid Connection		Electrical Connection		*				1/2"NPT			
44		Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67				
Accessories	45	Surge Prot Device		Action Direction		YES				*				
	46	Handwheel		Position		YES				*				
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*				
Solenoid Valve	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA				
	49	Solenoid Valve		Tag No.		NA				NA				
	50	Elec Protect. Class		Enclosure Protection		NA								
Position Detector	51	Position Detector		Tag No.		*				NA				
	52	Elec Protect. Class		Enclosure Protection		NA				NA				
Position Transmitter	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA				
	54	Elec Protect. Class		Enclosure Protection		NA				NA				
Purchase	55			Weight						*				
	56	Manufacturer		Model No.		*				*				
	57	Client Reference		Requisition No.		NA				*				
	58	Material Code				TC9765132581								
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST														
						INSTRUMENT SPECIFICATION Desuperheater								
						Code: 512				Dwg. No.: Ind Doc No.:				
						Sheet				Rev.:				
No.	By	Chk	Appr	Date	Revision									

General Data	1	Tag Number				TBA								
	2	Service				TBA								
	3	PID No.				TBA								
	4	ITEM				DSH 4"300RF-NON INTEG TCV 1"600RF-HART								
	5	SIL Level Required		Line Number		NA				TBA				
Operating and Mechanical Data	6					STEAM CONDITIONS								
	7					INLET STEAM				OUTLET STEAM				
	8	Line Size		Piping Class		4"		A2A		4"		A2A		
	9	Line Material		Line Schedule		A106 Gr.B		STD		A106 Gr.B		STD		
	10	Insulation Code												
	11	Molecular Weight		Density		18.015				18.015				
	12	Flow :	Min.	Norm.	Max.	Unit	400	500	750	KG/HR	*	*	*	KG/HR
	13	Temp. :	Min.	Norm.	Max.	Unit	300	300	343	DEG C	150	180	180	DEG C
	14	Press. :	Min.	Norm.	Max.	Unit	5	5	6.5	KG/CM2G	3	3.5	5	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility										
	16	Design Press.		Design Temp. Max.		16.5				320				
	17	Body Material		Stem / Plug Material		A216WCB				SS316				
	18	Size		Rating	Face	4"		300#		RF				
	19	Pipe Nozzle Length		Schedule		*				*				
	20	Insertion Length				*								
	21	Flange Material				A105N								
	22					WATER CONDITIONS								
	23	Line Size		Piping Class		1"				D2A				
	24	Line Material		Line Schedule		A106 Gr.B				XS				
	25	Insulation Code		Density										
	26	Flow :	Min.	Norm.	Max.	Unit	*		*		*		KG/HR	
	27	Temp. :	Min.	Norm.	Max.	Unit			105		170		DEG C	
	28	Press. :	Min.	Norm.	Max.	Unit			61		78		KG/CM2G	
	29	Design Temp.		Design Pressure		170				78.3				
	30	Body Material		Stem / Plug Material		A216WCB				SS316 (STELLITE)				
	31	Size		Rating	Face	1"		600		RF				
	32	Seat Leakage Class				IV								
	33													
	34													
	35	Area Classification		Req Safety Certification		IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type										
	37	Packing Material		Bonnet Type		*				*				
	38	Bellows	Cooling Fins	Extension		NA		NA		NA				
	39	Air-Fail Position				OPEN								
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*			
		41	Pneumatic Supply				3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)							
	Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*			
		43	Fluid Connection		Electrical Connection		*				1/2"NPT			
		44	Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67			
	45	Surge Prot Device		Action Direction		YES				*				
	46	Handwheel		Position		YES				*				
Accessories	47	Locking Device	Filter	Press Reduc Valve		NA		*		*				
	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA				
Solenoid Valve	49	Solenoid Valve		Tag No.		NA				NA				
	50	Elec Protect. Class		Enclosure Protection		NA								
Position Detector	51	Position Detector		Tag No.		*				NA				
	52	Elec Protect. Class		Enclosure Protection		NA				NA				
Position Transmitter	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA				
	54	Elec Protect. Class		Enclosure Protection		NA				NA				
Purchase	55			Weight						*				
	56	Manufacturer		Model No.		*				*				
	57	Client Reference		Requisition No.		NA				*				
	58	Material Code				TC9765132603								
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST														
						INSTRUMENT SPECIFICATION Desuperheater								
No.	By	Chk	Appr	Date	Revision				Code: 512	Dwg. No.: Ind Doc No.:			Sheet Rev.:	

General Data	1	Tag Number				TBA								
	2	Service				TBA								
	3	PID No.				TBA								
	4	ITEM				DSH 20"300RF-NON INTEG TCV 1"600RF-HART								
Operating and Mechanical Data	5	SIL Level Required		Line Number		NA				TBA				
	6					STEAM CONDITIONS								
	7					INLET STEAM				OUTLET STEAM				
	8	Line Size		Piping Class		20"		B2A		20"		B2A		
	9	Line Material		Line Schedule		A672GR.B60CL.12		XS		A672GR.B60CL.12		XS		
	10	Insulation Code												
	11	Molecular Weight		Density		18.015				18.015				
	12	Flow :	Min.	Norm.	Max.	Unit	34840	93950	180950	KG/HR	*	*	*	KG/HR
	13	Temp. :	Min.	Norm.	Max.	Unit	264	291	350	DEG C	250	275	330	DEG C
	14	Press. :	Min.	Norm.	Max.	Unit	10	11	13.2	KG/CM2G	*	*	*	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility										
	16	Design Press.		Design Temp. Max.		17				350				
	17	Body Material		Stem / Plug Material		A216WCB				SS316				
	18	Size		Rating	Face	20"		300#				RF		
	19	Pipe Nozzle Length		Schedule		*				*				
	20	Insertion Length				*								
	21	Flange Material				A105N								
	22					WATER CONDITIONS								
	23	Line Size		Piping Class		1"				D2A				
	24	Line Material		Line Schedule		A106 Gr.B				XS				
	25	Insulation Code		Density										
	26	Flow :	Min.	Norm.	Max.	Unit	*		*		*		KG/HR	
	27	Temp. :	Min.	Norm.	Max.	Unit			105		170		DEG C	
	28	Press. :	Min.	Norm.	Max.	Unit			61		78		KG/CM2G	
	29	Design Temp.		Design Pressure		170				78.3				
	30	Body Material		Stem / Plug Material		A216WCB				SS316 (STELLITE)				
	31	Size		Rating	Face	1"		600				RF		
	32	Seat Leakage Class				IV								
	33													
	34													
	35	Area Classification		Req	Safety Certification	IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type										
	37	Packing Material		Bonnet Type		*				*				
	38	Bellows	Cooling Fins	Extension		NA		NA		NA				
	39	Air-Fail Position				OPEN								
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*			
		41	Pneumatic Supply				3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)							
	Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*			
		43	Fluid Connection		Electrical Connection		*				1/2"NPT			
44		Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67				
Accessories	45	Surge Prot Device		Action Direction		YES				*				
	46	Handwheel		Position		YES				*				
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*				
Solenoid Valve	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA				
	49	Solenoid Valve		Tag No.		NA				NA				
	50	Elec Protect. Class		Enclosure Protection		NA								
Position Detector	51	Position Detector		Tag No.		*				NA				
	52	Elec Protect. Class		Enclosure Protection		NA				NA				
Position Transmitter	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA				
	54	Elec Protect. Class		Enclosure Protection		NA				NA				
Purchase	55			Weight						*				
	56	Manufacturer		Model No.		*				*				
	57	Client Reference		Requisition No.		NA				*				
	58	Material Code				TC9765132620								
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST														
						INSTRUMENT SPECIFICATION Desuperheater								
						Code: 512				Dwg. No.: Ind Doc No.:				
						Sheet				Rev.:				
No.	By	Chk	Appr	Date	Revision									

General Data	1	Tag Number					TBA								
	2	Service					TBA								
	3	PID No.					TBA								
	4	ITEM					DSH 12"300RF-INTEG TCV 1"300RF-HART								
	5	SIL Level Required		Line Number			NA				TBA				
Operating and Mechanical Data	6						STEAM CONDITIONS								
	7						INLET STEAM				OUTLET STEAM				
	8	Line Size		Piping Class			12"		A2A		12"		A2A		
	9	Line Material		Line Schedule			A106 Gr.B		STD		A106 Gr.B		STD		
	10	Insulation Code													
	11	Molecular Weight		Density			18.015				18.015				
	12	Flow :		Min.	Norm.	Max.	Unit	2500	14150	16380	KG/HR	*	*	*	KG/HR
	13	Temp. :		Min.	Norm.	Max.	Unit	216	262	263	DEG C	150	165	170	DEG C
	14	Press. :		Min.	Norm.	Max.	Unit	4.5	5	5.5	KG/CM2G	3.5	4	4.5	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility											
	16	Design Press.		Design Temp. Max.			16.5				280				
	17	Body Material		Stem / Plug Material			A216WCB				SS316				
	18	Size		Rating		Face	12"		300#				RF		
	19	Pipe Nozzle Length		Schedule			*				*				
	20	Insertion Length					*								
21	Flange Material					A105N									
Spray Water / Temperature Control Valve	22						WATER CONDITIONS								
	23	Line Size		Piping Class			1"				B2A				
	24	Line Material		Line Schedule			A106 Gr.B				XS				
	25	Insulation Code		Density											
	26	Flow :		Min.	Norm.	Max.	Unit	*	*	*	*	*	KG/HR		
	27	Temp. :		Min.	Norm.	Max.	Unit	105		105		110		DEG C	
	28	Press. :		Min.	Norm.	Max.	Unit	30		30		31		KG/CM2G	
	29	Design Temp.		Design Pressure			130				44				
	30	Body Material		Stem / Plug Material			A105N				SS316 (STELLITE)				
	31	Size		Rating		Face	1"		300				RF		
	32	Seat Leakage Class					IV								
	33														
	34														
	35	Area Classification Req		Safety Certification			IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type											
37	Packing Material		Bonnet Type			*				*					
38	Bellows		Cooling Fins		Extension	NA		NA		NA					
39	Air-Fail Position					OPEN									
Actuator	40	Actuator Type		Spring Range			SPRING & DIAPHRAGM				*				
	41	Pneumatic Supply					3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)								
Positioner	42	Type		Input Signal			4-20MA SMART HART 7				*				
	43	Fluid Connection		Electrical Connection			*				1/2"NPT				
	44	Elec Protect. Class		Enclosure Protection			Ex-ia				IP 67				
	45	Surge Prot Device		Action Direction			YES				*				
Accessories	46	Handwheel		Position			YES				*				
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*					
	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA					
Solenoid Valve	49	Solenoid Valve		Tag No.			NA				NA				
	50	Elec Protect. Class		Enclosure Protection			NA								
Position Detector	51	Position Detector		Tag No.			*				NA				
	52	Elec Protect. Class		Enclosure Protection			NA				NA				
Position Transmitter	53	Position Transmitter		Tag No.			INCLUDED WITH POSITIONER				NA				
	54	Elec Protect. Class		Enclosure Protection			NA				NA				
Purchase	55			Weight							*				
	56	Manufacturer		Model No.			*				*				
	57	Client Reference		Requisition No.			NA				*				
	58	Material Code					TC9765132646								
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST															
						INSTRUMENT SPECIFICATION Desuperheater									
						Code: 512				Dwg. No.: Ind Doc No.:				Sheet Rev.:	
No.	By	Chk	Appr	Date	Revision										

General Data	1	Tag Number				TBA									
	2	Service				TBA									
	3	PID No.				TBA									
	4	ITEM				DSH 18"300RF-INTEG TCV 1"300RF-HART									
	5	SIL Level Required		Line Number		NA				TBA					
Operating and Mechanical Data	6	STEAM CONDITIONS													
	7	INLET STEAM								OUTLET STEAM					
	8	Line Size		Piping Class		18"		B2A		18"		B2A			
	9	Line Material		Line Schedule		A106 Gr.B		40		A106 Gr.B		40			
	10	Insulation Code													
	11	Molecular Weight		Density		18.015				18.015					
	12	Flow :	Min.	Norm.	Max.	Unit	88300	88300	105650	KG/HR	*	*	*	KG/HR	
	13	Temp. :	Min.	Norm.	Max.	Unit	340	340.3	375.2	DEG C	215	220	225	DEG C	
	14	Press. :	Min.	Norm.	Max.	Unit	14.5	14.5	15	KG/CM2G	*	*	*	KG/CM2G	
	15	Temp. (after Desuperheater)		Flexibility											
	16	Design Press.		Design Temp. Max.		16.5				380					
	17	Body Material		Stem / Plug Material		A216WCB				SS316					
	18	Size	Rating	Face	18"		300#				RF				
	19	Pipe Nozzle Length		Schedule		*				*					
	20	Insertion Length				*									
	21	Flange Material				A105N									
	22	WATER CONDITIONS													
	23	Line Size		Piping Class		1"				B2A					
	24	Line Material		Line Schedule		A106 Gr.B				XS					
	25	Insulation Code													
	26	Flow :	Min.	Norm.	Max.	Unit	*	*	*				KG/HR		
	27	Temp. :	Min.	Norm.	Max.	Unit	105	105	110				DEG C		
	28	Press. :	Min.	Norm.	Max.	Unit	30	30	31				KG/CM2G		
	29	Design Temp.		Design Pressure		130				44					
	30	Body Material		Stem / Plug Material		A105N				SS316 (STELLITE)					
	31	Size	Rating	Face	1"		300				RF				
	32	Seat Leakage Class				IV									
	33														
	34														
	35	Area Classification		Req Safety Certification		IEC ZONE 1 IIC T3				Ex-ia					
	36	Instrument Type		Type											
	37	Packing Material		Bonnet Type		*				*					
	38	Bellows	Cooling Fins	Extension		NA		NA		NA					
	39	Air-Fail Position				OPEN									
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*				
41		Pneumatic Supply				3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)									
Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*					
	43	Fluid Connection		Electrical Connection		*				1/2"NPT					
	44	Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67					
	45	Surge Prot Device		Action Direction		YES				*					
Accessories	46	Handwheel		Position		YES				*					
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*					
	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM				NA			
	49	Solenoid Valve		Tag No.		NA				NA					
	50	Elec Protect. Class		Enclosure Protection		NA									
	51	Position Detector		Tag No.		*				NA					
	52	Elec Protect. Class		Enclosure Protection		NA				NA					
	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA					
	54	Elec Protect. Class		Enclosure Protection		NA				NA					
	55	Weight				*									
Purchase	56	Manufacturer		Model No.		*				*					
	57	Client Reference		Requisition No.		NA				*					
	58	Material Code				TC9765132662									
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST 5) Painting shall be as per attached Painting Specification															
						INSTRUMENT SPECIFICATION Desuperheater									
No.	By	Chk	Appr	Date	Revision	Code: 512				Dwg. No.: Ind Doc No.:					
										Sheet Rev.:					

General Data	1	Tag Number				TBA								
	2	Service				TBA								
	3	PID No.				TBA								
	4	ITEM				DSH 3"300RF-INTEG TCV 1"300RF-HART								
	5	SIL Level Required		Line Number		NA				TBA				
Operating and Mechanical Data	6	STEAM CONDITIONS												
	7	INLET STEAM								OUTLET STEAM				
	8	Line Size		Piping Class		3"		B2A		3"		B2A		
	9	Line Material		Line Schedule		A106 Gr.B		STD		A106 Gr.B		STD		
	10	Insulation Code												
	11	Molecular Weight		Density		18.015				18.015				
	12	Flow :	Min.	Norm.	Max.	Unit	750	900	1000	KG/HR	*	*	*	KG/HR
	13	Temp. :	Min.	Norm.	Max.	Unit	415	454	460	DEG C	350	350	350	DEG C
	14	Press. :	Min.	Norm.	Max.	Unit	13	13	13	KG/CM2G	*	*	*	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility										
	16	Design Press.		Design Temp. Max.		16.5				280				
	17	Body Material		Stem / Plug Material		A216WCB				SS316				
	18	Size		Rating		Face		3"		300#		RF		
	19	Pipe Nozzle Length		Schedule		*				*				
	20	Insertion Length				*								
	21	Flange Material				A105N								
	22	WATER CONDITIONS												
	23	Line Size		Piping Class		1"				B2A				
	24	Line Material		Line Schedule		A106 Gr.B				XS				
	25	Insulation Code												
	26	Flow :	Min.	Norm.	Max.	Unit	*	*	*				KG/HR	
	27	Temp. :	Min.	Norm.	Max.	Unit	105		105		110		DEG C	
	28	Press. :	Min.	Norm.	Max.	Unit	30		30		31		KG/CM2G	
	29	Design Temp.		Design Pressure		130				44				
	30	Body Material		Stem / Plug Material		A105N				SS316 (STELLITE)				
	31	Size		Rating		Face		1"		300		RF		
	32	Seat Leakage Class				IV								
	33													
	34													
	35	Area Classification		Req Safety Certification		IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type										
	37	Packing Material		Bonnet Type		*				*				
	38	Bellows	Cooling Fins	Extension		NA		NA		NA				
	39	Air-Fail Position				OPEN								
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*			
41		Pneumatic Supply				3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)								
Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*				
	43	Fluid Connection		Electrical Connection		*				1/2"NPT				
	44	Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67				
	45	Surge Prot Device		Action Direction		YES				*				
Accessories	46	Handwheel		Position		YES				*				
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*				
	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA				
	49	Solenoid Valve		Tag No.		NA				NA				
	50	Elec Protect. Class		Enclosure Protection		NA								
	51	Position Detector		Tag No.		*				NA				
	52	Elec Protect. Class		Enclosure Protection		NA				NA				
	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA				
	54	Elec Protect. Class		Enclosure Protection		NA				NA				
	55	Weight				*								
Purchase	56	Manufacturer		Model No.		*				*				
	57	Client Reference		Requisition No.		NA				*				
	58	Material Code				TC9765132689								

Notes:

- 1) TBA: To be Assigned.
- 2) *: Vendor to Assign.
- 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve.
- 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST
- 5) Painting shall be as per attached Painting Specification

					INSTRUMENT SPECIFICATION Desuperheater			
No.	By	Chk	Appr	Date	Revision	Code: 512	Dwg. No.: Ind Doc No.:	Sheet Rev.:

General Data	1	Tag Number					TBA								
	2	Service					TBA								
	3	PID No.					TBA								
	4	ITEM					DSH 24"300RF-NON INTEG TCV 1"600RF-HART								
	5	SIL Level Required		Line Number			NA				TBA				
Operating and Mechanical Data	6						STEAM CONDITIONS								
	7						INLET STEAM				OUTLET STEAM				
	8	Line Size		Piping Class			24"		B2A		24"		B2A		
	9	Line Material		Line Schedule			A672GR.B60CL.12		XS		A672GR.B60CL.12		XS		
	10	Insulation Code													
	11	Molecular Weight		Density			18.015				18.015				
	12	Flow :		Min.	Norm.	Max.	Unit	34840	93950	180950	KG/HR	*	*	*	KG/HR
	13	Temp. :		Min.	Norm.	Max.	Unit	264	291	350	DEG C	250	275	330	DEG C
	14	Press. :		Min.	Norm.	Max.	Unit	10	11	13.2	KG/CM2G	*	*	*	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility											
	16	Design Press.		Design Temp. Max.			17				350				
	17	Body Material		Stem / Plug Material			A216WCB				SS316				
	18	Size		Rating	Face		20"		300#		RF				
	19	Pipe Nozzle Length		Schedule			*				XS				
	20	Insertion Length					*								
21	Flange Material					A105N									
Spray Water / Temperature Control Valve	22						WATER CONDITIONS								
	23	Line Size		Piping Class			1"				D2A				
	24	Line Material		Line Schedule			A106 Gr.B				XS				
	25	Insulation Code		Density											
	26	Flow :		Min.	Norm.	Max.	Unit	*		*		*		KG/HR	
	27	Temp. :		Min.	Norm.	Max.	Unit			105		170		DEG C	
	28	Press. :		Min.	Norm.	Max.	Unit			61		78		KG/CM2G	
	29	Design Temp.		Design Pressure			170				78.3				
	30	Body Material		Stem / Plug Material			A216WCB				SS316 (STELLITE)				
	31	Size		Rating	Face		1"		600		RF				
	32	Seat Leakage Class					IV								
	33														
	34														
	35	Area Classification Req		Safety Certification			IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type											
37	Packing Material		Bonnet Type			*				*					
38	Bellows		Cooling Fins		Extension	NA		NA		NA					
39	Air-Fail Position					OPEN									
Actuator	40	Actuator Type		Spring Range			SPRING & DIAPHRAGM				*				
	41	Pneumatic Supply					3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)								
Positioner	42	Type		Input Signal			4-20MA SMART HART 7				*				
	43	Fluid Connection		Electrical Connection			*				1/2"NPT				
	44	Elec Protect. Class		Enclosure Protection			Ex-ia				IP 67				
Accessories	45	Surge Prot Device		Action Direction			YES				*				
	46	Handwheel		Position			YES				*				
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*					
Solenoid Valve	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA					
	49	Solenoid Valve		Tag No.			NA				NA				
Position Detector	50	Elec Protect. Class		Enclosure Protection			NA								
	51	Position Detector		Tag No.			*				NA				
Position Transmitter	52	Elec Protect. Class		Enclosure Protection			NA				NA				
	53	Position Transmitter		Tag No.			INCLUDED WITH POSITIONER				NA				
Purchase	54	Elec Protect. Class		Enclosure Protection			NA				NA				
	55	-		Weight			-				*				
	56	Manufacturer		Model No.			*				*				
	57	Client Reference		Requisition No.			NA				*				
	58	Material Code					TC9765132700								
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST															
						INSTRUMENT SPECIFICATION Desuperheater									
						Code: 512				Dwg. No.: Ind Doc No.:					
						Revision				Sheet Rev.:					



NUMALIGARH REFINERY LIMITED

Total pages: 38

NRL EXPANSION PROJECT JOB SPECIFICATION FOR PAINTING

			 Signatory's Name 2021.06.25 162523-0530	 Signatory's Name 2021.06.25 162523-0530	 Signatory's Name 2021.06.25 162523-0530			
D4	25/06/2021	Approved for Design	SS	MRM	JL			
D3	06/05/2021	Approved for Design	SS	MRM	JL			
D2	13/01/2021	Approved for Design	SS	MRM	VVS			
D1	01/12/2020	Approved for Design	SS	MRM	VVS			
A1	05/08/2020	Issued for Review	SS	MRM	VVS			
Rev.	Date	Reason for Issue	Prepared	Checked	Approved	Prepared	Review	Review
 TECHNIP INDIA LIMITED			Discipline Engineer	Discipline Lead	Contractor Representative	Discipline Engineer	Project Engineer	Department Head
			TPIL			NRL		
			Category		Code	Description		
			Facility Area Code		1Z	NR & MT Non Specific		
NRL's PO NO: 4300062833-AMA/16.06.2020			Document Type		SPE	Specification		
TP DOC REF: 082176C-ZZZ-JSD-2300-0001_D4			System Number		00	General		
			Life Cycle		01	Disk Ref.:		
This document is copyright and shall not be reproduced without permission of NRL			Originator/ Contractor	Asset Code	Discipline	Document Type	Sequence Number	Revision
			TP	1ZZZA	PI	SPE	0003	D4

TABLE OF CONTENTS

1. Introduction:.....	3
2. Definitions & Abbreviations;	3
3. Reference Documents:	4
4. Conflicts & Deviations:.....	4
5. Scope:	4
6. Terminology:.....	5
7. Exclusion:	5
8. Codes and Standards:	5
9. General Requirements:	7
10. Extent of Painting:	8
11. Standard Paint System for Packaged Equipment:	9
12. Surface Preparation:	9
12.1 Surface finish requirements:.....	10
12.2 Equipment for Surface Preparation:	11
12.3 Use of Dehumidifiers:	11
12.4 Air Blast Cleaning with abrasives:	12
12.5 Mechanical or Power Tool Cleaning:.....	12
12.6 Non-Compatible shop coat primer:	12
13. COATING PROCEDURE & APPLICATION:.....	13
14. SURFACE PREPARATION METHOD:.....	14
15. PAINTING / COATING MATERIALS:.....	15
15.1 List of Primers and Finish paints:.....	15
15.2 Detailed specification of Primer and Finish paint:	16
16. Painting System Tables and Selection Criteria:.....	20
16.1 Painting System Tables:.....	20
16.2 Table Selection Criteria:	21
17. Storage:.....	32
18. Approved Vendor List:	32
19. Quality Control, Inspection & Testing:	32
20. Guarantee:.....	35
21. Qualification Criteria of Painting Contractor / Sub-contractor:.....	35
22. Qualification /Acceptance Criteria for Paint Coating System:.....	35
23. Method of Sampling and Dispatch for Laboratory Testing:.....	37

ANNEXURE 1 – PAINTING SYSTEM

1. **Introduction:**

NUMALIGARH REFINERY LIMITED (NRL) has awarded a contract to M/s. Technip India Limited (TPIL) to provide Managing Project Management Consultancy services (MPMC) for the capacity expansion of their refinery from 3 to 9 MMTPA as part of Numaligarh Refinery Expansion Project (NREP) in Numaligarh, Golaghat District, Assam, India.

2. **Definitions & Abbreviations:**

Wherever used in this procedure, the following words shall have the meaning as given hereunder:

Abbreviation	Definition /Expanded form
NRL / OWNER / CLIENT	shall mean Numaligarh Refinery Limited
MPMC	Managing PMC (shall mean Technip India Limited)
CONSULTANT	Shall mean EPCM/ EPCM 1/ EPCM 2/ BOO
LICENSOR	Party responsible for process technology ownership for any UNIT
CONTRACTOR	Party whose services are obtained for performing the works specified as part of LSTK / EPC packages.
EPCM / EPsCMs	Engineering, Procurement & Construction Management Services to be performed for some process units, Utilities and Off-sites by CONSULTANT.
LSTK	Lump Sum Turn Key execution of EPC package by the CONTRACTOR
ENGINEER-IN-CHARGE (E-I-C)	NRL's representative authorized to provide all instructions related to execution of the PROJECT
SUB-CONTRACTOR	Party whose services are obtained by CONTRACTOR for performing the works specified as part of LSTK / EPC packages
VENDOR	Any third party supplying the equipment/materials for setting up the Plant
PROJECT / NREP	Indicates Numaligarh Refinery Expansion Project (NREP)
SITE	Indicates NRL's Refinery in Numaligarh, Golaghat District, Assam, India
UNIT	Indicates any particular portion of the NREP to be built which can be Process related or Utilities/Offsites related
EPCM-1	A portion of NREP for which NRL is engaging another Consultant to perform Engineering, Procurement, Construction management services as required within the battery limits of UNITS assigned to it
IM	Interface Management by MPMC with other consultants / CONTRACTORS engaged by NRL
EPCM-2	A portion of NREP for which NRL is engaging another Consultant to perform Engineering, Procurement, Construction management services as required within the battery limits of UNITS assigned to it
BOO	Build Own Operate – A portion of NREP for which NRL is engaging a CONTRACTOR on "Build Own Operate " mode to supply the output to be consumed in rest of the UNITS
OISD	Oil Industry Safety Directorate
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
P&ID / PID	Piping and Instrumentation Diagram
A/G	Above Ground
U/G	Under Ground
B/L	Battery Limit
ISBL	Inside Battery Limit
OSBL	Outside Battery Limit
EOT	Electrically-operated Overhead Travelling
MTO	Material Take Off
PESO	Petroleum and Explosives Safety Organisation
HPP	High Point of Paving
OWS	Oily Water Sewer
CRWS	Contaminated Rain Water Sewer

3. Reference Documents:

NR-0ZZZZ-PI-SPE-0001 - Job Specification for Design - Piping Material Specifications
NR-0ZZZZ-PI-SPE-0003 - Standard Specification for Hot Insulation of Vessels, Piping & Equipment
NR-0ZZZZ-PI-SPE-0004 - Standard Specification for Cold Insulation of Vessels, Piping & Equipment
NR-0ZZZZ-CV-SPE-0021 - Specification for Identification of Piping & Equipment

4. Conflicts & Deviations:

If conflicting statements exist within this document or between this document and other applicable specifications, Standard Drawings, Industry standards, codes, etc., it shall be brought to CLIENT / MPMC notice for clarification and proper approval shall be obtained before implementation. Decision of CLIENT / MPMC shall be final.

In case of contradiction between licensor specification, design basis and Job Supply Specifications / data Sheets, it has to be brought to the notice of CLIENT / MPMC and Decision of CLIENT / MPMC shall be binding on CONTRACTOR / VENDOR.

In general, order of priority of the documents shall be as follows,

- Local regulatory and statutory requirement.
- Licensor Requirements (as applicable)
- Project specification and datasheets, wherever applicable.
- This specification and relevant equipment/system specification.
- Codes and standard

5. Scope:

This specification defines the requirements of surface preparation, selection and application of paints and primers for all piping, equipment and structures etc.

The specification is applicable for supply of all paints, coatings, primers and other ancillary items etc. Method of surface preparation, supply and application of paints and primers shall suit given environment, location and temperature. Items requiring painting, field application procedures, inspection and testing of painting shall be governed by this specification.

This specification is suitable for use in normal and corrosive environment of various process, utility and other plants and offsite of refineries, petro-chemicals, onshore terminals and other chemical / industrial plants. Alternative paints / coatings would be specified if necessary for specific or more stringent requirements.

The painting specification covers every type of equipment such as tanks, vessels, drums, heat exchangers/coolers, air fin coolers, pumps, turbines, compressors, filters, engines, motors, boilers or heaters /furnaces their accessories, fans, stacks / chimney and package units etc.

The painting specification covers all types of process and utility piping services which can be non-insulated / insulated, jacketed or lined requiring painting. All types of pipe supports, hangers, spring boxes are also covered.

All types of structural steel members, platforms ladders, chequered plates, gratings, walkways, trolleys, monorails, davits, structural steel sheds and buildings are also covered under this painting specification.

The painting of equipment shall conform to equipment data sheets. Painting of piping shall conform to line schedule and piping isometrics etc.

6. **Terminology:**

RH	Relative Humidity
RCC	Reinforced Cement Concrete
PO	Purchase Order
CS	Carbon steel
LTCS	Low Temp. Carbon Steel
AS	Alloy Steel
SS	Stainless steel
MS	Mild Steel
GI.	Galvanized Iron / steel
HCL	Hydrochloric acid
ITP	Inspection Test Plan
TPI	Third Party Inspection
DFT	Dry Film Thickness
WFT	Wet Film Thickness
TSAC	Thermally Sprayed Aluminium Coating
Micr.	Micron

7. **Exclusion:**

The following surfaces and materials shall not require painting in general. However, if there is any specific requirement by the owner, the same shall be painted as per the relevant specifications:

- Austenitic stainless steel (Non-insulated)
- Plastics and or plastic coated surfaces
- Non-ferrous materials like Aluminum, Cu-Ni alloy, Monel, Incoloy
- RCC or cement lined surfaces except those specified
- Gaskets / seals
- Gauge Glasses
- Meter Faces
- Valve Stem and Threads
- Name Plates
- Insulation or Fireproofing
- Factory Finished Control Panels
- Factory Finished Instrument Cases and Meters

8. **Codes and Standards:**

The following codes shall be applicable, however purchaser may specify any other relevant code for any purpose at any time. The codes latest edition as on date of issue of material requisition shall be applicable.

Society for Protective Coatings (formerly Steel Structure Painting Council - SSPC)

SSPC-SP1	Steel Structures Painting Council Surface Preparation Specification – Solvent Cleaning
SSPC-SP10	Near-White Metal Blast Cleaning
SSPC-SP11	Surface preparation specification -Power tool cleaning to bare metal
SSPC-SP16	Brush-off Blast Cleaning Non-Ferrous Metals
SSPC-AB1	Mineral and Slag Abrasives

SSPC-AB 3	Ferrous Metallic Abrasives
SSPC-PA 1	Shop, Field, and Maintenance Painting of Steel.
SSPC-PA2	Procedure for determining conformance to Dry Coating thickness requirements
SSPC-VIS 1	Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning.

National Association of Corrosion Engineers (NACE)

NACE SP 0188	Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates
NACE SP 0287	Field Measurement of Surface Profile of Abrasive Blast Cleaned Steel Surfaces Using a Replica Tape.
NACE SP 0198	The control of corrosion under Thermal Insulation and Fireproofing Materials

International Organization for Standardization (ISO)

ISO 8501-1	Preparation of steel substrates before application of paints and related products – Visual assessment of surface cleanliness. Part 1: Rust grades and preparation grades of uncoated steel substrates and of steel substrates after overall removal of previous coatings
ISO 8501-2	Preparation of steel substrates before application of paints and related products – Visual assessment of surface cleanliness. Part 2: Visual assessment of surface cleanliness
ISO 8501-3	Preparation of steel substrates before application of paints and related products – Visual assessment of surface cleanliness. Part 3: Preparation grades of welds, cut edges and other areas with surface imperfections
ISO 8502-1 to 4	Preparation of steel substrates before application of paints and related products – Tests for the assessment of surface cleanliness
ISO 8502-4	Preparation of steel substrates before application of paints and related products – Tests for the assessment of surface cleanliness: Guidance on the Estimation of the Probability of Condensation Prior to Paint Application
ISO 8502-6	Preparation of steel substrates before application of paints and related products – Tests for the assessment of surface cleanliness: extraction of soluble contaminants for analysis-The Bresle method
ISO 8502-9	Preparation of steel substrates before application of paints and related products – Tests for the assessment of surface cleanliness: Field method for conductometric determination of water-soluble salts
ISO 8503-1 to 4	Preparation of steel substrates before application of paints and related products – Surface roughness characteristics of blast-cleaned surfaces
ISO 8504-1 to 3	Preparation of steel substrates before application of paints and related – Methods for surface preparation products
ISO 12944-2	Paints and varnishes – Corrosion protection of steel structures by protective paint systems - Part 2: Classifications of environments
ISO 12944-3	Paints and varnishes – Corrosion protection of steel structures by protective paint systems - Part 3: Design and constructive requirements
ISO 12944-4	Paints and varnishes – Corrosion protection of steel structures by protective paint systems - Part 4: Surface types and preparation
ISO 12944-5	Paints and varnishes – Corrosion protection of steel structures by protective paint systems - Part 5: Protective paint systems
ISO 12944-7	Paints and varnishes – Corrosion protection of steel structures by protective paint systems - Part 7: Execution and supervision of painting work
ISO 11126-2	Preparation of steel substrates before application of paints and related products -- Test methods for non-metallic blast-cleaning abrasives
ISO 11126-3	Preparation of steel substrate before application of paint and related product-specifications for non-metallic blast-cleaning abrasives

ISO 11127	Measurement of, and acceptance criteria for the thickness of dry films on rough surfaces
ISO 19840	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Measurement of, and acceptance criteria for, the thickness of dry films on rough surfaces
ISO 2409	Paints and varnishes – Cross Cut Test

American Society for Testing and Materials (ASTM)

ASTM D 3359	Measuring adhesion by tape test
ASTM D 4285	Standard test method for indicating oil or water in compressed air
ASTM D 4417	Test Method for Field Measurement of Surface Profile of Blasted Steel.
ASTM D 4752	Measuring MEK resistance of ethyl silicate (inorganic) zinc-rich primers by solvent rub
ASTM D 5402	Assessing the solvent resistance of organic coatings using solvent rubs
ASTM D 4940	Standard Test Method for Conductimetric Analysis of Water Soluble Ionic Contamination of Blast Cleaning Abrasives
ASTM D 6386	Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting
ASTM D 520	Standard Specification for Zinc Dust Pigment

Indian Standards (IS)

IS-5: 1994	Colours for ready mixed paints and enamels.
IS-2379: 1990	Colour codes for identification of pipe lines.
IS-2629: 1985	Recommended practice for hot-dip galvanizing on iron and steel.
IS-2633: 1986	Methods for testing uniformity of coating of zinc-coated articles.
IS-8629: 1977	Code of practice for protection of iron and steel structures from atmospheric corrosion.
IS:110	Specification for Ready Mixed Paint, Brushing, Grey Filler, for Enamels, for Over Primers
IS:101	Methods of test for ready mixed paints & enamels.

The Contractor shall perform the work in accordance with the following reference documents issued to them for execution of painting works.

- Bill of quantities for piping, equipment, machinery and structures etc
- Piping Line List
- Specifications for Painting
- Any Specific requirements from client

9. General Requirements:

- This specification shall govern all works covered by the contract, and without prejudice to the provisions of various Indian and international codes of practice, standard specifications etc. The Contractor shall carry out the works in all respects with the best quality of materials and workmanship and in accordance with the best engineering practices and instructions of Owner / Engineer in charge.
- All tools, brushes, rollers, spray guns, blast material, hand power tools for cleaning and all equipment,

scaffolding materials, shot / sand blasting equipment and air compressors etc. required to be used shall be suitable for the work and all in good order and shall be arranged by the Contractor at site and in sufficient quantity.

- The compressed air supply used for blasting shall be free of water and oil. Adequate separators and traps shall be provided and these shall be drained continuously. Pressure Gauges fitted to compressor shall be calibrated with necessary certificate.
- Blast cleaning equipment shall be in accordance with all applicable regulations. The spraying equipment to be used shall meet the recommendations and instructions set forth by the paint supplier for each specific paint or coating system.
- All mechanical equipment shall be earthed and all necessary precautions shall be taken to prevent the build-up of static electricity. Especially blasting equipment, its operators and the equipment being blasted shall be properly earthed to prevent the occurrence of electro-static discharges
- Mechanical mixing shall be used for paint mixing operations in case of two pack systems except that the Engineer in charge may allow the hand mixing of small quantities at his discretion.
- All painting materials and ancillary materials needed for completion of the contract shall conform to the prescribed specifications. Contractor shall procure these materials from specified manufacturers or their stockiest with proper marking and identification as proof of original materials.
- Any sub-standard or duplicate materials or lower grade/ brand materials shall not be used. Owner / Engineer in charge shall have the right to reject all such materials at any stage. Contractor shall seek prior approval from Owner before actual application to avoid rejection of works carried out with such sub-standard materials.
- The Contractor shall bring to the notice of Owner any discrepancy between this specification and codes specified herein. Contractor may request Owner for clarification of any of the applicable clause of this specification or about applicability of a particular painting system for any service / surface. Any deviation from this specification pertaining to supply or application without written permission of Owner shall be rejected by Engineer in charge.
- The items listed in the paint systems is indicative only, however Engineer in charge may decide about the applicability of the paint system for any of the works.
- The Contractor shall ensure all safety and protective apparatus are fully provided to their staffs.
- The contractor shall be fully responsible for carrying out all the necessary painting, coating and lining on external and internal surfaces as per the tender requirement.
- The paint manufacturer's instructions shall be followed as far as practicable at all times for best results. Particular attention shall be paid to the following:
 - Instructions for storage to avoid exposure as well as extremes of temperature.
 - Surface preparation prior to painting shall be followed as per the specification.
 - Mixing and thinning.
- Paint manufacturers shall furnish the characteristics of all paints materials on original printed literature, along with the test certificate for all specified characteristics given in this specification. All the paint materials shall be of first quality and conform to the general characteristics described in various tables.
- Contractor shall fully comply with the client specification for Colour Coding of Piping, Equipment and Structures issued during FEED stage of the project. This specification covers colour codes, identification marking on piping and equipment, recommended colours for paint systems and painting for "Civil Defence" requirements etc.
- Contractor shall ensure that the paint material supplied are fully within the validity period of the product and not exposed to open atmosphere.

10. **Extent of Painting:**

The following surfaces and materials shall require shop, pre-erection and field painting:

- All Non-insulated carbon steel and alloy steel piping as described under scope including valves, flanges, fittings, specialty items, in line items, and all supports (including painting of identification marks), flare lines ,furnace ducts and stacks.
- Titanium catalysed inorganic copolymer can be used commonly for all valves.

- All insulated parts of vessels, boilers, chimneys, stacks, piping and steam piping, and any other insulated items present.
- All items contained in a package unit requiring painting.
- All types of structural steel members, platforms ladders, chequered plates, gratings, walkways, trolleys, monorails, davits, structural steel sheds and buildings are also covered under this painting specification.
- External surfaces of MS chimney with or without refractory lining and internal surfaces of MS chimney without refractory lining. (If present)
- Representation of colour bands on all piping including insulated aluminium clad, galvanized, SS and nonferrous piping.
- Identification lettering / numbering on all painted surfaces of equipment / piping insulated aluminium clad, galvanized SS and non-ferrous piping.
- Marking / identification signs on painted surfaces of equipment / piping including hazardous service.
- Supply of all primers, paints and all other materials required for painting (other than Owner supplied materials)
- Metal Area over which insulation surface of equipment and pipes wherever required.
- Painting under insulation for carbon steel, alloy steel and stainless steel as per relevant NACE SP 0198 to prevent corrosion.
- Painting of pre-erection / fabrication and Shop primer.
- Repair work of damaged pre-erection / fabrication and shop primer and weld joints in the field / site before and after erection as required.
- All CS Piping, equipment, storage tanks and internal surfaces of RCC tanks in ETP plant.
- Quality control, testing and inspection during all stages of work (surface preparation, application of coating and testing of furnished coating).

11. **Standard Paint System for Packaged Equipment:**

Components (such as pumps, compressors turbines, blowers, electric motors, transformers, generators, hoists and cranes, instrumentation (except valve bodies and relief valves)) within equipment packages (not whole assemblies), electrical items (such as transformers, junction boxes, switchgear cabinets and enclosures), items, equipment for permanent storage and use in indoor dry and HVAC controlled areas, can be coated according to Manufacturers standard coating system and according to Internal Supplier's quality regime provided it complies with the following requirements:

- As minimum requirement, the standards shall comply with common standard IS, ISO and ASTM norms.
- The Manufacturer's coating system must meet the minimum requirements for corrosion protection, adapted to the climatic and corrosive conditions of the location of the part to be painted.
- Surface preparation should be abrasive blast cleaning to grade SSPC-SP10 minimum for carbon steel surfaces.
- The system shall have adequate proven track record for the conditions and items to be coated.

In this case, a painting procedure is to be submitted including the data sheets and a necessary approval to be obtained.

12. **Surface Preparation:**

Any one of following methods of surface preparation shall be followed, depending on condition of surface to be painted and as approved or instructed by Engineer in charge.

- Manual or hand tool cleaning

- Mechanical or power tool cleaning
- Dry abrasive blast cleaning

Before blasting salt contamination test to be carried out for metals & Testing for chloride and soluble salt concentrations and the pH level shall be done using a Bresle Sampler according to ISO 8502-6. The chloride and soluble salt concentrations shall be less than 20 mg/ m² and the pH shall be neutral (between 6 and 8). When these levels are exceeded, the surfaces shall be either steam cleaned or high pressure water washed as per SSPC SP1 or ISO 12944 before abrasive blasting. The cleaned surface shall be retested to verify that the contaminant levels are within the acceptable range. Checks shall be done on each component at least once per 200 m² of blasted surface and a minimum of 3 checks per shift. The test report shall be maintained recording the ambient and substrate temperature, relative humidity, abrasive medium, test obtained values etc., Measuring device shall be regularly calibrated.

12.1 Surface finish requirements:

- When surface is exposed to normal atmospheric conditions and where other methods cannot be adopted. May also be used for spot cleaning during maintenance.
 - Manual or hand tool cleaning to: SSPC-SP-2 or ST.2 Level
Remove loose rust / mill scale / loose paint thoroughly by chipping, scrapping, sanding and or wire brushing. Finished surface shall have a faint metallic sheen.
 - Mechanical or power tool cleaning to SSPC-SP-3 or ST.3 Level
Remove loose rust / mill scale / loose paint to degree specified by power tool chipping, de-scaling, sanding, wire brushing and grinding, after removal of dust, surface should have a pronounced metallic sheen.

Care to be taken where the welding / riveting portion of the joints.

- **Dry abrasive Blast cleaning:**

There are four different methods of dry abrasive blast cleaning as described below. Each method shall be selected depending on surface finish required for particular paint system. However Engineer in charge may instruct about any of the system to be followed for a particular job / item as deem necessary.

- **White metal to SSPC-SP-5 or SA.3 or NACE #1 Level**
Remove all visible rust / Mill scale / paint and foreign matter 100% to achieve desired surface profile with blast cleaning to white metal cleanliness in order to achieve extremely clean surface for prolonged life of paint system.
- **Near white metal to SSPC-SP-10 or SA.2 ½ or NACE # 2 Level**
Blast clean to near white metal cleanliness until at least 95% of each element of surface area is free of all visible residues with desired surface profile in order to have minimum acceptable clean surface. This is the minimum requirement for chemically resistant paint systems such as epoxy, vinyl, polyurethane based and inorganic zinc silicate paints, also for conventional paint systems used under fairly corrosive conditions to obtain desired life of paint system.
- **Commercial Blast to SSPC-SP-6 or SA.2 or NACE # 3 Level**
Blast clean until at least two-third of each element of surface area is free of all visible residues with desired surface profile. Used for steel required to be painted with conventional paints for exposure to mildly corrosive atmosphere for longer life of the paint systems.
- **Brush-off Blast to SSPC-SP-7 or SA.1 or NACE # 4 Level**
Blast cleaning to white metal cleanliness, removal of all visible rust, mill scale, paint and foreign matter where surface profile is not so important

12.2 Equipment for Surface Preparation:

- All tools, brushes, rollers, spray guns, blast material, hand power tools for cleaning and all equipment, scaffolding materials, shot & grit blasting equipment and air compressors etc. required to be used shall be suitable for the work and all in good order and shall be arranged by the Contractor at site and in sufficient quantity. The manufacturer's test certificates & data sheets for all the above items shall be reviewed by Engineer in charge at site before start of work.
- Mechanical mixer shall be used for paint mixing operations in case of two pack systems except that the Engineer in charge may allow the hand mixing of small quantities at his discretion in case of specific requirement for touch up work only.
- Mill scale, rust, rust scale and foreign matter shall be removed fully to ensure that a clean and dry surface is obtained. The minimum acceptable standard, in case of thermally sprayed metal coatings, in case of mechanical or power tool cleaning it shall be St. 3 or equivalent. In case of blast cleaning it shall be Sa 2-1/2 as per Swedish Standard SIS-055900 or SSPC-SP or ISO 8501-01. Blast cleaning shall be Sa 3 as per Swedish Standard in case thermally sprayed metal coatings.
- Before surface preparation by blast cleaning, the surface shall be degreased by aromatic solvent to remove all grease, oil etc.

12.3 Use of Dehumidifiers:

- Blast cleaning shall not be performed for internal or external surface, where dust can contaminate surfaces undergoing such cleaning or during humid weather conditions having humidity exceed 85%. In case of internal coating of storage tanks, dehumidifier shall be used, to control humidity level below 60%. Dehumidifier should depress the dew point of air in the enclosed space, enough to maintain it 3°C below the metal substrate temperature during entire period of blasting and coating application. During the interval time between application of primer coat and subsequent intermediate and top coats or between blast cleaning completion and start of application of primer coat, dehumidifier unit should be in continuous operation to ensure that no condensation occurs on substrate.
- Dehumidifier should be able to maintain grain drop (moisture removal) at the rate of 25 grains per pound of air per hour. Dehumidifier should have capacity of at least 2 air changes per hour of the enclosed space. All necessary psychometric data should be collected by contractor for the given site conditions before starting operation of dehumidifier to ensure that desired values of dew point, moisture content in enclosed scope is achieved.
- Dehumidification shall be maintained round the clock for surface preparation and painting till the total coating application is over.
- Dehumidifier shall not be stopped under any condition till the entire blasted surface is primed to the satisfaction of the technical representative of the paint manufacturer interested with quality assurance for the work. In case the dehumidifier breaks down in middle of the job, the same shall be replaced at the risk and the cost of the contractor and the entire unfinished work shall be repeated.
- The Engineer in charge shall have the right to disallow usage of dehumidifier if the performance is not meeting the specified requirements. Under such circumstances the contractor shall remove the equipment and replace the same with another equipment to provide satisfactory results without any additional cost to the owner.
- Irrespective of the method of surface preparation, the first coat of primer must be applied by airless spray / air assisted conventional spray if recommended by the paint manufacturer on dry surface. This should be done immediately and in any case within 4 hours of cleaning of surface. However, at times of unfavourable weather conditions, the Engineer in charge shall have the liberty to control the time period, at his sole discretion and/or to insist on recleaning, as may be required, before primer application is taken up. In general, during unfavourable weather conditions, blasting and painting shall be avoided as far as practicable.

- The external surface of R.C.C. chimney stack to be painted shall be dry and clean. Any loose particle of sand, cement, aggregate etc. shall be removed by scrubbing with soft wire brush. Acid etching with 10-15% HCL solution for about 15 minutes shall be carried and surface must be thoroughly washed with water to remove acid & loose particles and then dried completely before application of paint.

12.4 Air Blast Cleaning with abrasives:

- The surfaces shall be blast cleaned using one of the abrasives like angular chilled cast iron or steel grit, copper slag or Nickel slag, Aluminium oxide (Al_2O_3) particles at pressure of 7 kg/cm² at an appropriate distance and angle depending of nozzle size maintaining constant velocity and pressure. It is to be ensured that the crystalline free silica content is <1%.
- Chilled cast iron or steel shall be in the form of shot or grit of size in the range of G 16 - G42 conforming to SSPC AB1 and S250 grade size of steel shots (maximum) to obtain a desired surface profile of 35-50 microns trough to peak. For all other abrasives, size shall be in the range of G16 - G24. The combination of steel grits and shots shall be normally in the ratio of 3 : 1 . The quality of abrasives shall be free from contaminants and impurities and shall meet the requirements of SSPC AB1.
- Compressed air shall be free from moisture and oil. The blasting nozzles should be venturi style with tungsten carbide or boron carbide as the materials for liners. Nozzles orifice may vary from 3/16" to 3/4". On completion of blasting operation, the blasted surface shall be clean and free from any scale or rust and must show a grey white metallic luster. Primer / first coat of paint shall be applied within 4 hours of surface preparation. Blast cleaning shall not be done outdoors in bad weather without adequate protection or when there is dew on the metal, which is to be cleaned. Surface profile shall be uniform to provide good key to the paint adhesion (i.e. 35 to 50 microns). If possible vacuum collector shall be installed for collecting the abrasives and recycling.

12.5 Mechanical or Power Tool Cleaning:

Power tool cleaning shall be done by mechanical striking tools, chipping hammers, grinding wheels or rotating steel wire- brushes. Excessive burnish of surface shall be avoided as it can reduce paint adhesion. On completion of cleaning, the detached rust mill scale etc. shall be removed by clean rags and /or washed by water or steam and thoroughly dried with compressed air jet before application of paint.

12.6 Non-Compatible shop coat primer:

- For equipment on which application of total protective coating (Primer+ Intermediate + top coat) is carried out at shop, compatibility of finish coat with primer should be checked with paint manufacturer. Specific duration mentioned in the manufacturer specification shall be fully If the shop coat is in satisfactory condition showing no major defect upon arrival at site, the shop coat shall not be removed.
- Shop coated equipment (coated with Primer & finishing coat) should not be repainted unless paint is damaged. Repair shall be carried out as per **Table 10** of paint systems depending upon compatibility of paint.
- Shop primed equipment and surfaces will only be 'spot cleaned' in damaged areas by means of power tool brush cleaning or hand tool cleaning and then spot primed before applying one coat of field primer unless otherwise specified. If shop primer is not compatible with field primer then shop coated primer should be completely removed before application of selected paint system for particular environment.
- Coating application at field (field primer, intermediate and top coat) on equipment, structures, piping, etc. shall be carried out only after its erection and all welding, testing, steam purging (wherever carried out) have been completed.

13. **COATING PROCEDURE & APPLICATION:**

- All paint coatings shall be applied by airless spray excepting at the following special cases where application can be carried out by brush subject to suitability of the application of the paint product by brush.
 - Spot repair
 - Stripe coating on edges
 - Small bore parts not suitable for spray application
- Irregular surfaces such as sharp edges, welds, small brackets, and interstices may stripe coated to ensure specified DFT is achieved. Paint manufacturer's recommendation should be followed before deciding for brush application.
- Surface shall not be coated in rain, wind or in environment where injurious airborne elements exists, when the steel surface temperature is less than 5°F above dew point when the relative humidity is greater than 85% or when the temperature is below 40°F and when the ambient/substrate temp is below the paint manufacturer's recommended temperature of application and curing. De-humidifier equipment shall be used to control RH and Dew point. The paint application shall not be done when the wind speed exceeds 20 km per hour.
- Blast cleaned surface shall be coated with one complete application of primer as soon as practicable but in no case later than 4 hours the same day.
- To the maximum extent practicable, each coat of material shall be applied as a continuous film uniform thickness free of probes. Any spots or areas missed in application shall be recoated and permitted to dry before the next coat is applied. Applied paint should have the desired wet film thickness.
- Each coat shall be in proper state of cure or dryness before the application of succeeding coat. Material shall be considered dry for recoating when an additional coat can be applied without the development of any detrimental film irregularities, such as lifting or loss of adhesion of the under coat. Manufacturer instruction shall be followed for inter-coat interval.
- When the successive coat of the same colour have been specified, alternate coat shall be tinted, when practical, sufficiently to produce enough contrast to indicate complete coverage of the surface. The tinting material shall be compatible with the material and not detrimental to its service life and shall be recommended by the original paint manufacturer.
- Airless spray application shall be in accordance with the following procedure: as per steel structure paint Manual Vo. 1 & Vol. 2 by SSPC, USA, Air less spray relies on hydraulic pressure rather than air atomization to produce the desired spray. An air compressor or electric motor is used to operate a pump to produce pressures of 1000 to 6000 psi. Paint is delivered to the spray gun at this pressure through a single hose within the gun, a single paint stream is divided into separate streams, which are forced through a small orifice resulting in atomization of paint without the use of air. This results in more rapid coverage with less over spray.
- Airless spray equipment is mounted on wheels, and paint is aspirated in a hose that sucks paint from any container, including drums. The unit shall have in built agitator that keep the paint uniformly mixed during the spraying. The unit shall consist of in built strainer. Usually very small quantity of thinning is required before spray. In case of high build epoxy coating (two pack). 30:1 pump ratio and 0.020-0.023" tip size will provide a good spray pattern. Ideally fluid hoses should not be less than 3/8" ID and not longer than 50 ft to obtain optimum results. In case of gun choking, de-choking steps shall be followed immediately.
- Brush application of paint shall be in accordance with the following:
 - Brushes shall be of a style and quality that will enable proper application of paint.

- Round or oval brushes are most suitable for rivets, bolts, irregular surface, and rough or pitted steel. Wide flat brushes are suitable for large flat areas, but they shall not have width over five inches.
- Paint shall be applied into all corners.
- Any runs or sags shall be brushed out.
- There shall be a minimum of brush marks left in the applied paint
- Surfaces not accessible to brushes shall be painted by spray, daubers, or sheepskin.
- For each coat the painter should know the WFT corresponding to the specified DFT and standardize the paint application technique to achieve the desired WFT. This has to be ensured in the qualification trial.
- No coat shall be applied until the preceding coat has dried. The material shall be considered dry for re-coating when another coat can be applied without the development of any film irregularities such as lifting or loss of adhesion of undercoats. Drying time of the applied coat should not exceed maximum specified for it as a first coat; if it exceeds the paint material has possibly deteriorated or maxing is faulty.
- No paint shall be force dried under conditions which will cause chalking, wrinkling, blistering formation of pores, or detrimentally affect the conditions of the paint.
- No drier shall be added to paint on the job unless specifically called for in the manufacturer's specification for the paint.
- Paint shall be protected from rain, condensation, contamination, snow and freezing until dry to the fullest extent practicable.

14. **SURFACE PREPARATION METHOD:**

The table below describes the methods for surface preparation and the standards used for cleanliness and surface preparation for painting /coating works.

Sr. No.	Description	International Standards (Equivalent)			Remarks
		ISO 8501-1 / SIS-05 59 00	SSPC-SP,USA	NACE, USA	
1.	Manual or hand tool cleaning: Removal of loose rust, loose mill scale and loose paint, chipping, scrapping, standing and wire brushing. Surface shall have a faint metallic sheen.	ST.2	SSPC-SP-2	-	This method is applied when the surface is exposed to normal atmospheric conditions when other methods cannot be adopted and also for spot cleaning during maintenance painting.
2.	Mechanical or power tool cleaning: Removal of loose rust loose mill scale and loose paint to by power tool chipping, de-scaling, sanding, wire brushing and grinding, after removal of dust, surface shall have a pronounced metallic sheen.	ST.3	SSPC-SP-3	-	
3	Dry abrasive Blast cleaning: There are four common grades of blast cleaning White metal				

3.1	Blast cleaning to white metal cleanliness: Removal of all visible rust. Mill scale, paint & foreign matter 100% cleanliness with desired surface profile	SA 3	SSPC-SP-5	NACE #1	Where extremely clean surface can be expected for prolong life of paint system.
3.2	Near white metal: Blast cleaning to near white metal cleanliness, until at least 95% of each element of surface area is free of all visible residues with desired surface profile.	SA 2 ½	SSPC-SP-10	NACE #2	For chemically resistant paint systems such as epoxy, vinyl, polyurethane based and inorganic zinc silicate paints, and for paint systems used under fairly corrosive conditions to obtain desired life of paint system.
3.3	Commercial Blast: Blast cleaning until at least two-third of each element of surface area is free of all visible residues with desired surface profile.	SA 2	SSPC-SP-6	NACE #3	For steel required to be painted with conventional paints for exposure to mildly corrosive atmosphere for longer life of the paint systems.
3.4	Brush-off Blast: Blast cleaning to white metal cleanliness, removal of all visible rust, mill scale, and paint foreign matter. Surface profile is not so important	SA 1	SSPC-SP-7	NACE #4	-

15. PAINTING / COATING MATERIALS:

15.1 List of Primers and Finish paints:

A broad list of primers and finish paints to be used for painting and coating works is as given below:

PRIMERS	
PR-1	Chlorinated Rubber Zinc Phosphate Primer
PR-2	Etch Primer / Wash Primer
PR-3	Two component Epoxy Zinc Phosphate Primer cured with polyamine hardener
PR-4	Single pack, cold galvanizing compounds containing minimum 92% Electrolytic Zinc in dry film. make ZINGA, LOCKTITE (of HENKEL) or ZRC
FINISH COATS / PAINTS	
FP-1	Two component Acrylic – Polyurethane finish paint
FP-2	Chlorinated Rubber finish paint
FP-3A	High Build Epoxy finish coating cured with polyamine hardener
FP-3B	High Build Epoxy finish coating cured with polyamide hardener
FP-3C	Solvent less Epoxy Coating cured with polyamine hardener
FP-4	High build Coal Tar Epoxy coating cured with polyamine hardener
FP-5	Self-priming surface Tolerant High Build Epoxy coating. cured with polyamine hardener

FP-6	Two component Inorganic Zinc Silicate coating
FP-7	Heat resistant synthetic medium based Aluminium paint
FP-8	Two component Heat resistant Silicone Aluminium paint.
FP-9	Specially formulated Coal Tar Epoxy coating. cured with polyamine hardener
FP-10	Two component Epoxy Phenolic coating cured with Polyamine adduct hardener system
FP-11	Engineered Epoxy Poly Siloxane Coating or High Build cold applied inorganic Co-polymer based Aluminium coating
FP-12	Two component solvent free type High Build Epoxy Phenolic / Novalac Epoxy Phenolic coating cured with Polyamine adduct hardener system

15.2 Detailed specification of Primer and Finish paint:

The following are the various parameters of primers, finish paints and coating materials to be used for carrying out various painting / coating works. Refer general notes at the end of the table.

PRIMERS

Sr. No.	Description	PR-1	PR-2	PR-3	PR-4
1.	Technical Name	Chlorinated Rubber Zinc Phosphate Primer	Etch Primer / Wash Primer	Epoxy Zinc Phosphate Primer	Zinga, Locktite or ZRC Cold Galvanized
2.	Pack Type	Single Pack	Two Pack	Two Pack	Single Pack
3	Composition	Air Drying Chlorinated, Rubber based medium Plasticized with unsaponifiable Plasticizer, pigmented with zinc phosphate	Polyvinyl butyral resin medium cured with phosphoric acid solution. pigmented with zinc tetroxy chromate	Polyamine cured epoxy resin, medium, pigmented with zinc phosphate	Synthetic resin based zinc galvanizing containing min 92% of electrolytic zinc dust of 99.95% purity.
4.	Vol. Solids %	40±3	10±1	50±1	37%
5.	DFT (Micron) / Coat	40-45	8-10	40-50	40-50fl
6.	Covering M ² / Coat / Litre	8-10	8-10	8-10	4 m ² /kg
7.	Wt. Kg. / Litre	1.3±0.05	1.2±0.05	1.4±0.05	2.67 kg at 15°C
8.	Touch Dry at 30°C Min.	30 minutes	2hrs.	After 30 min.	10 minutes
9.	Hard Dry at 30°C Max.	8 hrs.	24 hrs.	8 hrs.	24 hrs.
11.	Over-coat Interval at 30°C	Min.: 8 hrs.	Min.: 4-6 hrs.	Min.:8hrs.	Min.:4 hrs
12.	Pot Life at 30°C	Not applicable	Not applicable	6-8 hrs.	Unlimited
13.	Temperature. Resistance (min) (ASTM D2485 method B)	60°C Dry service	NA Dry service	80°C Dry service	50°C Dry service

FINISH PAINTS

Sr. No.	Description	FP-1	FP-2	FP-3A / 3B	FP-3C	FP-4
1.	Technical Name	Acrylic Polyurethane finish paint	Chlorinated Rubber based finish paint	Epoxy-High Build coating	Solvent less Epoxy coating	High Build Coal Tar Epoxy coat.
2.	Pack Type	Two pack	Single pack	Two pack	Two pack	Two pack
3	Composition	Aliphatic isocyanate cured acrylic finish paint with Glossy-High Glossy finish	Plasticized chlorinated rubber based medium with chemical and weather resistant pigments.	FP-3A: Aromatic amine cured epoxy resin medium suitably pigmented. FP-3B: polyamide cured epoxy resin medium suitably pigmented	Cured with Amine Adduct; catalysed epoxy resin suitably pigmented.	Polyamide cured epoxy resin blended with coal tar medium, suitably pigmented
4.	Vol. Solids %	40±3	38±2	60±3	99±1	65±3
5.	DFT (Micron) / Coat	30-40	30-40	100-125	200-500	100-125
6.	Covering M ² / Coat / Litre	11-15	11-15	5-6	2-3	5.2-6.5
7.	Wt. Kg. / Litre	1.15±0.03	1.15±0.03	1.42±0.03	1.40±0.03	1.40±0.03
8.	Touch Dry at 30°C Max.	30 Min.	30 Min.	3 Hrs.	3 Hrs.	4 Hrs..
9.	Hard Dry at 30°C Max. Full Cure at 30°C for Immersion	8 Hrs.	8 Hrs..	16 Hrs.	16 Hrs.	48 Hrs.
		NA	NA	5 days	NA	5 days
10.	Over-coat Interval at 30°C (Min)	12 Hrs.	Overnight.	Overnight. Max.: 5 days	8 Hrs.. Max.: 48 hrs	24 Hrs. Max.:5 days
11.	Pot Life at 30°C for paints -two components	6-8 Hrs.	NA	4-6 Hrs.	30 Min.	4-6 Hrs.
12.	Temperature. Resistance (min) (ASTM D2485 method B)	80°C Dry Service	60°C Dry Service	80°C Dry Service	120°C Dry Service 50°C Immersion	80°C Dry Service

FINISH PAINTS... CONT'D

Sr. No.	Description	FP-5	FP-6 (Primer)	FP-7	FP-8
1.	Technical Name	High Build Epoxy , self-priming type surface tolerant coating (for complete rust control)	Inorganic zinc silicate coating	Aluminium Paint Heat resistant synthetic medium based suitable up to 250°C dry temp	Silicone Aluminium Paint Heat resistant suitable up to 540°C dry temp.
2.	Pack Type	Two pack	Two pack	Two pack	Single pack
3	Composition	Epoxy resin based suitable pigmented and capable of adhering to manually prepared surface and old coating.	Air drying self-curing solvent based inorganic zinc silicate coating with minimum 80% zinc content on dry film. The final cure of the dry film shall pass the MEK rub test. Zinc purity shall be Type-II of ASTM D520	Heat resistant synthetic medium based Aluminium paint suitable up to 250°C.	Silicon resin based medium with Aluminium flakes.
4.	Vol. Solids %	78±3	60±3	38±0.03	20±2
5.	DFT (Micron) / Coat	100-125	65-75	15-20	15-20
6.	Covering M ² / Coat / Litre	6.0-7.2	8-9	10-12	8-10
7.	Wt. Kg. / Litre	1.41 ± 0.03	2.3 ± 0.03	0.95 ± 0.03	1.0 ± 0.03
8.	Touch Dry at 30°C Max.	3 Hrs.	30 Min.	3 Hrs.	30 Min.
9.	Hard Dry at 30°C Max. Full Cure 30°C for Immersion	24 Hrs. 5 days	12 Hrs. NA	12 Hrs. NA	24 Hrs. NA
10.	Over-coat Interval Min.	10 hrs.	12 hrs. at 20°C & 50% RH.	24 hrs.	24 hrs.
11.	Pot Life at 30°C	90 Min.	4-6 Hrs.	NA	NA
12.	Temperature. Resistance (min) Dry service (ASTM D2485 method B)	80°C	400°C.	250°C	540°C .

FINISH PAINTS... CONT'D

Sr. No.	Description	FP-9	FP-10	FP-11	FP-12
1.	Technical Name	Coal Tar Epoxy Polyamine cured	Epoxy Phenolic coating Two-component cured with Polyamine adduct hardener system. (primer + intermediate coat + finish paint)	Poly Siloxane Coating - ambient temperature curing / High build inorganic copolymer based Aluminium coating, cold applied suitable for under insulation coating of CS and SS.	High Build Epoxy phenolic based -Two components solvent free type / Novalac Epoxy Phenolic coating
2.	Pack Type	Single pack	Two pack	Two pack	Single pack
3	Composition	Specially formulated polyamine cured coal tar epoxy suitable for application under insulation	Temperature curing epoxy phenolic coating system suitable for application under insulation of CS/AS/SS piping	Amercoat 738 from PPG Protective & Marine Coatings or Intertherm 751 CSA of International (Akzo Nobel). Note: 6	High build epoxy phenolic / Novalac Epoxy phenolic coating cured with Polyamine adduct hardener system
4.	Vol. Solids %	70 ± 3	70 ± 3	60 ± 2	98 -100
5.	DFT (Micron) / Coat	100-125	75-100	75-100	125-150
6.	Covering M ² / Coat / Litre	5-8	4-5	7-9	6.5-8
7.	Wt. Kg. / Litre	1.45 ± 0.03	1.65 ± 0.03	1.3	1.7
8.	Touch Dry at 30°C Min.	4 Hrs.	3 Hrs.	1 Hrs.	2 Hrs.
9.	Hard Dry at 30°C Max.Full Cure 30°C for Immersion	24 Hrs. 168 Hrs.(7days)	24 Hrs. 168 Hrs.(7days)	16 Hrs. -	24 Hrs. 168 Hrs.7days)
10.	Over-coat Interval Min,	6 Hrs. Max.: 5 days	16 Hrs. Max.: 21 days	16 Hrs.. Max.: NA	16 Hrs. Max.: 21 days
11.	Pot Life at 30°C	4 Hrs.	4-6 Hrs.	1 Hr.	1 Hr.
12.	Temperature. Resistance (min) Dry service (ASTM D2485 method B)	-45°C to 125°C under insulation & immersion	-45°C to 150°C under insulation & immersion. (Note: 5)	- Up to 400 °C for CS & SS for Intertherm 751 CSA - Up to 480 °C for CS and up to 600 °C for SS for Amercoat 738 (Note 6)	-45°C to 150°C for immersion service

Notes:

1. Covering capacity and DFT depends on method of application. Covering capacity specified above is theoretical. Allowing the losses during application, min. specified DFT shall be maintained.
2. All primers and finish coats shall be cured at ambient temperature and air drying unless otherwise specified.
3. All paints shall be applied in accordance with manufacturer's instructions for surface preparation, intervals, curing and application. The surface preparation, quality and workmanship shall be ensured. In case of conflict between this specification and manufacturer's recommendation, the same shall be clarified with Engineer in charge.
4. Technical data sheets for all paints shall be supplied at the time of submission of quotations.
5. Theoretical covering capacity, hard dry, full cure period & over coating interval shall be as per paint manufacturer's datasheets and no testing is required. Slight variation in the value of these parameters may be permissible with the discretion of engineer in-charge
6. FP-10 Two-component Epoxy phenolic coating cured with Polyamine adduct hardener system (primer + intermediate coat + finish paint) suitable up to 225°C (Intertherm 228 from M/s Akzo Nobel Coatings India Pvt Ltd. Bangalore). For all other companies, the temperature resistance shall be a maximum of 150°C.
7. FP-11 Ambient temperature curing epoxy poly siloxane Coating or high build cold applied inorganic co-polymer based aluminium coating. Amercoat 738 from PPG Protective & Marine coatings, Mumbai is suitable up to 480°C for CS surfaces and 600°C for SS surfaces. Intertherm 751 from Akzo Nobel Coatings India Pvt Ltd., Bangalore, Inorganic co- polymer cold applied Aluminium spray coating is suitable up to 400°C of CS & SS surfaces.

16. Painting System Tables and Selection Criteria:

16.1 Painting System Tables:

There are 11 painting system tables in this specification covering most of painting and coating works. However, new table may be added based on project requirement for any specific painting works as necessary. The tables are as under:

Table-01	Painting systems for uninsulated piping, equipment in process units, power plant, DM plant, cooling tower, chimney / stack, and any other equipment in process units also including offsites
Table-02	Painting system for insulated equipment and piping (under insulation) in process units and off sites (Carbon steel, LTCS, SS & low alloy steel)
Table-03	Painting system for uninsulated storage tanks in process units and off sites (Carbon steel & low alloy steel)
Table-04	Painting system for internal surface of storage tanks in process units and off sites (Carbon steel & low alloy steel)
Table-05	Painting system for external surface of underground piping and vessels in units and off sites (Carbon steel)
Table-06	Painting system for internal protection of components of coolers / condensers in fresh water service in units and off sites (Carbon steel)
Table-07	Painting system for internal protection of components of coolers / condensers in fresh water service in units and off sites (Stainless steel, duplex stainless steel, non-ferrous materials & galvanized steel)
Table-08	Painting system for effluent treatment plants (ETP)
Table-09	Coating systems for gratings, rolling & stationery ladders, spiral stairways and hand rails in all location

Table-10 Repair of pre-erection / pre-fabrication or shop primer after erection / welding of uninsulated piping and equipment in all environments. (CS, LTCS & low allow steel)

Table-11 Painting System for Structures in process units also including offsite

16.2 Table Selection Criteria:

The painting Table shall be selected based on following broad parameters given below. The selection criteria shall also be in the order below. All necessary precaution shall be taken in selecting the applicable table. In case of any difficulty Contractor may seek clarification before starting the works from Engineer in charge whose decision shall be final and binding on the Contractor.

Sr. No.	Criteria	Description	Details
1	Plant Location	Non Coastal / Inland	More than 50 KM from Sea shore
2	Environment	Industrial	For Non Coastal / Inland
3	Type of facility	- Units - Offsite	Process Units, Power Plant, Cooling Towers, DM Plant, pipe Rack in units, Package units, chimney/ stack, any other equipment in units Offsite- pipe racks, Piping on Sleepers
4	Temperature Ranges	(-) 180°C to 600°C	Temperature varies for case to case. Selection of painting systems according to the operating temperatures of the line.
5	Material of Const. (MOC)	Carbon Steel (CS) Low Temp Alloy Steel (LTCS), Stainless Steel (SS)	For aluminium, copper , Monel, Incoloy, Nickel, No painting is required
6	Insulation	Equipment / Piping Equipment / Piping	See Under insulation table
7	Aboveground Underground	Equipment / Piping Equipment / Piping	Equipment /piping in pit consider underground
8	Surface	- External - Internal	Equipment / piping Equipment only

NOTES (for all tables)

- The list of items specified in tables is not exhaustive. More items may be included for a particular Contract as necessary. The Contractor shall complete painting including prefabrication primer for all the items in his scope of work as per tender documents and instructions of Engineer in charge.

2. If the pre-erection / prefabrication and shop primer has already been completed, the same shall not be repeated again in the field. In case the damages of primer are severe and spread over large areas, the Engineer in charge may decide and advise re-blasting and priming again. Repair of pre- fabrication / pre-erection primer, as instructed, shall be carried out by Contractor.
3. All coating system including surface preparation, primer and finish coat for piping shall be done at site / field only.
4. Finish coating is not permitted at equipment manufacture shop.

TABLE -01

PAINTING SYSTEMS FOR UNINSULATED PIPING, EQUIPMENT IN PROCESS UNITS, POWER PLANT, DM PLANT, COOLING TOWER, CHIMNEY / STACK, AND ANY OTHER EQUIPMENT IN PROCESS UNITS ALSO INCLUDING OFFSITES (Carbon Steel, LTCS & Low Alloy Steel)

Sl. No.	Temp.in °C	Surface Preparation & Pre erection / Shop Primer	Painting System (Post-erection / Field)		Total Final DFT in Micr. (min.)
			Primer	Finish Coat	
1.1	-90 to -15	SSPC-SP-10; Inorganic Zinc Silicate (FP-6), 1 coat 65 to 75 micr	None	None	65 - 75
1.2	-14 to 100	SSPC-SP-10 Inorganic Zinc Silicate (FP-6), 1 coat, 65 - 75 µm Epoxy High build (FP-3A), 1 coat, 170 µm Acrylic Polyurethane (FP-1), 1 coat, 40 µm Total DFT at shop = 285 µm	Acrylic Polyurethane (FP-1) - 40 µm will apply after pressure water wash & surface rubbing Cumulative DFT = 325		325
1.3	101 to 400	SSPC-SP-10; Inorganic Zinc Silicate (FP-6), 1 coat, 65 to 75 µm	None	Silicone Aluminium Paint (FP-8), 2 coats 2x20 =40 µm	105 - 115
1.4	401 to 550	SSPC-SP-10; 1 coat of Silicone Aluminium Paint (FP-8), 1 coat, 20 µm	None	Silicone Aluminium Paint (FP-8), 2 coats 2x20 =40 µm	60

- No over coating to be done on FP-6 as it will lead to mud cracking.
- Silicone Aluminium Paint (FP-8) shall be ambient temperature curing type.
- Finish coat including primer compatible with finish coat. (i.e. field primer) shall be applied at site only
- For external surface of MS chimney with or without refractory lining and for internal surface without refractory lining, paint system at Sl. No.1.3 of the above table shall be followed.
- For external surface of RCC Chimney 2 coats of FP-3A @ 100 Micr. DFT/ coat to obtain total DFT of 200 Micr. shall be applied after proper surface preparation as per Clause 9.3.7
- In case of paint systems as per Sl. Nos. 1.3 and 1.4, the colour bands shall be applied over the Aluminium paint as per the Colour coding system requirement for specific service of piping.
- For Sl. No. 1.3 and 1.4 finish coat at field may be applied at shop itself and touch-up will be done at field.

TABLE -02

PAINTING SYSTEM FOR INSULATED EQUIPMENT AND PIPING (UNDER INSULATION) IN PROCESS UNITS AND OFFSITES (Carbon Steel, LTCS, SS & Low Alloy Steel)

Sl. No.	Temp.in °C	Surface Preparation & Pre erection / Shop Primer	Painting System		Total Final DFT in Micr. (min.)
			(Post-erection / Field)		
			Primer / Intermediate	Finish Coat	
2.1	Equipment & Piping - Carbon steel, LTCS and low Alloy steel				
2.1.1	- 45 to 200	SSPC-SP-10 Epoxy Phenolic (FP-10), 1 coat, 125 micr.	None	Epoxy Phenolic (FP-10), 1 coat, 125 micr.	250
2.1.2	201 to 540	SSPC-SP-10; Titanium catalysed inorganic ceramic copolymer, 1 coat, 150 micr.	None	Titanium catalysed inorganic ceramic copolymer, 1 coat, 150 micr.	300
2.2	Piping, Vessels & Equipment - Stainless Steel including Alloy-20 (Note:2)				
2.2.1	- 45 to 200	For SS, SSPC-SP-6 Commercial Blast/ For SS SSPC-SP-1 Epoxy Phenolic (FP-10), 1 coat, 125 micr.	None	Epoxy Phenolic (FP-10), 1 coat, 125 micr.	250
2.2.2	201 to 600	For SS, SSPC-SP-6 Commercial Blast/ For SS SSPC-SP-1 With non-chloride solvent 1 coat of Titanium catalysed inorganic ceramic copolymer, 1 coat, 150 micr	None	1 coat of Titanium catalysed inorganic ceramic copolymer, 1 coat, 150 micr	300
2.3	No painting is required for insulated Monel, Incoloy and Nickel lines.				

- **"Cyclic Service"** is characterized by rapid temperature fluctuation.
- The blast cleaning abrasives for SS and Alloy steel surfaces shall be Aluminium oxide grits/shots or garnet.
- Surface shall be thoroughly degreased using an appropriate emulsion cleaner and abrasive cleaned (sweep blasting) to create a sufficient anchor profile. Abrasive for blast cleaning of stainless steel surfaces shall be performed with a suitable non-metallic abrasive such as aluminium oxide. When hand or power tool cleaning is required on stainless steel, only stainless steel wire brushes that have not been previously used on carbon steel surface must be used. All coatings and solvents for use on stainless steel shall be free of substances

such as chlorides and other halides, sulphur, and shall be free of low melting point metals (zinc, aluminium, tin and lead).

- For 2.1.1, 2.1.2 & 2.2.1 finish coat at field may be applied at shop itself and touch-up will be done at field.

TABLE -03

**PAINTING SYSTEM FOR UNINSULATED STORAGE TANKS IN PROCESS UNITS AND OFFSITES
(Carbon Steel & Low Alloy Steel)**

Sl. No.	Temp.in °C	Surface Preparation (Field)	Painting System (In field after welding & erection)		Total DFT in Microns (min.)
			Primer	Finish Coat	
3.1	All external surfaces of shell, wind girders, appurtenances, roof tops of all above ground tank including top side of external and internal floating roof and associated external structural works.				
3.1.1	-14 to 100	SSPC-SP-10	Inorganic Zinc Silicate (FP-6), 1 coat, 65-75 micr. + Epoxy Zinc Phosphate (PR-3), 1 coat, 40 micr.	High build epoxy (FP3A), 2 coats, 100 micr/coat. + Acrylic Polyurethane (FP-1). 1 coat, 70 micr.	345 - 355
3.1.2	101 to 150	SSPC-SP-10	Epoxy Phenolic (FP-10), 1 coat, 80 micr. + Epoxy Phenolic (FP-10), 1 intermediate coat, 80 micr.	Epoxy Phenolic (FP-10), 1 coat, 80 micr. + Acrylic Polyurethane (FP-1), 1 coat, 40 micr.	280
3.1.3	151 to 500	SSPC-SP-10	Inorganic Zinc Silicate (FP-6), 1 coat, 65-75 micr.	Silicone Aluminium Paint (FP-8), 2 coats @20 micr./coat (or) Polysiloxane (FP-11), 1 coat, 50 micr.	105 - 115
3.2	External surfaces of bottom plate (soil side) for all storage tanks.				
3.2.1	-14 to 80	SSPC-SP-10	Inorganic Zinc Silicate (FP-6), 1 coat, 65-75 micr.	High glass flake epoxy, 2 coats, 200 micr DFT/coat (2x200=400)	475
3.2.2	81 to 150	SSPC-SP-10	Epoxy Phenolic (FP-10), 1 coat, 80 micr. + Epoxy Phenolic (FP-10), 1 intermediate coat, 80 micr. (80+80=160)	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	240
3.2.3	151 to 550	SSPC-SP-10	Polysiloxane (FP-11), 1 coat, 125 micr.	Polysiloxane (FP-11), 1 coat, 125 micr.	250

3.3	For underside of the bottom plate (in case tank is not lifted during PWHT) (see Note 2c)				
3.3.1	-180 to 650	For CS, SSPC-SP-6 Commercial Blast	Inert polymeric matrix coating, 1 coat, 125 microns.	Inert polymeric matrix coating, 2 coats, 125 micr DFT/coat.	350-400
		For SS, SSPC-SP-1 With non-chloride solvent			

- For 3.1.1, High Build Epoxy (FP3A) should be suitable for occasional water immersion
 - For 3.3.1, Products from Jotun or High-temp coating or SK formulation recommended.
 - All paint coating application including primer for tankage shall be carried out at field after erection and completion of all welding.
 - For underside of bottom plate, painting shall be carried out before laying of bottom plate for tanks with Non-Post Weld Heat Treatment (PWHT).
 - For tanks with PWHT, painting shall be carried out after PWHT.
 - In case tank is not lifted during PWHT then painting shall be applied before laying of bottom plate, SI no. 3.3.1 shall be followed.
- Caution:** PWHT temperature shall not exceed 650°C.
- Painting for safety requirements:
 - The following uninsulated items shall be painted for camouflaging, as per site safety requirement.
 - All columns
 - All tanks in offsites
 - All Tank-tops diameter 3 m and more On-site
 - Large vessels
 - Two coats of selected finish paint as per requirement shall be applied in a particular pattern as per (c) and as per the instructions of NRL.
 - Method of camouflaging: Disruptive painting for camouflaging shall be done in three colours in the ratio of 5:3:2 (all matt finish). 5 – Dark Green, 3 – Light Green, 2 – Dark Medium Brown.
 - The patches should be asymmetrical and irregular.
 - The patches should be inclined at 30° and 60° to the horizontal.
 - The patches should be continuous where two surfaces meet at an angle and the patches should be coincide with corners.
 - Slits and holes shall be painted in dark shades.
 - Width of patches should be 1 to 2 meters.

TABLE -04

**PAINTING SYSTEM FOR INTERNAL SURFACE OF STORAGE TANKS IN PROCESS UNITS AND OFFSITES
(Carbon Steel & Low Alloy Steel)**

Sl. No.	Temp.in °C	Surface Preparation (Field)	Painting System (In field after welding & erection)		Total DFT in Microns (min.)
			Primer	Finish Coat	
4.1	Crude oil, ATF, Turpentine oil, Lubricating oil and Vegetable oil				
	Underside of floating roof, internal surface of cone roof, inside of bottom plate, Internal surfaces of Shell - including wetted and free board height, oil side surfaces of deck plates, oil side surfaces of pontoons, roof structures, structural steel, ladders and other carbon steel internals.				
4.1.1	-14 to 90	SSPC-SP-10	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	Epoxy Phenolic (FP-10), 1 intermediate coat, 80micr. + Epoxy Phenolic (FP-10), 1 finish coat, 80 micr.	240 - 300

4.2	Petroleum products & Intermediates Like LDO, HSD, Gas oil, Feeds of FCC -PC, FCC-LCO, VGU-HDT, ISOM, DHDT, Reformate, DCU, NHT & Gasoline, Naphtha, Isomerate and Kerosene. Underside of Floating roofs, internal surface of cone roof, inside of bottom plate, internal surfaces of Bare shell for full height, underside of floating roof, oil side surfaces of deck plates, oil side surfaces of pontoons, support structures and ladders etc.				
4.2.1	-14 to 45	SSPC-SP-10	Inorganic Zinc Silicate (FP-6), 1 coat, 75 micr.	-	75
4.2.2	46 to 90	SSPC-SP-10	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	Epoxy Phenolic (FP-10), 1 intermediate coat, 80micr. + Epoxy Phenolic (FP-10), 1 finish coat, 80 micr.	240-300
4.3	Raw / Fresh water, Potable water and Fire water All internal surfaces, accessories and roof structures of cone and dome roof tanks				
4.3.1	-14 to 60	SSPC-SP-10	Epoxy Zinc Phosphate (PR-3), 1 coat, 100 micr.	High build epoxy (FP3A), 2 coats, 100 micr./coat. (2x100=200)	300-350
4.4	De-mineralized water (DM) All internal surfaces, accessories and roof structures of cone and dome roof tanks				
4.4.1	-14 to 60	SSPC-SP-10	Epoxy Zinc Phosphate (PR-3), 1 coat, 100 micr.	Solvent less epoxy (FP-3C), 2 coats, 200 micr./coat. (2x200=400)	500
4.4.2	61 to 150	SSPC-SP-10	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	Epoxy Phenolic (FP-10), 1 intermediate coat, 80micr. + Epoxy Phenolic (FP-10), 1 finish coat, 80 micr.	240 - 300
4.5	Hydrochloric Acid (HCl) 10% All internal surfaces, accessories and roof structures of cone and dome roof tanks				
4.5.1	-14 to 60	SSPC-SP-10	None	Natural Rubber Lining	4.5 mm
4.6	Aggressive Solvents like Hexane, Hexene, Benzene, Xylene and Toluene All internal surfaces, accessories and roof structures of cone and dome roof tanks.				
4.6.1	-14 to 65	SSPC-SP-10	Inorganic Zinc Silicate (FP-6), 1 coat, 75 micr.	-	75
4.7	Ethylene Glycol (EG) Tanks Internal shell-full height, bottom plate, underside of roof and all accessories				
4.7.1	ALL	SSPC-SP-10	-	Vinyl chloride Co-polymer Amercoat 23, 3 coats, 75micr./coat (3x75=225)	225

4.8	Inside pontoon and inside of double deck of all tanks floating roofs				
4.8.1	-14 to 80	SSPC-SP-3	High build epoxy (FP-5), 1 coat, 100micr.	High build epoxy (FP-5), 1 coat, 100micr.	200
4.9	Wet Slops, Amine Solutions, Sour water, Water draw off All internal surfaces, accessories and roof structures of Cone and Dome roof tanks.				
4.9.1	-14 to 90	SSPC-SP-10	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	Epoxy Phenolic (FP-10), 1 intermediate coat, 80micr. + Epoxy Phenolic (FP-10), 1 finish coat, 80 micr.	240
4.10	Vacuum Residue, Fuel oil, Dry Slop, Bitumen and other High Temperature Hydrocarbon Liquids. Underside of floating roof, internal surface of cone roof, bottom plate, inside of bare shell - including wetted and non-wetted surfaces, oil side surfaces of deck plates, oil side surfaces of pontoons, roof structures, structural steel and ladders.				
4.10.1	Up to 150	SSPC-SP-10	High build epoxy phenolic/Novalac Epoxy (FP-12), 1 coat, 125micr.	High build epoxy phenolic/Novalac Epoxy (FP-12), 1 intermediate coat, 125micr. + High build epoxy phenolic/Novalac Epoxy (FP-12), 1 intermediate coat, 125micr (125+125=250)	375
4.11	Alkalis up to 50 % Concentration All internal surfaces accessories and roof structures of cone and dome roof tanks				
4.11.1	Up to 60	SSPC-SP-10	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	High build epoxy (FP3A), 2 coats, 100 micr./coat. (2x100=200)	280

Notes:

- For 4.2.1, Inorganic Zinc Silicate (FP-6) shall be suitable and resistant for immersion service for the respective Hydrocarbons.
- For 4.3.1, High build epoxy (FP-3A) shall be suitable for drinking water service and should have competent authority certification.
- For 4.4.1, Single coat of Solvent less epoxy (FP-3C) is also acceptable.
- For 4.10.1, This system can be used where maximum operating temperature is below 150°C and design temperature is up to 200°C. For operating temperature >150°C, corresponding paint product manufacturer datasheet of High build epoxy phenolic/Novalac epoxy or Solvent free glass flake epoxy/ceramic epoxy to be followed further to approval from owner.

TABLE -05

PAINTING SYSTEM FOR EXTERNAL SURFACE OF UNDERGROUND PIPING AND VESSELS IN UNITS AND OFFSITES (Carbon Steel)

Sl. No.	Temp.in °C	Surface Preparation & Shop Primer	Coating System (Field)		Total Final DFT in Microns (min.)
			Surface Preparation & Primer	Finish Coat	
5.1	External surface of non-insulated underground piping				
5.1.1	25 to 110	-	Upto 80 deg C, 3LPE coating. Upto 110 deg C, 3LPP coating, thickness as per JSS, NR-0ZZZZ-CV-SPE-0027	Upto 80 deg C, 3LPE coating, Upto 110 deg C, 3LPP coating, thickness as per JSS, NR-0ZZZZ-CV-SPE-0027	
5.1.2	111 to 150	-	SSPC-SP-10 High build epoxy phenolic/Novalac Epoxy (FP-12), 1 coat, 400micr.	High build epoxy phenolic/Novalac Epoxy (FP-12), 1 coat, 400micr.	800
5.1.3	151 to 400	-	SSPC-SP-10; Polysiloxane (FP-11), 1 coat, 125micr.	SSPC-SP-10; Polysiloxane (FP-11), 1 coat, 125micr.	250
5.2	External surface of non-insulated underground storage vessels				
5.2.1	-14 to 80	SSPC-SP-10; Inorganic Zinc Silicate (FP-6), 1 coat, 65-75 micr	-	High Glass Flake Epoxy, 2 coats, 200 micr. DFT/coat. (2x200=400)	400
5.2.2	81 to 150	SSPC-SP-10	High build epoxy phenolic/Novalac Epoxy (FP-12), 1 coat, 400micr.	High build epoxy phenolic/Novalac Epoxy (FP-12), 1 coat, 400micr.	800
5.2.3	151 to 400	SSPC-SP-10; Polysiloxane (FP-11), 1 coat, 125micr.	-	Polysiloxane (FP-11), 1 finish coat, 125micr.	250

Note:

- For 5.1.3, 5.2.3, the system is suitable up to 550 °C.

TABLE -06

PAINTING SYSTEM FOR INTERNAL PROTECTION OF COMPONENTS OF COOLERS / CONDENSERS IN FRESH WATER SERVICE IN UNITS AND OFFSITES (Carbon Steel)

Sl. No.	Temp.in °C	Surface Preparation & Shop Primer	Coating System (Field)		Total Final DFT in Microns (min.)
			Surface Preparation & Primer	Finish Coat	
6.1	Fresh water boxes, channels, partition plates, end covers and tube sheets etc.				
6.1.1.	Up to 80	SSPC-SP-10;	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	Epoxy Phenolic (FP-10), 2 coats, 80 micr./coat	240
6.1.2.	80 to 140	SSPC-SP-10;	-	Glass Fibre Reinforced Novolac epoxy , 1 coat, 1500 micr.	1500

TABLE -07

PAINTING SYSTEM FOR INTERNAL PROTECTION OF COMPONENTS OF COOLERS / CONDENSERS IN FRESH WATER SERVICE IN UNITS AND OFFSITES (Stainless Steel, Duplex Stainless Steel, Non-ferrous materials & Galvanized Steel)

Sl. No.	Temp.in °C	Surface Preparation & Shop Primer	Coating System (Field)		Total Final DFT in Microns (min.)
			Surface Preparation & Primer	Finish Coat	
7.1	Up to 80	Sweep Blasting	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	Epoxy Phenolic (FP-10), 1 coat, 80 micr.	160
7.2.	80 to 140	Sweep Blasting	-	Glass Fibre Reinforced Novolac epoxy , 1 coat, 1500 micr.	1500

TABLE -08

PAINTING SYSTEM FOR EFFLUENT TREATMENT PLANTS (ETP)

Sl. No.	Temp.in °C	Surface Preparation	Coating System		Total
			Primer	Finish Coat	DFT in Microns (min.)
8.1	External Surface of C.S./ M.S. items screens, walk way, bridges, baffles, dual media filters, Vertical pumps, piping in treated effluent sump, bio sludge pump, Screw pump and pump house, CS tank, sump and vessels				
8.1.1	-14 to 80	SSPC-SP-10	Inorganic Zinc Silicate (FP-6), 1 coat, 65-75 micr.	High Build Epoxy (FP-3A), 2 coats, 100 micr. + Acrylic Polyethylene (FP-1), 1 coat, 40 micr. (2x100+40=240)	305 - 315

8.2	Internal surfaces of CS/MS Items: Bio-sludge sump, Filter feed sump, Process sump, Sanitary sump, Transfer sump, Sludge, Slop oil tank, scrapping mechanism in Clarifier				
8.2.1	-14 to 80	SSPC-SP-10	Inorganic Zinc Silicate (FP-6), 1 coat, 65-75 micr.	High Build Epoxy (FP-3A), 3 coats, 100 micr/coat. (3x100=300)	365 - 375
8.3	R.C.C./concrete surfaces exposed to effluent water / liquid such as tanks, structures, drains etc. in process sump, TPI separator (Process and oil), Aeration tank and Transfer sump etc.				
8.3.1	-14 to 80	Blast cleaning to SSPC-SP guide lines and Acid etching with 10-15% HCl acid followed by thorough water washing.	Epoxy Screed lining		3000
8.4	C.S/ M.S Dual media filters (Internal), Chemical dosing tanks(internal) such as Di Ammonium Phosphate (DAP) and Urea				
8.4.1	Up to 60	SSPC-SP-10	Natural Rubber Lining (As per IS 4682, Part I)		4500

- The paint /coating manufacturers shall provide their Quality control test certificate of coating materials High build epoxy (FP-3A) for immersion service of the exposed effluent.

TABLE -09

COATING SYSTEMS FOR GRATINGS, ROLLING & STATIONERY LADDERS, SPIRAL STAIRWAYS, HAND RAILS AND GI ITEMS IN ALL LOCATION

Sl. No.	Temp.in °C	Coating System	Total DFT in Microns (min,)
9.1	Up to 60	Hot Dip Galvanizing to 80 - 85 microns (600 - 610 gm./m ²) as per IS 4759, 2629, 4736, 2633 +1 coat of Epoxy Zinc Phosphate (PR-3) @ 40 microns DFT/coat + 1 coat of Polyurethane (FP-1) @ 40 microns DFT/coat	80 microns of finish coat (excluding the thickness of galvanizing)

NOTES:

- No galvanized specimen shall have thickness less than 80 microns.
- Repair of the damaged area of galvanized coatings due to welding during erection shall be carried out as per recommended practice IS 11759 using cold galvanizing spray process. Organic Paint systems are not acceptable for repair.
- After repair of damaged galvanized coating by Cold Galvanized, the repaired area shall be top coated with paint system as given in Table 10.0 above (i.e., 1 coat of PR-3 @ 40 microns. DFT/coat + 1 coat of FP-1 @ 40 microns. DFT/coat).

4. Approved Cold Galvanizing manufacturers are **ZINGA, LOCKTITE** or **Z.R.C.**
5. In general Galvanized gratings do not require painting until otherwise specified elsewhere or as per the requirements from Client.

TABLE -10

**REPAIR OF PRE-ERECTION / PRE-FABRICATION OR SHOP PRIMER AFTER ERECTION / WELDING OF UNINSULATED PIPING AND EQUIPMENT IN ALL ENVIRONMENTS.
(CS, LTCS & low allow steel)**

Sl. No.	Temp.in °C	Surface Preparation	Coating System	Total DFT in Micr (min.)	Remarks
1	-90 to 400	SSPC-SP-3	Inorganic Zinc Silicate (FP-6), 1 coat	65-75	See note below and clause 5.9.3
2	401 to 550	SSPC-SP-3	High build epoxy phenolic based	20	

- The repair of pre-erection / pre-fabrication or Shop Primer given above shall be done for all items requiring repairs. In case the damages of primer are severe and spread over large area, entire primer shall be removed by blasting to achieve SSPC-SP-10 and surfaces to be primed again with FP-6 or FP-8 as applicable.
- The primer shall be quickly removed from damaged area by mechanical scraping and emery paper conforming to SSPC-SP-3 to expose the white metal. Blast cleans the surface, if possible. Feather the primed surface over the intact adjacent surface surrounding the damaged area by emery paper.

TABLE -11

Painting System for Structures in process units also including offsite

Sl. No.	Temp.in °C	Surface Preparation & Pre erection / Shop Primer	Painting System (Post-erection / Field)	Total Final DFT in Micr. (min.)
1	< 100	SSPC-SP-10 Inorganic Zinc Silicate (FP-6), 1 coat, 65 - 75 µm Epoxy High build (FP-3A), 1 coat, 170 µm Acrylic Polyurethane (FP-1), 1 coat, 40 µm Total DFT at shop = 285 µm	None	285

- For the structures with fireproofing, considering the risk of corrosion under fireproofing (CUF), as per NACE SP0198, suitable protective coating system should be applied prior to the fireproofing application and a necessary approval needs to be obtained. Applied coating should be compatible with the Fireproofing material.

17. Storage:

All paints and painting materials shall be stored only in rooms to be arranged by contractor and approved by Engineer in charge for the purpose. All necessary precautions shall be taken to prevent fire. The storage building shall preferably be separate from adjacent building. A signboard bearing the word "**Paint Storage –No Naked Light - Highly Inflammable**" shall be clearly displayed outside. Manufacturer's recommendation shall be followed for storage of paint materials.

18. Approved Vendor List:

Approved list of paint manufacturer to be referred from master vendor list

19. Quality Control, Inspection & Testing:

- All painting materials including primers and thinners brought to site by contractor for application shall be procured directly from manufactures as per specifications and shall be accompanied by manufacturer's test certificates. Paint formulations without certificates are not acceptable
- The contractor must produce Test Certificate from Pre-Qualified Paint Manufacturer for various tests as specified in this document, for each batch and for each category of product. The Engineer in charge shall have the right to test wet samples of paint from each batch at random for verifying quality of paint supplied. Contractor shall arrange to have such tests, when called for by Engineer in charge, performed at his cost any one of the independent laboratories listed in this document.

Samples for the test will be drawn at random in presence of Engineer in charge or his representations. Following tests to be carried out if called for by Engineer in charge:

- Specific Gravity
- % solids by weight (% zinc content in case of inorganic or organic zinc primer)
- Drying time (touch dry & full curing)
- Adhesion
- Flexibility
- Hardness
- Storage stability (pot life)

Test methods for above tests shall be as per relevant ASTM or ISO Standard.

- The painting work shall be subject to inspection by Engineer in charge at all times. In particular, following stage-wise inspection will be performed and contractor shall offer the work for inspection and approval of every stage before proceeding with the next stage. The record of inspection shall be maintained in the registers. Stages of inspection are as follows:
 - Surface preparation
 - Primer application
 - Each coat of paint

During surface preparation, following tests are to be carried out:

- Test for absence oil and grease after degreasing before blasting.
- Tests for surface finish of blasted surface shall be done by visual inspection using SSPC-VISI. Clear cellophane tape test as per ISO 8502-9 shall be used to confirm absence of dust on blasted surface. Checks shall be done on each component at least once per 200 m2 of blasted surface and minimum of 3 checks per shift.

- Test for presence of soluble salt as per method ISO 8502-9. Maximum allowable salt content shall be considered 20 mg /m². Checks shall be done on each component at least once per 200 m² of blasted surface and minimum of 3 checks per shift. In case salt exceeds specified limit, the contaminated surface shall be cleaned by method as per Annexure-C of ISO: 12944-4 (water cleaning). After cleaning surface shall be retested for salt after drying.
- Blast profile measurement
- Test for blasting Media and Blasting air- In addition to above, record should include type of shop primer already applied on equipment e.g. zinc silicate, or zinc rich epoxy, or zinc phosphate.

Any defect noticed during the various stages of inspection shall be rectified by the contractor to the entire satisfaction of Engineer in charge before proceeding further. Irrespective of the inspection, repair and approval at intermediate stages of work, contractor shall be responsible for making good any defects found during final inspection/guarantee period/defect liability period as defined in general condition of contract. Dry film thickness (DFT) shall be checked and recorded after application of each coat and extra coat of paint should be applied to make- up the DFT specified without any extra cost to owner, the extra coat should have prior approval of Engineer in charge.

- **Primer Application:** After surface preparation, the primer should be applied to cover the crevices, comers, sharp edges etc. in the presence of inspector nominated by Engineer in charge.
- The shades of successive coats should be slightly different in colour in order to ensure application of individual coats, the thickness of each coat and complete coverage should be checked as per provision of this specification. This should be approved by Engineer in charge before application of successive coats.
- The contractor shall provide standard thickness measurement instrument with appropriate range(s) for measuring.
 - Dry film thickness of each coat,
 - Surface profile gauge for checking of surface profile in case of sand blasting.
 - Holiday detectors and pinhole detector and protector whenever required for checking in case of immersion conditions.
- Prior to application of paints on surfaces of chimneys, the thickness of the individual coat shall be checked by application of each coat of same paint on M.S.test panel. The thickness of Paint on test panels shall be determined by using gauge such as 'Elkometer'. The thickness of each coat shall be checked as per provision of this specification. This shall be approved by Engineer in charge before application of paints on surface of chimney.
- At the discretion of Engineer in charge, the paint manufacturer must provide the expert technical service at site as and when required. This service should be free of cost and without any obligation to the owner, as it would be in the interest of the manufacturer to ensure that both surface preparation and application are carried out as per their recommendations. The contractor is responsible to arrange the same.
- Final inspection of finished coating shall consist of measurement of:
 - Paint dry film thickness (DFT),
 - Adhesion, and,
 - Holiday detection check as well as for finish and workmanship.
- Coating DFT measurement shall be as per ISO: 2808. Type II electromagnetic gauges should be used for ferrous substrates. DFT gauge calibration, number of measurement shall be as per SSPC-PA 2. Measured DFT shall be within + 10% of the dry film thickness, specified in the specifications.

- Adhesion of the primer to the steel substrate and intercoat adhesion of the subsequent coat(s) after curing for at least a week after application of the topcoat shall be examined by a knife-test in accordance with ASTM D6677. For the knife test, if the rating is better than 8, the adhesion is considered acceptable. The adhesion is defective and:-tested areas shall be repaired afterward using the spot repair procedure. Alternatively, the applicator may perform the adhesion test on a steel coupon coated using the same surface preparation and coating application procedure as the work piece. Adhesion testing shall be carried out for each component at least once per 200 m² (2000 ft²) of coated surface.
- Holiday testing shall be conducted in accordance with NACE SP0188. For immersion services, 100% of coated area shall be inspected for holidays. For atmospheric exposure, 10% of coated area which must include weld seams, corners and edges to be holiday tested. Voltage at which test is to be carried out will depend upon DFT of coating being tested and shall be as per NACE SP0188. Any holiday is unacceptable and should be marked and repaired immediately.
- The contractor shall arrange for spot checking of paint materials for Specific gravity, glow time (ford cup) and spreading rate.

- **Final Inspection of coating system:**

A final inspection shall be conducted prior to the acceptance of the work. The Contractor and the Owner / Engineer in charge shall both be present and they shall sign an agreed inspection report. Such reports shall include:

- General
 - Names of the painting Contractor and the responsible personnel
 - Dates when work was performed
- Painting Materials
 - Information on painting materials being applied
 - Condition of painting materials received
- Environmental Conditions
 - Weather and ambient conditions
 - Painting periods
- Surface Preparation
 - Condition of surface before preparation
 - Tools and methods used to prepare surface
 - Condition of surface after preparation
- Painting Application
 - Equipment used
 - Mixing procedure prior to application
 - Coating application techniques used
- Testing
 - Type and calibration of inspection instruments used
 - Type of quality control tests performed, and results

20. Guarantee:

The Contractor shall guarantee that the chemical and physical properties of paint material used are in accordance with the specifications contained herein / to be provided during execution of work.

21. Qualification Criteria of Painting Contractor / Sub-contractor:

Painting contractor who is awarded the contract for painting by the Owner, must have necessary equipment's, machinery, tools and tackles for surface preparation, paint application and inspection. The contractor must have qualified, trained and experienced surface preparator, paint applicator, inspector and supervisors. The Contractor supervisor, inspector, surface preparator and paint applicator must be conversant with the standards referred in this specification.

22. Qualification /Acceptance Criteria for Paint Coating System:

- Pre-Qualification of Paint Coating Manufacturer and his Products

Paint / coating manufacture meeting the following requirements shall be considered for supply of their products. Contractor is advised to select coating manufacturer. Only after obtaining prequalification from Owner for the manufacturer based on following requirements. Even those manufacturers, whose names are appearing elsewhere in the tender document, under the list of "Owner Recommended or Approved Vendors", will also be required to meet the following prequalification requirements.

- Manufacturer should have been in continuous business of paint / coating formulation and manufacturer for at least past 5 years.
- Manufacturer should possess past experience of supplying his products to hydrocarbon processing industry or offshore platforms in the past 5 years.
- Coating manufacturer should have supplied at least 10000 litre of an individual product to hydrocarbon processing industry or offshore platform.
- The manufacturer's manufacturing procedure & QA/QC system shall meet ISO 9001 Requirements and preferably should possess ISO 14000 certificate.
- The Quality control set up should be manned by qualified paint technologists whose bio data should be sent along with quality control organization chart.

- Pre-Qualification Testing:

Manufacturer should have got his products tested at least one time in last 3 years at a reputed independent laboratory for the following test items. Test certificates which are more than 3 years old will not be considered.

Test	Test Method
Specific gravity	ASTM D1475
Dipping properties	ASTM D823
Film characteristics	-
Solid content by weight	ASTM D2369
Drying Time	ASTM D1640
Flexibility	ASTM D1737 / D522
Hardness	ASTM D3363
Adhesion	ASTM D2197
Abrasion resistance	ASTM D968/ D1044
DFT/coat	As per SSPC guidelines

Storage Stability	ASTM D1849
Resistance to moisture vapour permeability for 2000 hrs	ASTMD2247
Cyclic Test for the duration of 4200 h (25 cycles a 168 hours)	ISO 7253, ASTM G53
% Zn in Dry film for Inorganic Zinc Silicate primer	-
Chemical Resistance test - 10% & 40% NaOH (applicable only for F-6 & F-15) - 10% H ₂ SO ₄ (applicable only for F-6 & F-15) - 10% Nitric Acid test (applicable only for F-6 & F-15) - Benzene / Toluene (applicable only for F-6 & F-15) - Kerosene (applicable only for F-6 & F-15) - Sea water (applicable only for F-6 & F-15) - MIBK test (applicable only for F-6 & F-15)	ASTM D543
Resistance to water using water immersion (applicable only for F6-, F-7, F-8, F-14 & F-15)	ASTM D870
Dry Heat Resistance test (applicable only for F-9, F-6AIB, F-2, F-15, F-16, Polysiloxane, heat;:resistance Al silicone)	
Thermal shock resistance test (only for F-9, F-6, Polysiloxane)	ASTM D2485 - 91

- Each coating product to be qualified shall be identified by the following
 - An infrared scan (fingerprint), for Part A and B, each component as per ASTM D2621
 - Specific gravity of Base and curing agent (Ref. ISO 2811)
 - Ash content (ASTM 01650), volatile and non-volatile matters (ISO 3251) of each component
- The identification shall be carried out on the batch, which is used for the Pre-qualification testing. Pre-qualification of the products shall be carried out at an independent laboratory.
- Test shall be carried out at any one of the following laboratories and tests to be witnessed & certified by third party inspection agency (TUV, BY, DNV)
 - IICT, Hyderabad
 - HBTI, Kanpur
 - DMSRDE, Kanpur
 - BIS Laboratories
 - UICT, Matunga, Mumbai
 - UTES, Kolkata
 - PDIL,-Sindri
 - NTH, Kolkata
- Contractor shall furnish to Owner for approval / acceptance of all necessary documents / information including test certificates to prove that the paint manufacturers, from whom he intends to procure paint products, meet the various requirements for fulfilling the pre-qualification criteria as given above. The paint manufacturer shall be qualified and approved for supply after review / assessment of the submission made by the contractor.
- Contractor along with delivery of paint material has to furnish following information from paint manufacturer to Owner for acceptance / approval of products.

a) Batch test certificates (Batch Testing)

Contractor has to produce test certificate from paint manufacturer for each batch and for each category of product for the following test items. Test to be witnessed & certified by third party inspection agency. All test results must mention clearly the batch no. and category of product tested. Tests to be conducted for following properties:

- Infrared scan for Part A and B, each component
- Specific Gravity
- % solids by weight (% zinc content in case of inorganic or organic zinc primer)

b) Product information sheet Technical data sheet for each category of product.

23. Method of Sampling and Dispatch for Laboratory Testing:

(Pre-Qualification tests, Batch testing and Inspection testing)

Samples of coating materials should be submitted to the Govt. laboratory in sealed containers with batch no. and test certificate on regular format of manufacturer's testing laboratory. The sampling shall be certified and sealed by a certifying agency.

All test panels should be prepared by Govt. testing agency coloured photographs of test panels should be taken before and after the test and should be enclosed along with test report.

Sample batch no. and manufacturer's test certificate should be enclosed along with the report. Test report must contain details of observation and rusting if any, as per the testing code.

Manufacturers should intimate the company, details of sample submitted for testing, name of Govt. testing agency, date, and contact personnel of the govt. testing agency. At the end of the test the manufacturer should submit the test reports to the company for approval.

Coating systems for panel test shall be decided after discussion with Owner

ANNEXURE 1 – PAINTING SYSTEM

Substrate	Scope	Insulated/Uninsulated	Operating Temp	Painting System	Table No
CS, LTCS, AS	Piping, Equipment in process and offsites	Uninsulated	(-90 to -15)	CS 1	Table No - 01
			(-14 to 100)	CS 2	
			(101 to 400)	CS 3	
			(401 to 550)	CS 4	
CS, LTCS & LAS	Piping, Equipment in process and offsites	Insulated	(-45 to 200)	CS 5	Table No - 02
			(201 to 540)	CS 6	
CS & LAS	Storage tanks external surface	Uninsulated	(-14 to 100)	CS 7	Table No - 03
			(101 to 150)	CS 8	
			(151 to 500)	CS 9	
	Storage tanks external surface of bottom plate soil side	Uninsulated	(-14 to 80)	CS 10	
			(81 to 150)	CS 11	
			(151 to 550)	CS 12	
CS & LAS	Underside of bottom plate	Uninsulated	(-180 to 650)	CS 13	Table No - 04
	Crude Oil, ATF, Turpentine oil, Lubricating and Vegetable oil	Uninsulated	(-14 to 90)	CI 1	
	Petroleum products & intermediates (LDO, HSD, Gas oil, Feeds of FCC-PC, FCC-LCO, VGU-HDT, ISOM, DHDT, Reformate, DCU, NHT & Gasoline, Naphtha, Isomerate & kerosene)	Uninsulated	(-14 to 45)	CI 2	
	Raw / Fresh water, Potable water and Fire water	Uninsulated	(-14 to 60)	CI 4	
	De-mineralized water	Uninsulated	(-14 to 60)	CI 5	
	Hydrochloric Acid (HCl) 10%	Uninsulated	(-14 to 60)	CI 6	
	Aggressive Solvents like Hexane, Hexene, Benzene, Xylene and Toluene	Uninsulated	(-14 to 65)	CI 8	
	Ethylene Glycol (EG)	Uninsulated	All	CI 9	
	Inside pontoon and Inside of double deck of all tanks floating roofs	Uninsulated	(-14 to 80)	CI 10	
	Wet slops, Amine solutions, Sour water, Water draw off	Uninsulated	(-14 to 90)	CI 11	
	Vacuum residue, Fuel oil, Dry slop, Bitumen and other high temp Hydrocarbon liquids	Uninsulated	Up to 150	CI 12	
	Alkalis up to 50 % concentration	Uninsulated	Up to 60	CI 13	
CS	Underground piping	Uninsulated	(25 to 65)	CB 1	Table No - 05
			(66 to 150)	CB 2	
			(151 to 400)	CB 3	
	Underground storage vessels	Uninsulated	(-14 to 80)	CB 4	
			(81 to 150)	CB 5	
			(151 to 400)	CB 6	
CS	Internal protection of components of coolers / condensers in fresh water service in process units and offsites	Uninsulated	Up to 80	CC 1	Table No - 06
			80 to 140	CC 2	
CS/MS, RCC/Concrete surface	External Surface - CS / MS items screens, walk way, bridges, baffles, dual media filters, vertical pumps, piping in treated effluent sump, bio sludge pump, screw pump and pump house, CS tank, sump and vessels	Uninsulated	(-14 to 80)	CE 1	Table No - 08
	Internal Surface - CS / MS items, Bio-sludge sump, Filter feed sump, Process sump, Sanitary sump, Transfer sump, Sludge, Slop oil tank, scrapping mechanism in clarifier	Uninsulated	(-14 to 80)	CE 2	
	RCC / Concrete surface exposed to effluent water / liquid such as tanks, structures, drains etc., in process sump, TPI separator (Process and oil), aeration tank and Transfer sump etc.,	Uninsulated	(-14 to 80)	CE 3	
	CS / MS Dual media filters (internal), Chemical dosing tanks (internal) such as Di Ammonium Phosphate (DAP) and Urea	Uninsulated	Up to 60	CE 4	
CS, LTCS & LAS	Repair of pre-erection / pre-fabrication or shop primer after erection / welding in all environments	Uninsulated	(-90 to 400)	CR 1	Table No - 10
			(401 to 550)	CR 2	
SS	Piping, Equipment in process and offsites	Insulated	(-45 to 200)	SS 1	Table No - 02
			(201 to 600)	SS 2	
SS, Duplex SS, Non ferrous & GI	Internal protection of components of coolers / condensers in fresh water service in process units and offsites	Uninsulated	Up to 80	SC 1	Table No - 07
			(80 to 140)	SC 2	
GI	Gratings, Rolling & Stationary ladders, Spiral stairways, Hand rails and GI items	Uninsulated	Up to 60	CG 1	Table No - 09
Structural Steel	Structures in process units and offsite	Uninsulated	Up to 100	CST 1	Table No - 11

Steam/Condensate - Final Shade shall be Grey Aluminum (RAL 9007)

		BHARAT HEAVY ELECTRICALS LIMITED R.C.PURAM, HYDERABAD				STANDARD QUALITY PLAN FOR VENDOR ITEMS ITEM: DESUPERHEATER WITH SPRAY WATER TEMPERATURE CONTROL VALVE BHEL SPEC: As per PO / Enquiry				QP. NO: HYQA/SQP/TC/DESUPERHEATER/2 Rev. No. : 00 DATE: 22.07.2021 PAGE 1 OF 6			
Sl No	Component & Operations	Characteristics	Class	Type Of Check	Quantum Of Check	Ref Document	Acceptance Norms	Format Of Record	*	Agency			Remarks
									D	P	W	V	
1.0 RAW MATERIALS & BOUGHT OUT ITEMS													
1.1	Body, Bonnet, Pipe, Nozzle, Trim, Flanges etc.	Chemical composition & Mechanical Properties, Heat Treatment	Major	Chemical & Mech.	One per melt	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	MTC	✓	2		1	- Review of TCs by IBR - PMI Report for SS/AS & Trim material will be provided for review.
1.2	Body & Bonnet (Casting) (As applicable)	Non Destructive Test	Critical	RT, MPI	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	RT Report, MPI Report	✓	2		1	
1.3	Forging & Bars of Valves (As applicable)	Non Destructive Test	Major	UT	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	UT Report	✓	2		1	
1.4	Actuator	Function	Major	Functional Check	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	Mfr's Test Report	✓	2		1	
1.5	Accessories (Positioner, Air Filter Regulator, Limit Switch, Solenoid Valve etc.	Make & Model	Major	Verification	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	Mfr's Test Report		2		1	
LEGEND: P:- PERFORM, W:- WITNESS, V:- VERIFICATION, INDICATING 1:- BHEL / BHEL NOMINATED INSPECTION AGENCY, 2:- VENDOR / SUB VENDOR AS APPROPRIATE AGAINST EACH COMPONENT / CHARACTERISTICS UNDER THE COLUMNS P, W & V. * D: RECORDS IDENTIFIED WITH TICK (✓) SHALL BE ESSENTIALLY INCLUDED IN QA DOCUMENTATION.				PREPARED BY  Sachin Katiyar Sr. Engineer / QA		REVIEWED BY  B. Ashok Kumar AGM/QA		APPROVED BY  B. Ashok Kumar AGM/QA					
Format no. : HYQA/QP/VSQP Rev.02													

			STANDARD QUALITY PLAN FOR VENDOR ITEMS ITEM: DESUPERHEATER WITH SPRAY WATER TEMPERATURE CONTROL VALVE BHEL SPEC: As per PO / Enquiry				QP. NO: HYQA/SQP/TC/DESUPERHEATER/2 Rev. No. : 00 DATE: 22.07.2021 PAGE 2 OF 6						
Sl No	Component & Operations	Characteristics	Class	Type Of Check	Quantum Of Check	Ref Document	Acceptance Norms	Format Of Record	* D	Agency P W V			Remarks
2.0	INPROCESS INSPECTION												
2.1	MPI on Trim Components	Non Destructive Test	Major	MPI	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	MPI Report	✓	2		1	
2.2	LPI on machined seating surface, Spindle, Butt welds ends of valves	Non Destructive Test	Major	LPI	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	LPI Report	✓	2		1	
2.3	PMI on SS & AS components	PMI Test	Major	PMI Test	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	PMI Report	✓	2		1	
2.4	Availability of approved Weld procedure Specification, Procedure Qualification, Welder Qualification Records.		Major	Review	100%	ASME Sec IX	ASME Sec IX	WPS, PQR, WQR	✓	2		1	
2.5	RT for Butt weld Joint (if any)	Non Destructive Test	Major	RT	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	RT Report	✓	2		1	
2.6	LPI/MPI Test for Fillet Weld Joint (if any)	Non Destructive Test	Major	LPI/MPI	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	LPI/MPI Test Report	✓	2		1	
2.7	Hardness Test on Welds (as applicable)	Hardness Test	Major	Hardness Test	As per spec	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	Test Report	✓	2		1	
LEGEND: P:- PERFORM, W:- WITNESS, V:- VERIFICATION, INDICATING 1:- BHEL / BHEL NOMINATED INSPECTION AGENCY, 2:- VENDOR / SUB VENDOR AS APPROPRIATE AGAINST EACH COMPONENT / CHARACTERISTICS UNDER THE COLUMNS P, W & V. * D: RECORDS IDENTIFIED WITH TICK (✓) SHALL BE ESSENTIALLY INCLUDED IN QA DOCUMENTATION.			PREPARED BY  Sachin Katiyar Sr. Engineer / QA		REVIEWED BY  B. Ashok Kumar AGM/QA		APPROVED BY  B. Ashok Kumar AGM/QA						
Format no. : HYQA/QP/VSQP Rev.02													

 BHARAT HEAVY ELECTRICALS LIMITED R.C.PURAM, HYDERABAD			STANDARD QUALITY PLAN FOR VENDOR ITEMS					QP. NO: HYQA/SQP/TC/DESUPERHEATER/2					
			ITEM: DESUPERHEATER WITH SPRAY WATER TEMPERATURE CONTROL VALVE					Rev. No. : 00					
			BHEL SPEC: As per PO / Enquiry					DATE: 22.07.2021					
								PAGE 3 OF 6					
Sl No	Component & Operations	Characteristics	Class	Type Of Check	Quantum Of Check	Ref Document	Acceptance Norms	Format Of Record	* D	Agency			Remarks
3.0 FINAL INSPECTION & TESTING													
3.1	Final Assembly	Visual, Dimensional Check, End connection & completeness wrt approved Drg/Data sheet	Major	Visual & Measrt.	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	IR	✓	2	1		
3.2	Control Valve	Hydro Test	Major	Hydro Test	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	TC	✓	2	1		
3.3		Seat Leak Test	Major	Seat Leak Test	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	TC	✓	2	1		
3.4	Hydro Test for Desuperheater	Hydro Test	Critical	Hydro Test	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	TC	✓	2	1		
3.5	Actuator Leak Test	Pressure Test on Actuator	Major	Testing	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	TC	✓	2	1		
3.6	Functional Test	Calibration, Valve Travel, Opening & Closing time, other applicable tests as applicable	Major	Functional Test	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	TC	✓	2	1		

LEGEND: P:- PERFORM, W:- WITNESS, V:- VERIFICATION, INDICATING T:- BHEL / BHEL NOMINATED INSPECTION AGENCY, 2:- VENDOR / SUB VENDOR AS APPROPRIATE AGAINST EACH COMPONENT / CHARACTERISTICS UNDER THE COLUMNS P, W & V. * D: RECORDS IDENTIFIED WITH TICK (✓) SHALL BE ESSENTIALLY INCLUDED IN QA DOCUMENTATION.	PREPARED BY  Sachin Katiyar Sr. Engineer / QA	REVIEWED BY  B. Ashok Kumar AGM/QA	APPROVED BY  B. Ashok Kumar AGM/QA
	Format no. : HYQA/QP/VSQP Rev.02		

		BHARAT HEAVY ELECTRICALS LIMITED R.C.PURAM, HYDERABAD		STANDARD QUALITY PLAN FOR VENDOR ITEMS				QP. NO: HYQA/SQP/TC/DESUPERHEATER/2					
				ITEM: DESUPERHEATER WITH SPRAY WATER TEMPERATURE CONTROL VALVE				Rev. No. : 00					
				BHEL SPEC: As per PO / Enquiry				DATE: 22.07.2021					
								PAGE 4 OF 6					
Sl No	Component & Operations	Characteristics	Class	Type Of Check	Quantum Of Check	Ref Document	Acceptance Norms	Format Of Record	*	Agency			Remarks
									D	P	W	V	
3.7	Cv Test	Flow Characteristic	Critical	Type Test	One/ Type	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	Cv Test Report	✓	2	1*	1	*witness is applicable as per approved Datasheet else review of type test report. Refer Note-3
3.8	Certificates	IBR Certification along with stamping, CCOE/PESO, BIS Certification etc. as applicable	Major	Verification	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	IR	✓	2	1		
4.0	PRESERVATION & PACKING												
4.1	Painting	Surface Preparation, Shade, DFT	Major	Visual & Measrt.	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet	IR	✓	2		1	
4.2	Packing & Marking	Packing & Marking	Major	Visual	100%	BHEL Spec. / Approved Drawing/Data Sheet	BHEL Spec. / Approved Drawing/Data Sheet			2		1	

THIS QUALITY PLAN DESCRIBES TYPICAL STANDARD TEST REQUIREMENTS OF DESUPERHEATER. VENDOR MAY BE REQUIRED TO PREPARE & SUBMIT QUALITY PLAN SPECIFIC TO ENQUIRY WITH ADDITIONAL REQUIREMENTS AS PER APPROVED ENGINEERING DOCUMENTS (DRAWING / DATA SHEET).

ABBREVIATIONS:			
MTC – MILL TEST CERTIFICATE	MPI – MAGNETIC PARTICLE INSPECTION	TC – TEST CERTIFICATE, TR – TEST REPORT	
HT – HEAT TREATMENT	UT – ULTRASONIC TEST	TPIA - THIRD PARTY INSPECTION AGENCY APPOINTED BY BHEL	
IR - INSPECTION REPORT	RT – RADIOGRAPHY TEST	COC – CERTIFICATE OF CONFORMITY	
MEASRT - MEASUREMENT	WPS – WELDING PROCEDURE SPECIFICATION	PQR – WELDING PROCESS QUALIFICATION RECORD	
WQR – WELDER QUALIFICATION RECORDS	PO – PURCHASE ORDER	LPI – LIQUID PENETRANT INSPECTION	
LEGEND: P:- PERFORM, W: -WITNESS, V: -VERIFICATION, INDICATING 1:- BHEL / BHEL NOMINATED INSPECTION AGENCY, 2:- VENDOR / SUB VENDOR AS APPROPRIATE AGAINST EACH COMPONENT / CHARACTERISTICS UNDER THE COLUMNS P, W & V. * D: RECORDS IDENTIFIED WITH TICK (✓) SHALL BE ESSENTIALLY INCLUDED IN QA DOCUMENTATION.		PREPARED BY <i>Sachin Katiyar</i> Sachin Katiyar Sr. Engineer / QA	REVIEWED BY <i>B. Ashok Kumar</i> B. Ashok Kumar AGM/QA
		APPROVED BY <i>B. Ashok Kumar</i> B. Ashok Kumar AGM/QA	

Format no. : HYQA/QP/VSQP Rev.02

 BHARAT HEAVY ELECTRICALS LIMITED R.C.PURAM, HYDERABAD		STANDARD QUALITY PLAN FOR VENDOR ITEMS ITEM: DESUPERHEATER WITH SPRAY WATER TEMPERATURE CONTROL VALVE BHEL SPEC: As per PO/ Enquiry				QP. NO: HYQA/SQP/TC/DESUPERHEATER/2 Rev. No. : 00 DATE: 22.07.2021 PAGE 5 OF 6								
		Sl No	Component & Operations	Characteristics	Class	Type Of Check	Quantum Of Check	Ref Document	Acceptance Norms	Format Of Record	Agency			Remarks
											D	P	W	

NOTE:

1. ALL MATERIAL OF CONSTRUCTION (MOC) SHALL BE AS PER APPROVED DATA SHEET / BOM / DRG AND BOUGHT OUT ITEMS (BOIs) SHALL BE PROCURED FROM BHEL AGREED SOURCES / APPROVED DATA SHEET / BOM / DRG/ SPECIFICATION.
2. SPARES PROCURED ALONG WITH MAIN EQUIPMENT/ITEM SHALL BE INSPECTED FOR RELEVANT / APPLICABLE CHECKS AS INDICATED IN THE SQP.
3. VERIFICATION OF TYPE TEST REPORT OF SIMILAR VALVE FOR WHICH TYPE TEST ALREADY CARRIED OUT. TYPE TEST REPORTS SUBMITTED SHALL NOT BE OLDER THAN 5 YEARS FROM DATE OF PURCHASE ORDER.
4. ANY OTHER TESTS/ CHECKS INDICATED IN SPECIFICATION, P.O., OR DRAWING & ANY ADDITIONAL CHECKS ENVISAGED BY BHEL/TPI TO ENSURE WORKMANSHIP, FINISH, AESTHETICS, ETC. SHALL ALSO BE CONDUCTED AND WITNESSED/VERIFIED BY BHEL /TPI / CUSTOMER AS REQUIRED.
5. ANY PROJECT / CUSTOMER SPECIFIC REQUIREMENT, LIKE QP APPROVAL & CUSTOMER/CONSULTANT INSPECTION, WHICH SHALL BE NOTIFIED HAVE TO BE FULFILLED BY THE VENDOR AT THE TIME OF EXECUTION OF ORDER.

Refer Annexure-I for Notes.

LEGEND: P:- PERFORM, W: -WITNESS, V: - VERIFICATION, INDICATING T:- BHEL / BHEL NOMINATED INSPECTION AGENCY, 2:- VENDOR / SUB VENDOR AS APPROPRIATE AGAINST EACH COMPONENT / CHARACTERISTICS UNDER THE COLUMNS P, W & V. * D: RECORDS IDENTIFIED WITH TICK (✓) SHALL BE ESSENTIALLY INCLUDED IN QA DOCUMENTATION.	PREPARED BY  Sachin Katiyar Sr. Engineer / QA	REVIEWED BY  B. Ashok Kumar AGM/QA	APPROVED BY  B. Ashok Kumar AGM/QA
	Format no. : HYQA/QP/VSQP Rev.02		

 BHARAT HEAVY ELECTRICALS LIMITED R.C.PURAM, HYDERABAD		STANDARD QUALITY PLAN FOR VENDOR ITEMS				QP. NO: HYQA/SQP/TC/DESUPERHEATER/2					
		ITEM: DESUPERHEATER WITH SPRAY WATER TEMPERATURE CONTROL VALVE				Rev. No. : 00					
		BHEL SPEC: As per PO / Enquiry				DATE: 22.07.2021					
						PAGE 6 OF 6					
Sl No	Component & Operations	Characteristics	Class	Type Of Check	Quantum Of Check	Ref Document	Acceptance Norms	Format Of Record	* D	Agency P W V	Remarks

Annexure – I

Notes:

1. PRE-DESPATCH INSPECTION PHOTOGRAPHS OF THE EQUIPMENT/ITEM SHALL BE INCLUDED IN QUALITY DOCUMENTATION.
2. LATEST VERSION OF STANDARDS/ DRAWINGS/ TOLERANCES ETC TO BE MENTIONED IN QUALITY PLAN/ DRAWING. THIS QP SHOULD BE READ ALONG WITH BHEL SPEC, BHEL DRAWINGS/ APPROVED DRAWINGS, DATA SHEET, BOM AND PO.
3. DRAWING/ DATA SHEET/ SPECIFICATION SHALL PREVAIL OVER QUALITY PLAN IN CASE OF ANY CONTRADICTION.
4. BHEL RESERVES THE RIGHT FOR CONDUCTING REPEAT TEST, IF REQUIRED.
5. BHEL APPROVED INSPECTION ENGINEERS TO BE DEPLOYED FOR INSPECTION.
6. ONLY LEVEL II & ABOVE QUALIFIED PERSON IN RESPECTIVE NDE TO VERIFY OR WITNESS THE NDT TEST REPORT/ RESULTS
7. INSPECTION TO BE OFFERED ONLY AFTER ENSURING THAT ALL DOCUMENTS (QUALITY PLAN, DRAWINGS, DATA SHEET, PURCHASE SPECIFICATIONS, ETC) ARE AVAILABLE AS PER PURCHASE ORDER
8. VENDOR TO OFFER ORIGINAL TEST CERTIFICATES ISSUED BY THIRD PARTY LABORATORIES OR SUPPLIERS
9. VENDOR TO ENSURE WITH TPIA THAT A NOTE 'COMPARED WITH ORIGINAL TEST CERTIFICATE. REVIEWED, VERIFIED AND FOUND IN ORDER' SHALL CONTAIN WITH EVERY INSPECTION REPORT.
10. ONLY VALID AND CALIBRATED MEASURING INSTRUMENTS AND EQUIPMENT SHALL BE USED – TPIA TO VERIFY.
11. VENDOR TO ENSURE WITH TPIA THAT MATERIAL TEST CERTIFICATE & TRACEABILITY RECORDS ARE AVAILABLE FOR USE OF CORRECT MATERIAL
12. QUALIFICATION OF EQUIPMENT, PROCESS & PERSONNEL FOR SPECIAL PROCESSES LIKE WELDING, BRAZING, PAINTING & METAL COATING ETC. (AS APPLICABLE AS PER PO) SHALL BE ENSURED.
13. VENDOR TO ENSURE THAT ALL CERTIFICATES ARE ENDORSED BY TPIA WITH COMMENTS (WITNESSED OR VERIFIED) AS PER QUALITY PLAN.
14. VENDOR SHALL OFFER LOG SHEETS CONTAINING ACTUAL MEASURED VALUES INSTEAD OF SAYING OK/ NOT OK TO TPIA.
15. VENDOR SHALL SUBMIT COMPLETE INSPECTION AND TEST DOCUMENTATION WHICH EVER IS IDENTIFIED WITH (v) UNDER COLUMN 'D' OF APPROVED QUALITY PLAN SHALL BE ENCLOSED WITH THE INSPECTION REPORT.
16. VENDOR SHALL SUBMIT ORIGINAL COPIES OF ALL INSPECTION AND TEST DOCUMENTS AUTHENTICATED BY TPIA

LEGEND: P:- PERFORM, W:- WITNESS, V:- VERIFICATION, INDICATING 1:- BHEL / BHEL NODMINATED INSPECTION AGENCY, 2:- VENDOR / SUB VENDOR AS APPROPRIATE AGAINST EACH COMPONENT / CHARACTERISTICS UNDER THE COLUMNS P, W & V. * D: RECORDS IDENTIFIED WITH TICK (✓) SHALL BE ESSENTIALLY INCLUDED IN QA DOCUMENTATION.	PREPARED BY  Sachin Katiyar Sr. Engineer / QA	REVIEWED BY  B. Ashok Kumar AGM/QA	APPROVED BY  B. Ashok Kumar AGM/QA
	Format no. : HYQA/QP/VSQP Rev.02		



**Pre-Qualification Requirements (PQR) of Bidder:
for Field instruments/items.**

Ref.: TCEI-FIELD-ITEMS-PQR-2022
Rev.No.: 00

Pre-Qualification Requirements (PQR) of Bidders for Field instruments/items:

- 1.0 Bidder shall be Original Equipment Manufacturer or OEM's authorized distributor/channel partner having offices in India.
- 2.0 In case the OEM authorizes their distributors/channel partner for representing them in totality:
 - a. Authorized distributor/channel partner shall submit authorization certificate from their OEM to quote for complete job, for authorized distributors/channel partner the expiry date of validity of distributorship should be clearly indicated along with documentary evidences
 - b. OEM shall declare that in the event of discontinuation of their partnership, at any point of time during the warranty period, OEM will take the total responsibility for meeting all the commitments made earlier by the authorized distributor/channel partner.
 - c. The responsibility of complete item to be supplied as per BHEL specification requirements including engineering and selection of its components shall be with OEM only. For this OEM shall submit a letter in original.
 - d. Authorized distributors/channel partner shall have association with the OEM for the past five years and should have supplied and commissioned with the OEM supplied items in India.
- 3.0 Bidder to confirm that they will provide spares and services support for the offered items for at least ten years from the date of supply. In case of authorized distributor/channel partner, "After Sales Service" and availability of spares to be guaranteed by OEM for at least ten years.
- 4.0 Bidder shall have an established facility in India for engineering documentation, after sale service for the offered items.
- 5.0 OEM shall offer a proven type design supplied for in any power plant / refinery application.

Bidder shall submit a certificate of satisfactory performance of the offered items from their clients in India (client details to be provided), working satisfactorily at least 2 similar separate executed in last 10 years and out of that at least one shall be in successful operation for 1 year.

Note: If BHEL is unable to verify the PTR furnished, with end user contact details provided above, the offer will be rejected.

- 6.0 The Bidder shall be registered vendor for at least one of the following major Engineering consultants in India:
 - a. NTPC EOC Noida
 - b. Engineers India Limited (IOCL/HPCL/ONGC/BPCL/NRL/TOYO/TPL/TKIS/TECHNIP)

Bidder must submit the documentary evidence/proof in support of vendor registration by submitting the valid vendor registration letter from the respective agency.

Note: Submitting PO copy of supply against particular project is not acceptable.

- 7.0 All correspondence, Documentation, catalogs and Manuals shall be in English language.
- 8.0 Bidder shall furnish the necessary documentary evidence/proof in support of claim for meeting the above Pre-qualification requirements, failing which their offer will be liable for rejection.



BUYER SPECIFIC - ADDITIONAL TERMS & CONDITIONS (ATC)		
NOTE: Bidder to confirm in affirmative by typing "YES" or "Applicable Data" in the response column.		
Sl. No.	DETAILED TERMS & CONDITIONS	BIDDER RESPONSE
1	GENERAL INSTRUCTIONS:	
A	The quotation should be neatly typed and free from over writing/ erasures. Any correction or addition must be authenticated. The offer including annexures and brochures should be submitted in English. All Pages of Techno Commercial Bids (Main Pages), ATC should be signed and Stamped. Prices shall be quoted both in figures and words. In case of any discrepancy in value, the prices quoted in words shall be considered for evaluation and establishing L1 Status.	
B	Bidders to please note that the Terms & conditions contained in this document and ATC are to be read fully before submission of quotations.	
C	Bidders are advised to comply with ATC, should there be any deviations (where deviations are permitted), it shall be entered in the deviation column. BHEL reserves the right to reject such offers.	
2	Documentation for Payment	
A	Indigenous Purchase	
	Following documents shall be submitted immediately on dispatch of material to BHEL HPEP Site a. Original Tax Invoice b. Packing List - clearly showing number of packages, gross weight and net weight. c. Test/Warranty/Guarantee certificates, O&M Manual (If specified in ANNEXURE) d. Insurance intimation/declaration certificate e. Pre-dispatch Inspection report /Third Party Inspection Certificates. f. Consignee copy of LR signed & stamped by Customer/Site representative for DD Items g. e-waybill.	
3	Payment Terms: Following shall be the terms of Payment.	
	Indigenous: a. Micro & Small Enterprises (MSEs) - 100% Direct EFT payment within 45 days b. Medium Enterprises - 100% Direct EFT payment within 60 days c. Non MSME Bidders - 100% direct EFT Payment within 90 Days Note A. Above due date is reckoned from the date of Receipt of material or 15 days from the date of submission of complete set of documents as per PO whichever is later. Payment will be made on acceptance of Material. B. MSEs (covered under MSME Act) need to register and renew periodically and update the same with BHEL C. The taxes that are reimbursed are limited to applicable taxes as on the Purchase Order delivery date or the amount actually paid whichever is less. D. Adherence to the above time schedule of payment is contingent upon Bidder complying with GST provisions and availment of Input Tax Credit by BHEL before the date of payment. E. In case of packaged items, 10% of supply value will be retained till completion of total supplies.	
4	Excess materials supplied beyond tolerance limit as specified in PO will not be paid and bidder may raise credit note for the excess/unaccepted material as per GST law.	
5	Rejected materials , if any, shall be collected by the bidder within 90 days of such communication to the bidder. Beyond this period the bidder forfeits their right to the materials.	
6	PERFORMANCE BANK GUARANTEE (PBG) (Applicable in case mentioned in Annexure G) In case enquiry specifically spells out PBG requirement, PBG is to be submitted by Bidder in requisite format as per Annexure VII. Further detailing on PBG as specified in Annexure G. The PBG shall be for the performance of the goods and shall remain binding notwithstanding such variations, alterations or extensions of item as may be made, give, conceded or agreed to between the Bidder and BHEL under these Terms and conditions or otherwise.	
7	Procurement directly from the manufacturers/ suppliers shall be preferred. However, no agent shall be allowed to represent more than one manufacturer/ supplier in the same tender. Moreover, either the agent could bid on behalf of the manufacturer/ supplier or the manufacturer / supplier could bid directly but not both. In case bids are received from both from the manufacturer/ supplier and the agent, bid received from the agent shall be ignored.	
8	RIGHT OF REJECTION /NON- PLACEMENT OF PO: BHEL reserves the right to accept or reject any or all bid/s in full or part without assigning any reason whatsoever.	



9	INTEGRITY PACT Bidders shall have to enter into Integrity Pact with BHEL as per Annexure VI - for Tender value of rupees two crores and above and shall be signed by the authorized signatory along with the offer, failing which Bidder's offer will be rejected.
	BHEL HPEP is registered with RXIL (TReDS) platform. MSME bidders are requested to get registered with RXIL (TReDS) platform to avail the facility as per the GOI guidelines.
10	Inspection Measuring and Test Equipment (IMTE) used by the Bidder/ Contractor or sub-contractor shall be calibrated, maintained and controlled. Calibration shall be valid and IMTE maintained in sound condition during usage.
11	ISO-9001, ISO14001 and OHSAS 18001 shall be complied
12	Risk Purchase clause: In case bidder fails/delays to supply whole or part of the ordered items or supplies defective items or fails to fulfil any other terms and conditions given in Purchase Order/Contract, BHEL has the right to terminate the order/contract or withdraw balance scope of work/supply and make the purchase of such material / services from elsewhere at the risk and cost of the defaulted bidder. The bidder is liable for the additional expenditure / difference in Cost, if any, including consequential losses which BHEL may sustain by reason of risk purchase in addition to the applicable LD as per the order/contract. Non-performance of contract attracts penal provisions in line with BHEL guidelines for Suspension of Business Dealings (SBD).
13	Any other terms and conditions of the bidder attached / referred against the tender enquiry will not be considered.
14	All drawings, patterns and tools supplied by BHEL or made at BHEL's expense are BHEL's property. These cannot be used or referred to any other party and must be used only in the execution of BHEL's orders.
15	Any amount payable by the bidder under any of the conditions of this contract shall be liable to be adjusted against any amount payable to the bidder under any other work / contract awarded by BHEL HPEP or any other BHEL Units. This is without prejudice to any other action as may be deemed fit by BHEL.
16	The bids of the bidders who are on the banned list and also the bids of the bidders, who engage the services of the banned firms, will be rejected. The list of firms banned by BHEL is available on BHEL web site: www.bhel.com
17	Execution The whole contract is to be executed in the most workman like manner, substantial and approved as per the contracted terms.
18	Progress Report The bidder shall render such report as to the progress of work and in such form as may be called for by the Buyer from time to time. The submission and acceptance of such reports shall not prejudice the rights of the buyer in any manner. Bidder shall communicate to BHEL immediately, the change of address, ownership, contact person(s), the mobile numbers and e-mail of the dealing person concerned. Milestones shall be periodically updated by bidder through PRADAN Portal (https://web.bhelhyd.co.in/mm/). Non updation will adversely affect service rating of bidder performance.
19	Non-disclosure Obligations Drawings, technical documents or other technical information received by one party shall not without the consent of the other party, be used for any other purpose than that for which they were provided. They may not, without the consent of the submitting party, otherwise be used or copied, reproduced, transmitted or communicated to third parties. All information and data contained in general product documentation, whether in electronic or any other form, are confidential and binding only to the extent that they are by reference expressly included in the contract. The bidder shall, as per agreed date/s but not later than the date of delivery, provide free of charge any information and/or drawings which are necessary to permit the Buyer to erect, commission, operate and maintain the product. Such information and drawings shall be supplied as specified in technical specification. All intellectual properties, including designs, drawings and product information etc. exchanged during the formation and execution of the contract shall continue to be the property of the submitting party. The bidder shall provide Buyer with all information pertaining to the delivery in so far as it could be of importance to Buyer. The bidder shall not reveal confidential information to its own employees not involved with the tender/contract and its execution and delivery or to third parties, unless Buyer has agreed to this in writing beforehand. The bidder shall not be entitled to use the Buyer's name in advertisements and other commercial publications including website without prior written permission from Buyer. In the event of violation of the confidentiality as agreed, BHEL will take legal action as deemed fit. Non-disclosure agreement to be entered as per Annexure- II wherever applicable.
20	Inspection and Testing
A	The goods and stores shall be manufactured by approved quality system and each part/component may be inspected and tested by the Buyer prior to shipment and shall comply with relevant requirements. Buyer has the right to inspect at any stage during manufacture/ delivery.

B	Buyer or his authorized representative shall be entitled at all reasonable times during execution to inspect, examine and test at the bidder's premises the material and workmanship of all stores to be supplied under the contract, and if the part of the stores are being manufactured at other premises, the bidder shall obtain for buyer or his authorized representative permission to inspect, examine and test as if the said stores are being manufactured at the bidder's premises. Such inspection, examination and testing, if made shall not release the bidder from any obligation under the contract. For indigenous bidders all costs related to first inspection request shall be borne by the buyer and the cost of subsequent inspections due to non-readiness of material/rework/ rejections shall be borne by the bidder. In case of imports all inspection charges including third party inspections if any shall be borne by the bidder. The cost of inspection staff/third party specified by the Buyer shall be borne by bidder unless otherwise specifically agreed. If the contract provides for tests on the premises of the bidder or any of his sub-contractor/s, bidder shall be responsible to provide such assistance, labor, materials, electricity, fuels, stores, apparatus, instruments as may be required and as may be reasonably demanded to carry out such tests efficiently. Cost of any type test or such other special tests shall be borne by the bidder unless otherwise specifically agreed in the contract. The Bidder shall give the authorized representative of the buyer reasonable notice in writing of the date on and the place at which any stores will be ready for inspection/ testing as provided in the Contract. Annexure – I, may strictly be complied with for the time lines. Any delay in submission of the documents by the bidder will not alter the delivery date.
21	Quality and Condition of the Deliverables
	The bidder shall be responsible for compliance with applicable technical, safety, quality, environmental requirements and other regulations in relation to products, packaging and raw and ancillary materials.
22	Packaging and Dispatch
	The bidder shall package the deliverables safely and carefully and pack them suitably in all respects considering the peculiarity of the material for normal safe transport by sea/air/rail/road to its destination suitably protected against loss, damage, corrosion in transit and the effect or tropical salt laden atmosphere. The packages shall be provided with fixtures/hooks and sling marks as may be required for easy and safe handling by mechanical means. Special packaging conditions/ environmental conditions as defined in the NIT shall be fully complied. Each package must be marked with consignee name, address, P.O. number, Package Number, gross weight & net weight, dimensions (Lx B x H) and bidder's name. The packing shall allow for easy removal and checking of goods on receipt and comply with carrier's conditions of packing or established trade practices. Packing list for goods inside each package with P.O. item No. & quantity must also be fixed securely outside the box to indicate the contents. If any consignment needs special handling instruction, the same shall be clearly marked with standard symbols/instructions. Hazardous material should be notified as such and their packing, transportation and other protection must conform to relevant regulations.
23	Rejected/Short shipments/ warranty/guarantee replacements
	In case of any short shipment during initial supply which is subsequently dispatched by the bidder or any guarantee / warranty replacements shall be dispatched on "DDP-Delivered duty paid BHEL stores" basis for imported items and "FOR-BHEL Stores/designated destination" basis for indigenous items.
24	Non-waiver of Defaults
	If any individual provision of the contract is invalid, the other provisions shall not be affected.
25	Settlement of Disputes
	Except as otherwise specifically provided in the contract, all disputes concerning questions of the facts arising under the contract, shall be decided by the Buyer, subject to written appeal by the bidder to the buyer, whose decision shall be final. Any disputes of differences shall to the extent possible be settled amicably between the parties thereto, failing which the disputed issues shall be settled through arbitration The bidder shall continue to perform the contract, pending settlement of disputes(s).
26	Conciliation clause
	CONCILIATION CLAUSE FOR CONDUCTING CONCILIATION PROCEEDINGS UNDER THE BHEL CONCILIATION SCHEME, 2018: The Parties agree that if at any time (whether before, during or after the arbitral or judicial proceedings), any Disputes (which term shall mean and include any dispute, difference, question or disagreement arising in connection with construction, meaning, operation, effect, interpretation or breach of the agreement, contract or the Memorandum of Understanding, penalty deduction, time extension), which the Parties are unable to settle mutually, arise inter-se the Parties, the same may, be referred by either party to Conciliation to be conducted through Independent Experts Committee to be appointed by competent authority of BHEL from the BHEL Panel of Conciliators. The proceedings of Conciliation shall broadly be governed by Part-III of the Arbitration and Conciliation Act 1996 or any statutory modification thereof and as provided in Procedure in http://www.bhel.com/index.php/story_details?story=2454 . The Procedure together with its Formats will be treated as if the same is part and parcel hereof and shall be as effectual as if set out herein in this ATC.
27	ARBITRATION (WITH SOLE ARBITRATOR)
	Except as provided elsewhere in this Contract, in case amicable settlement is not reached between the Parties, in respect of any dispute or difference; arising out of the formation, breach, termination, penalty deduction, validity or execution of the Contract; time extension, or, the respective rights and liabilities of the Parties; or, in relation to interpretation of any provision of the Contract; or, in any manner touching upon the Contract, then, either Party may, by a notice in writing to the other Party refer such dispute or difference to the sole arbitration . Sole arbitrator to be appointed by Head of the Unit - BHEL , HPEP .
	The Arbitrator shall pass a reasoned award and the award of the Arbitrator shall be final and binding upon the Parties.
	Subject as aforesaid, the provisions of Arbitration and Conciliation Act 1996 (India) or statutory modifications or re-enactments thereof and the rules made thereunder and for the time being in force shall apply to the arbitration proceedings under this clause. The seat of arbitration shall be Sangareddy / Hyderabad, Telangana. The language of arbitration shall be English and the documents shall be submitted in English.
	The cost of arbitration shall initially be borne equally by the Parties subject to the final apportionment of the cost of the arbitration in the award of the Arbitrator.
	Subject to the arbitration in terms of clause 27, the courts at Sangareddy, Telangana State shall have exclusive jurisdiction over any matter arising out of or in connection with this contract. Notwithstanding the existence or any dispute or differences and/or reference for the arbitration, the Contractor shall proceed with and continue without hindrance the performance of its obligations under this Contract with due diligence and expedition in a professional manner except where the Contract has been terminated by either Party in terms of this Contract.



	ARBITRATION FOR CONTRACT WITH PUBLIC SECTOR ENTERPRISE (PSE) OR A GOVERNMENT DEPARTMENT
	In the event of any dispute or difference relating to the interpretation and application of the provisions of commercial contract(s) between Central Public Sector Enterprises (CPSEs/ Port Trusts inter se and also between CPSEs and Government Departments/Organizations (excluding disputes concerning Railways, Income Tax, Customs & Excise Departments), such dispute or difference shall be taken up by either party for resolution through AMRCD as mentioned in DPE OM No 4(1)/2013-DPE(GM/FTS 1835 dated 22-05-2018
28	Applicable Laws and jurisdiction of Courts
	This agreement shall be construed and interpreted in accordance with the laws of India and shall have exclusive jurisdiction of Sangareddy/Hyderabad courts, Telangana, India.
29	BHEL-Fraud prevention policy shall be adhered to.
	The Bidder along with its associate/ Collaborators/ Sub-contractors/ sub-bidders/ consultants/ service providers shall strictly adhere to BHEL Fraud Prevention policy displayed on BHEL Website http://www.bhel.com and shall immediately bring to the notice of BHEL management about any fraud or suspected fraud as soon as it comes to their notice. List of nodal officers is hosted on BHEL Hyderabad website https://hpep.bhel.com/ .
30	Suspected Cartel Formation
	The Bidder declares that they will not enter into any illegal or undisclosed agreement or understanding, whether formal or informal with other Bidder(s). This applies in particular to prices, specifications, certifications, subsidiary contracts, submission or non-submission of bids or any other actions to restrict competitiveness or to introduce cartelization in the bidding process. In case , the Bidder is found having indulged in above activities, suitable action shall be taken by BHEL as per extant policies / guidelines .

Note: Purchase officer has to fill Annexure-I while sending enquiry

Annexure G

Special Conditions of Contract (SCC) for GEM

Sl.No	Buyer ATC Clause No.	Clause	Available Options	BHEL requirements	Bidder response
1		Bank Guarantee against milestone payment	(If envisaged to be detailed with %)		
2	Others	Contract execution Bank Guarantee	__% of PO Value to be submitted after receipt of PO / Not Applicable		
3	17	Performance Bank Guarantee			
4		Guarantee / Warranty Period	a. Guarantee period shall be 12 months from the date of commissioning or 18 months from the date of supply whichever is earlier. b. In case erection & commissioning is involved, Guarantee period shall be 12 months from the date of commissioning. c. In case of equipment bought as a package which are intended to be incorporated in installations or systems, the guarantee period shall be 12 months from the date of commissioning of such equipment.		
5	48	Detailed Billing Breakup (BBU)	Applicable / Not applicable		
6			(If applicable , Bidder shall provide detailed billing break-up in respect of the major items/components (stipulated in the tender) as part of offer .)		
7	Others	Customer approval of vendor	Applicable / Not applicable		
8		Drawing approval	Applicable / Not applicable		
9		QAP approval	Applicable / Not applicable		
10	36	Inspection by	BHEL / BHEL TPI / Vendor TPI / Customer		

Annexure-I			
Major Activity timelines shall be considered for indigenous purchases			
S No	Activity	Agency	Timeline
1	PO acknowledgement	Vendor	_____ days from PO
2	Submission of Drawings and QP	Vendor	_____ days from PO
3	Approval of Drawings and QP	BHEL/Customer	_____ days from PO
4	Raising of Inspection Call	Vendor	_____ days from PO
5	Inspection completion	Self/BHEL/Third party inspection agency	_____ days from inspection call date
6	Despatch Instructions	BHEL	_____ days from inspection report
7	Reciept of Material	Vendor	_____ days from Despatch instructions

Above is illustrative only. Purchase department can add more activities depending on nature of prodcut/contracts

However absence of this annexure in NIT will entail non processing of delivery extention cases in case of delay in supplies of goods owing to reason attributable to BHEL.

(To be executed on Non- Judicial Stamp Paper for an appropriate value.
To be stamped as an agreement)

(For Suppliers on Unit's / Division's PMD)

ANNEXURE-II

Framework Confidentiality Agreement Cum Undertaking

This Agreement made on this the _____ day of (month) _____ 20 ____ ("Effective Date") by and between M/s. BHARAT HEAVY ELECTRICALS LIMITED, having registered office at "BHEL House", Siri Fort, New Delhi – 110049 (India), acting through its _____ Unit (hereinafter may be referred to as "BHEL" or "the company").

And

M/s. _____ (address) _____
represented by authorized representative Sri _____ (herein after referred to as the "Supplier").

The supplier and the company may, unless the context otherwise requires, hereinafter be collectively referred to as "Parties" or singly as the "Party".

RECITALS

Whereas, BHEL is engaged in the design, engineering, manufacturing, construction, testing, commissioning and servicing of a wide range of products, systems and services for the core sectors of the economy, viz. Power, Transmission, Industry, Transportation, Renewable energy, Oil & Gas and Defence and providing associated services to varied customers in relation to which BHEL / its affiliates own valuable information of a secret and confidential nature.

Whereas the Company may, in connection with contract(s) (as defined hereunder) placed or to be placed upon the supplier, or otherwise, from time to time, make available, Technical Information as is defined hereunder.

And Whereas BHEL is willing to provide such Technical Information to the Supplier from time to time and the Supplier understands and acknowledges that such Technical Information is valuable for the Company and as such is willing to protect confidentiality of such information, subject to the terms and conditions set out hereunder.

Now therefore, in view of the foregoing premises and in consideration of the mutual covenants and agreements hereinafter set forth, the parties agree as under:

1. Definitions:

Unless the context so requires, in this Agreement, the following terms will bear the meaning ascribed to the said term in this clause.

- A. **“Contract”** means the contract entered into with a supplier and includes a Purchase Order, or a Work Order for procurement of any goods or for provision of any services.
- B. **“Effective Date”** means the date of this Agreement as mentioned in the preamble of this Agreement.
- C. **“Supplier”** includes a Contractor or a Vendor of the Company whether for supplying of goods or for providing any services under a Contract or both.
- D. **“Technical Information”** includes Drawings, and / or Product Standards and / or Specifications and / or Corporate / Plant Specifications and / or Technological Process Sheets and / or Technical Data Sheets and / or Jigs & Fixtures and / or Pattern & Dies and / or Special Gauges and / or Tools etc. Belonging to or wherein the Company has acquired from a third party a right of user and includes any improvement thereto from time to time whether carried out by the Company or by the Suppliers.
- E. **“Intended Purpose”** means the purpose for which the Technical Information is provided to the supplier under or in connection with a contract.
- F. **“Improvement”** includes any modification made to, or adaptation of, the Technical Information which enhances or is calculated to enhance the performance (Whether in terms of effectiveness or in terms of efficiency or both) of the product and / or the service to be provided by the Supplier under a Contract.

2. This Agreement shall come into force / deemed to have come into force, as the case may be, on the Effective Date; or, on the first date when the Technical Information or any part thereof is provided by BHEL to the supplier; whichever is earlier.

3. **Agreement deemed to be incorporated in each contract:** Unless and to the extent otherwise stipulated in the Contract, the conditions of this Agreement are deemed to be incorporated in all Contracts which may be entered into between the Company and the Supplier. Further, unless otherwise stipulated, the obligations under this Agreement are and will be independent of the obligations under the Contracts and such obligations of the Supplier hereunder will remain of full effect and validity notwithstanding that the period of validity of the Contract has expired by efflux of time stipulated therein; or, the contract has been discharged by performance or breach; or, the termination of the Contracts for any reason whatsoever.

4. **Ownership:**

4.1 The Company may, from time to time, make available to the Supplier, Technical Information on a non-exclusive basis by way of loan.

4.2 The Supplier acknowledges and agrees that all Technical Information and copies thereof that are or may be provided by the Company to the Supplier, are and shall remain the property of

BHEL or that of the concerned entity from whom BHEL has obtained the Technical Information and such Technical Information are and shall constitute trade secrets of the BHEL. Nothing in this Agreement or in any disclosures made hereunder by or on behalf of the Company shall be construed as granting upon the Supplier any patent, copyright or design or any other intellectual property rights of whatsoever description that subsists or may hereinafter exist in the Technical Information. Furthermore, nothing in this Agreement or in any disclosures made hereunder by or on behalf of the Company shall be construed as granting upon the supplier any license or right of use of such patent, copyright or design or any other intellectual property rights of whatsoever description which may now or hereafter exist in the Technical Information except for use of the Technical Information strictly in accordance with this Agreement and the contract and / or as directed in writing by the Company, solely for the Intended Purpose under the Contract.

- 4.3 Neither party is obligated by or under this Agreement to purchase from or provide to the other party any service or product and that any such purchase / sale of any product and / or service by one party to the other party will be governed by the Contract if any, that may be entered into by and between the Company and the Supplier.
- 4.4 The Supplier is / has been made well aware and acknowledges that the Technical Information being / which may be shared with it by the Company has been either generated by the Company by incurring huge investment and cost or obtained from foreign collaborators under Technical Collaboration Agreement (TCA) with stringent confidentiality conditions.
- 4.5 The supplier agrees and undertakes to adhere to confidentiality requirements as applicable to BHEL under a TCA and also ensure that the confidentiality requirements are adhered to by all its concerned employees or sub-contractors /suppliers (where permitted to be engaged by BHEL). Any damages, losses, expenses of any description whatsoever, arising out of or in connection with a breach of the confidentiality requirements under a TCA owing to any act or omission on the part of the supplier or its employees or sub-contractors / suppliers that is claimed by a foreign collaborator from the Company shall be wholly borne by the Supplier and it shall keep BHEL fully indemnified in this behalf. The demand by the Company shall be conclusive upon the Supplier who shall thereupon forthwith pay to the Company without demur, dispute or delay the amount as demanded without demanding any further proof thereof.
- 4.6 The Supplier agrees and undertakes that unless so decided and advised by the Company in writing all rights / title to any Improvement to the Technical Information shall vest in the Company. The Supplier undertakes and agrees to inform forthwith to the Company of any such Improvement made to the Technical Information and transfer all drawings / documents or other materials connected with such Improvement to the Company and also agrees to fully cooperate with the Company for protecting the Company's interests in such Improvements

in the Technical Information including but not limited to obtaining necessary protection for the intellectual property rights in such improvement, if so desired by the Company. If a question arises whether a modification amounts to improvement to the Technical Information, the same shall be decided by the Company and such decision shall be final and binding upon the supplier.

5. Use and Non – Disclosure:

5.1 Unless otherwise stipulated by the Company, all Technical Information made available to the supplier, by the Company shall be treated as Confidential irrespective of whether the same is marked or otherwise denoted to be Confidential or not.

5.2 The Supplier undertakes and agrees that the Technical Information in its possession shall be held in strict confidence and will be used strictly in accordance with this Agreement and solely for the Intended Purpose under the Contract. Use of the Technical Information for any other purpose other than Intended Purpose is prohibited.

5.3 In particular, the Supplier shall not use Technical Information or any Improvement in its possession for the manufacture or procurement of the product(s) or components or parts thereof or use the Technical Information or any portion thereof or any modification or adaptation thereof in any form to provide any product and / or service to any third party, without the prior written consent of the Company.

5.4 The Supplier shall not disclose any of such Technical Information to any third party without the prior written consent of the Company. The Supplier agrees that without prior written consent of the Company, the supplier shall not disclose to a third party about the existence of this Agreement, or of the fact that it is / was in possession of or has experience in the use of any Technical Information nor shall the Supplier share in any manner whatsoever, with a third party, the name or details of any Contract(s) awarded by the Company to it or performed by the Supplier or the scope of work thereof or share any document or correspondence by and between the Company and the supplier in or in connection with this Agreement or such Contract(s). Notwithstanding what is stated elsewhere, the overall responsibility of any breach of the confidentiality provisions under this Agreement shall rest with the Supplier.

5.5 This Supplier undertakes and agrees not to make copies or extracts of and not to disclose to other any or all of the Technical Information in its possession, except as follows:

(a) The Supplier may disclose the Technical Information to such of its officers and employees strictly to the extent as is necessary for such officer or employee for the Intended Purpose, provided that the Confidential Information (or copies thereof) disclosed shall be marked

clearly as the confidential and proprietary information of Company and that such officers and employees shall similarly be bound by undertakings of confidence, restricted use and non-disclosure in respect of the Technical Information. The Supplier shall be responsible for any breach of such confidentiality provisions by such officers and employees.

- (b) With the prior written consent of Company, the supplier may disclose for the Intended Purpose such Technical Information as is provided for in such consent to such of its professional advisers: consultants, insurers and subcontractors who shall be similarly bound by undertakings of confidence, restricted use and non-disclosure in respect of such Technical Information.
- (c) The Supplier shall not be prevented to make any disclosure required by (i) order of a court of competent jurisdiction or (ii) any competent regulatory authority or agency where such disclosure is required by law, provided that where the supplier intends to make such disclosure, it shall first consult Company and take all reasonable steps requested by it to minimize the extent of the Technical Information disclosed and to make such disclosure in confidence and also shall cooperate with the Company in seeking any protective order or any other remedy from proper authority in this matter.

6. Exceptions:

The Obligations of the Supplier pursuant to the provisions of this agreement shall not apply to any Confidential Information that:

- a) was / is known to, or in the possession of the Supplier prior to disclosure thereof by the Company;
- b) is or becomes publicly known, otherwise than as a result of a breach of this agreement by the Supplier.
- c) is developed independently of the Disclosing party by the Supplier in circumstances that do not amount to a breach of the provisions of this Agreement or the Contract;
- d) is received from a third party in circumstances that do not result in a breach of the provisions of this Agreement.

- 7. The Obligation of maintaining confidentiality of the Technical Information on each occasion, shall subsist for the entire duration during which the Technical Information / equipment is in possession of the Supplier and shall thereafter subsist for a further period of _____ years from the date when the complete Technical Information has been returned in portions on different dates, the period of ____ years will be reckoned from the date when the last portion of the Technical Information has been returned. Notwithstanding the expiry of the confidentiality obligation, the obligation of the Supplier under clause 5.4 shall continue to subsist for a further period of _____ years.

8. Warranties & Undertakings:

- a) The Supplier undertakes to ensure the due observance of the undertakings of confidence, restricted use and non-disclosure by its persons to whom it discloses or releases copies or extracts of the Technical Information.
- b) The Supplier shall keep the Technical Information or improvement made therein properly segregated and not mix up the same with any other material / documents belonging to him / it or to any other third party.
- c) The Supplier further undertakes that he / it shall not hypothecate or give on lease or otherwise alienate or do away with any of the Technical Information and / or equipment of the Company, made available to him / it, and undertakes that he / it shall hold the same as a trustee, in capacity of custodian thereof and use / utilise the same solely for the purpose of executing the contract awarded by the Company.
- d) The Supplier further undertakes that he / it shall return all the equipment and / or Technical Information as far as practicable in the same condition in which the same was made available to him / it by the Company together with any Improvement thereon and the documents connected with such Improvement, to the Company forthwith upon completion of the scope of work or contract for which such Technical Information was provided by the Company to it or as directed by the Company together with a confirmation by way of an affidavit or in such manner as directed by the Company that it has not retained any equipment and / or Technical Information / improvement thereof. In case any such equipment and / or Technical Information or thereof shall remain in his possession or is not capable of being returned, the retention and use of such Technical Information or improvement thereto shall continue to be governed by this Agreement.
- e) The Supplier undertakes to indemnify the Company for all the direct, indirect and / or consequential losses, damages, expenses whatsoever including any consequential loss of business, profits suffered by the Company owing to breach by the Supplier of its obligations under this Agreement and / or the confidentiality requirements, if any, contained in the Contract and that the Supplier hereby agrees that the decision of the Company in all such or any such matter/s shall be final and binding on the Supplier. On mere written demand of the Company, the Supplier shall forthwith and without demur or delay pay to the Company any such sum as determined by the Company as the amount of loss or damage or expense which has been suffered by the Company. The Supplier agrees that the Company shall be entitled to withhold and appropriate any amount payable to the Supplier under any Contract then existing between the Company and the Supplier, in case the Supplier fails to make payment, in terms of the written demand, within 7 days thereof. Without prejudice to the forgoing actions, in respect to any breach of this Agreement, the Company shall be entitled to take

any other action against the Supplier as per applicable laws, the Contract, Company's applicable policies, guidelines rules, procedures, etc.

9. Without prejudice to any other mode of recovery as may be available to the Company for recovery of the amount determined as due as per Clause 9 (f) hereinabove, the Company shall have a right to withhold, recovery and appropriate the amount due towards such losses, damages, expenses, from any amount due to the Supplier in respect of any other Contract (s) placed on him / it by any department / office / unit/ division of the said Company.

10. Arbitration & Conciliation:

Except as provided elsewhere in this Contract, in case amicable settlement is not reached between the Parties, in respect of any dispute or difference; arising out of the formation, breach, termination, penalty deduction, validity or execution of the Contract; time extension, or, the respective rights and liabilities of the Parties; or, in relation to interpretation of any provision of the Contract; or, in any manner touching upon the Contract, then, either Party may, by a notice in writing to the other Party refer such dispute or difference to the sole arbitration. Sole arbitrator to be appointed by Head of the Unit - BHEL, HPEP.

The Arbitrator shall pass a reasoned award and the award of the Arbitrator shall be final and binding upon the Parties.

Subject as aforesaid, the provisions of Arbitration and Conciliation Act 1996 (India) or statutory modifications or re-enactments thereof and the rules made thereunder and for the time being in force shall apply to the arbitration proceedings under this clause.

The seat of arbitration shall be Sangareddy / Hyderabad, Telangana. The language of arbitration shall be English and the documents shall be submitted in English.

The cost of arbitration shall initially be borne equally by the Parties subject to the final apportionment of the cost of the arbitration in the award of the Arbitrator.

Subject to the arbitration in terms of clause 46 of ITB (clause 27 of ATC GeM), the courts at Sangareddy, Telangana State shall have exclusive jurisdiction over any matter arising out of or in connection with this contract.

Notwithstanding the existence or any dispute or differences and/or reference for the arbitration, the Contractor shall proceed with and continue without hindrance the performance of its obligations under this Contract with due diligence and expedition in a professional manner except where the Contract has been terminated by either Party in terms of this Contract.

ARBITRATION FOR CONTRACT WITH PUBLIC SECTOR ENTERPRISE (PSE) OR A GOVERNMENT DEPARTMENT

In the event of any dispute or difference relating to the interpretation and application of the provisions of commercial contract(s) between Central Public Sector Enterprises (CPSEs/ Port Trusts inter se and also between CPSEs and Government Departments/Organizations (excluding disputes concerning Railways, Income Tax, Customs & Excise Departments), such dispute or difference shall be taken up by either party for resolution through AMRCD as mentioned in DPE OM No 4(1)/2013-DPE(GM/FTS 1835 dated 22-05-2018

CONCILIATION CLAUSE FOR CONDUCTING CONCILIATION PROCEEDINGS UNDER THE BHEL CONCILIATION SCHEME, 2018: The Parties agree that if at any time (whether before, during or after the arbitral or judicial proceedings), any Disputes (which term shall mean and include any dispute, difference, question or disagreement arising in connection with construction, meaning, operation, effect, interpretation or breach of the agreement, contract or the Memorandum of Understanding, penalty deduction, time extension), which the Parties are unable to settle mutually, arise inter-se the Parties, the same may, be referred by either party to Conciliation to be conducted through Independent Experts Committee to be appointed by competent authority of BHEL from the BHEL Panel of Conciliators.

The proceedings of Conciliation shall broadly be governed by Part-III of the Arbitration and Conciliation Act 1996 or any statutory modification thereof and as provided in Procedure in http://www.bhel.com/index.php/story_details?story=2454 . The Procedure together with its Formats will be treated as if the same is part and parcel hereof and shall be as effectual as if set out herein in this ITB

11. Governing Law & Jurisdiction:

This agreement shall be construed and interpreted in accordance with the laws of India and shall have exclusive jurisdiction of Sangareddy/Hyderabad courts, Telangana, India.

SIGNATURE

WITNESSES

1

Name:

Address:

2

Name:

Address:

Annexure - III

Proforma for self-certification by Supplier for minimum local content on their letter head for tender value less than Rs 10 Crore

"We _____ (Name of Manufacturer) undertake that we meet the mandatory minimum Local Content (LC) requirement i.e. _____ (to be filled as notified in the policy) for claiming Purchase Preference linked with Local Contents under the Govt. policy against tender no. _____."

Auditor's certification with respect to minimum local content on the letter head of Statutory Auditor for tender value above Rs.10 crore

"We _____ the statutory auditor of M/s _____ (name of the bidder) hereby certify that M/s _____ (name of manufacturer) meet the mandatory Local Content requirements of the Goods and/or Services i.e. _____ (to be filled as notified in the policy) quoted vide offer No. _____ dated _____ against BHEL's tender No. _____ by M/s _____ (Name of the bidder)."

Annexure - IV

Proforma for self-certification by Supplier for Compliance to Clause No 20 (B)

I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India and I certify that M/s.... (Name of firm) is **not from such a country/is from such a country** (delete whichever is NOT applicable) and has been duly registered with the Competent authority (delete if NOT applicable) . I hereby certify M/s fulfills all requirements in this regard and is eligible to be considered . (where applicable , valid registration by the competent authority shall be attached)

Sd/-

Authorised Signatory with Stamp

INTEGRITY PACT

Between

Bharat Heavy Electricals Ltd. (BHEL), a company registered under the Companies Act 1956 and having its registered office at "BHEL House", Siri Fort, New Delhi – 110049 (India) hereinafter referred to as "The Principal", which expression unless repugnant to the context or meaning hereof shall include its successors or assigns of the ONE PART

and

_____, (description of the party along with address), hereinafter referred to as "The Bidder/ Contractor" which expression unless repugnant to the context or meaning hereof shall include its successors or assigns of the OTHER PART

Preamble

The Principal intends to award, under laid-down organizational procedures, contract/s for

_____. The Principal values full compliance with all relevant laws of the land, rules and regulations, and the principles of economic use of resources, and of fairness and transparency in its relations with its Bidder(s)/ Contractor(s).

In order to achieve these goals, the Principal will appoint Independent External Monitor(s), who will monitor the tender process and the execution of the contract for compliance with the principles mentioned above.

Section 1 – Commitments of the Principal

- 1.1 The Principal commits itself to take all measures necessary to prevent corruption and to observe the following principles:-
 - 1.1.1 No employee of the Principal, personally or through family members, will in connection with the tender for, or the execution of a contract, demand, take a promise for or accept, for self or third person, any material or immaterial benefit which the person is not legally entitled to.
 - 1.1.2 The Principal will, during the tender process treat all Bidder(s) with equity and reason. The Principal will in particular, before and during the tender process, provide to all Bidder(s) the same information and will not provide to any Bidder(s) confidential / additional information through which the Bidder(s) could obtain an advantage in relation to the tender process or the contract execution.
 - 1.1.3 The Principal will exclude from the process all known prejudiced persons.
- 1.2 If the Principal obtains information on the conduct of any of its employees which is a penal offence under the Indian Penal Code 1860 and Prevention of Corruption Act 1988 or any other statutory penal enactment, or if there be a substantive suspicion in this regard, the Principal will inform its Vigilance Office and in addition can initiate disciplinary actions.

Section 2 – Commitments of the Bidder(s)/ Contractor(s)

- 2.1 The Bidder(s)/ Contractor(s) commit himself to take all measures necessary to prevent corruption. He commits himself to observe the following principles during his participation in the tender process and during the contract execution.
 - 2.1.1 The Bidder(s)/ Contractor(s) will not, directly or through any other person or firm, offer, promise or give to the Principal or to any of the Principal's employees involved

in the tender process or the execution of the contract or to any third person any material, immaterial or any other benefit which he / she is not legally entitled to, in order to obtain in exchange any advantage of any kind whatsoever during the tender process or during the execution of the contract.

- 2.1.2 The Bidder(s)/ Contractor(s) will not enter with other Bidder(s) into any illegal or undisclosed agreement or understanding, whether formal or informal. This applies in particular to prices, specifications, certifications, subsidiary contracts, submission or non-submission of bids or any other actions to restrict competitiveness or to introduce cartelization in the bidding process.
- 2.1.3 The Bidder(s)/ Contractor(s) will not commit any penal offence under the relevant IPC/ PC Act; further the Bidder(s)/ Contractor(s) will not use improperly, for purposes of competition or personal gain, or pass on to others, any information or document provided by the Principal as part of the business relationship, regarding plans, technical proposals and business details, including information contained or transmitted electronically.
- 2.1.4 The Bidder(s)/ Contractor(s) will, when presenting his bid, disclose any and all payments he has made, and is committed to or intends to make to agents, brokers or any other intermediaries in connection with the award of the contract.
- 2.2 The Bidder(s)/ Contractor(s) will not instigate third persons to commit offences outlined above or be an accessory to such offences.

Section 3 – Disqualification from tender process and exclusion from future contracts

If the Bidder(s)/ Contractor(s), before award or during execution has committed a transgression through a violation of Section 2 above, or acts in any other manner such as to put his reliability or credibility in question, the Principal is entitled to disqualify the Bidder(s)/ Contractor(s) from the tender process or take action as per the separate "Guidelines on Banning of Business dealings with Suppliers/ Contractors". framed by the Principal.

Section 4 – Compensation for Damages

- 4.1 If the Principal has disqualified the Bidder from the tender process prior to the award according to Section 3, the Principal is entitled to demand and recover the damages equivalent Earnest Money Deposit/Bid Security.
- 4.2 If the Principal has terminated the contract according to Section 3, or if the Principal is entitled to terminate the contract according to section 3, the Principal shall be entitled to demand and recover from the Contractor liquidated damages equivalent to 5% of the contract value or the amount equivalent to Security Deposit/Performance Bank Guarantee, whichever is higher.

Section 5 – Previous Transgression

- 5.1 The Bidder declares that no previous transgressions occurred in the last 3 years with any other company in any country conforming to the anti-corruption approach or with any other Public Sector Enterprise in India that could justify his exclusion from the tender process.
- 5.2 If the Bidder makes incorrect statement on this subject, he can be disqualified from the tender process or the contract, if already awarded, can be terminated for such reason.

Section 6 – Equal treatment of all Bidders/ Contractors/ Sub-contractors

- 6.1 The Bidder(s)/ Contractor(s) undertake(s) to obtain from all subcontractors a commitment consistent with this Integrity Pact and report Compliance to the Principal. This commitment shall be taken only from those sub-contractors whose contract value is more than 20 % of Bidder's/ Contractor's contract value with the Principal. The Bidder(s)/ Contractor(s) shall continue to remain responsible for any default by his Sub-contractor(s).
- 6.2 The Principal will enter into agreements with identical conditions as this one with all Bidders and Contractors.
- 6.3 The Principal will disqualify from the tender process all bidders who do not sign this pact or violate its provisions.

Section 7 – Criminal Charges against violating Bidders/ Contractors /Sub-contractors

If the Principal obtains knowledge of conduct of a Bidder, Contractor or Subcontractor, or of an employee or a representative or an associate of a Bidder, Contractor or Subcontractor which constitutes corruption, or if the Principal has substantive suspicion in this regard, the Principal will inform the Vigilance Office.

Section 8 –Independent External Monitor(s)

- 8.1 The Principal appoints competent and credible Independent External Monitor for this Pact. The task of the Monitor is to review independently and objectively, whether and to what extent the parties comply with the obligations under this agreement.

- 8.2 The Monitor is not subject to instructions by the representatives of the parties and performs his functions neutrally and independently. He reports to the CMD, BHEL.
- 8.3 The Bidder(s)/ Contractor(s) accepts that the Monitor has the right to access without restriction to all contract documentation of the Principal including that provided by the Bidder(s)/ Contractor(s). The Bidder(s)/ Contractor(s) will grant the monitor, upon his request and demonstration of a valid interest, unrestricted and unconditional access to his contract documentation. The same is applicable to Sub-contractor(s). The Monitor is under contractual obligation to treat the information and documents of the Bidder(s)/ Contractor(s) / Sub-contractor(s) with confidentiality.
- 8.4 The Principal will provide to the Monitor sufficient information about all meetings among the parties related to the contract provided such meetings could have an impact on the contractual relations between the Principal and the Contractor. The parties offer to the Monitor the option to participate in such meetings.
- 8.5 As soon as the Monitor notices, or believes to notice, a violation of this agreement, he will so inform the Management of the Principal and request the Management to discontinue or take corrective action, or heal the situation, or to take other relevant action. The Monitor can in this regard submit non-binding recommendations. Beyond this, the Monitor has no right to demand from the parties that they act in a specific manner, refrain from action or tolerate action.
- 8.6 The Monitor will submit a written report to the CMD, BHEL within 8 to 10 weeks from the date of reference or intimation to him by the Principal and, should the occasion arise, submit proposals for correcting problematic situations.
- 8.7 The CMD, BHEL shall decide the compensation to be paid to the Monitor and its terms and conditions.
- 8.8 If the Monitor has reported to the CMD, BHEL, a substantiated suspicion of an offence under relevant IPC / PC Act, and the CMD, BHEL has not, within reasonable time, taken visible action to proceed against such offence or reported it to the Vigilance Office, the

Monitor may also transmit this information directly to the Central Vigilance Commissioner, Government of India.

8.9 The number of Independent External Monitor(s) shall be decided by the CMD, BHEL.

8.10 The word 'Monitor' would include both singular and plural.

Section 9 – Pact Duration

9.1 This Pact begins and shall be binding on and from the submission of bid(s) by bidder(s). It expires for the Contractor 12 months after the last payment under the respective contract and for all other Bidders 6 months after the contract has been awarded.

9.2 If any claim is made / lodged during this time, the same shall be binding and continue to be valid despite the lapse of this pact as specified as above, unless it is discharged/ determined by the CMD, BHEL.

Section 10 – Other Provisions

10.1 This agreement is subject to Indian Laws and jurisdiction shall be registered office of the Principal, i.e. New Delhi.

10.2 Changes and supplements as well as termination notices need to be made in writing. Side agreements have not been made.

10.3 If the Contractor is a partnership or a consortium, this agreement must be signed by all partners or consortium members.

10.4 Should one or several provisions of this agreement turn out to be invalid, the remainder of this agreement remains valid. In this case, the parties will strive to come to an agreement to their original intentions.

10.5 Only those bidders/ contractors who have entered into this agreement with the Principal would be competent to participate in the bidding. In other words, entering into this agreement would be a preliminary qualification.

For & On behalf of the Principal

(Office Seal)

For & On behalf of the Bidder/ Contractor

(Office Seal)

Place-----

Date-----

Witness: _____

(Name & Address) _____

Witness: _____

(Name & Address) _____

BANK GUARANTEE FOR PERFORMANCE SECURITY

Bank Guarantee No:

Date:

To

NAME

& ADDRESSES OF THE BENEFICIARY

Dear Sirs,

In consideration of the Bharat Heavy Electricals Limited ¹ (hereinafter referred to as the 'Employer' which expression shall unless repugnant to the context or meaning thereof, include its successors and permitted assigns) incorporated under the Companies Act, 1956 and having its registered office at _____ through its Unit at.....(name of the Unit) having awarded to (Name of the Vendor / Contractor / Supplier) having its registered office at _____ ² hereinafter referred to as the 'Contractor/Supplier', which expression shall unless repugnant to the context or meaning thereof, include its successors and permitted assigns), a contract Ref No.....dated³ valued at Rs.....⁴ (Rupees -----)/FC.....(in words.....) for⁵ (hereinafter called the 'Contract') and the Contractor having agreed to provide a Contract Performance Guarantee, equivalent to% (.... Percent) of the said value of the Contract to the Employer for the faithful performance of the Contract,

we,, (hereinafter referred to as the Bank), having registered/Head office at and inter alia a branch at being the Guarantor under this Guarantee, hereby, irrevocably and unconditionally undertake to forthwith and immediately pay to the Employer a maximum amount Rs ----- (Rupees -----) without any demur, immediately on a demand from the Employer, . Any such demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this guarantee. However, our liability under this guarantee shall be restricted to an amount not exceeding Rs. _____.

We undertake to pay to the Employer any money so demanded notwithstanding any dispute or disputes raised by the Contractor/ Supplier in any suit or proceeding pending before any Court or Tribunal relating thereto our liability under this present being absolute and unequivocal.

The payment so made by us under this Guarantee shall be a valid discharge of our liability for payment thereunder and the contractors/supplier shall have no claim against us for making such payment.

We thebank further agree that the guarantee herein contained shall remain in full force and effect during the period that would be taken for the performance of the said Contract and that it shall continue to be enforceable till all the dues of the Employer under or by virtue of the said Contract have been fully paid and its claims satisfied or discharged.

We BANK further agree with the Employer that the Employer shall have the fullest liberty without our consent and without affecting in any manner our obligations hereunder to vary any of the terms and conditions of the said Contract or to extend time of performance by the said Contractor/Supplier from time to time or to postpone for any time or from time to time any of the powers exercisable by the Employer against the said Contractor/Supplier and to forbear or enforce any of the terms and conditions relating to the said Agreement and we shall not be relieved from our liability by reason of any such variation, or extension being granted to the said Contractor/Supplier or for any forbearance, act or omission on the part of the Employer or any indulgence by the Employer to the said Contractor/Supplier or by any such matter or thing whatsoever which under the law relating to sureties would but for this provision have effect of so relieving us.

The Bank also agrees that the Employer at its option shall be entitled to enforce this Guarantee against the Bank as a principal debtor, in the first instance without proceeding against the Contractor and notwithstanding any security or other guarantee that the Employer may have in relation to the Contractor's liabilities.

This Guarantee shall remain in force upto and including.....⁶ and shall be extended from time to time for such period as may be desired by Employer.

This Guarantee shall not be determined or affected by liquidation or winding up, dissolution or change of constitution or insolvency of the Contractor/Supplier but shall in all respects and for all purposes be binding and operative until payment of all money payable to the Employer in terms thereof.

Unless a demand or claim under this guarantee is made on us in writing on or before the⁷we shall be discharged from all liabilities under this guarantee thereafter.

We, BANK lastly undertake not to revoke this guarantee during its currency except with the previous consent of the Employer in writing.

Notwithstanding anything to the contrary contained hereinabove:

- a) The liability of the Bank under this Guarantee shall not exceed.....⁸
- b) This Guarantee shall be valid up to⁹

- c) Unless the Bank is served a written claim or demand on or before _____¹⁰ all rights under this guarantee shall be forfeited and the Bank shall be relieved and discharged from all liabilities under this guarantee irrespective of whether or not the original bank guarantee is returned to the Bank.

We, _____ Bank, have power to issue this Guarantee under law and the undersigned as a duly authorized person has full powers to sign this Guarantee on behalf of the Bank.

For and on behalf of
(Name of the Bank)

Dated.....

Place of Issue.....

¹ NAME AND ADDRESS OF EMPLOYER I.e Bharat Heavy Electricals Limited

² NAME AND ADDRESS OF THE VENDOR /CONTRACTOR / SUPPLIER.

³ DETAILS ABOUT THE NOTICE OF AWARD/CONTRACT REFERENCE

⁴ PROJECT/SUPPLY DETAILS

⁵ BG AMOUNT IN FIGURES AND WORDS

⁶ VALIDITY DATE

⁷ DATE OF EXPIRY OF CLAIM PERIOD

⁸ BG AMOUNT IN FIGURES AND WORDS.

⁹ VALIDITY DATE

¹⁰ DATE OF EXPIRY OF CLAIM PERIOD

Note:

1. Units are advised that expiry of claim period may be kept 2/3 months after validity date.

2. In Case of Bank Guarantees submitted by Foreign Vendors-

- a. **From Nationalized/Public Sector / Private Sector/ Foreign Banks (BG issued by Branches in India)** can be accepted subject to the condition that the Bank Guarantee should be enforceable in the town/city or at nearest branch where the Unit is located i.e. Demand can be presented at the Branch located in the town/city or at nearest branch where the Unit is located.
- b. **From Foreign Banks (wherein Foreign Vendors intend to provide BG from local branch of the Vendor country's Bank)**
 - b.1 In such cases, in the Tender Enquiry/ Contract itself, it may be clearly specified that Bank Guarantee issued by **any of the Consortium Banks only** will be accepted by BHEL. As such, Foreign Vendor needs to make necessary arrangements for issuance of Counter- Guarantee by Foreign Bank in favour of the Indian Bank (BHEL's Consortium Bank). It is advisable that all charges for issuance of Bank Guarantee/ counter- Guarantee should be borne by the Foreign Vendor. The tender stipulation should clearly specify these requirements.

- b.2** In case, Foreign Vendors intend to provide BG from Overseas Branch of our Consortium Bank (e.g. if a BG is to be issued by SBI Frankfurt), the same is acceptable. However, the procedure at **sl.no. b.1** will required to be followed.
- b.3** The BG issued may preferably be subject to Uniform Rules for Demand Guarantees (URDG) 758 (as amended from time to time). In case, of Foreign Vendors, the BG Format provided to them should clearly specify the same.
- b.4** The BG should clearly specify that the demand or other document can be presented in electronic form.